



US012404227B2

(12) **United States Patent**  
**Copp et al.**

(10) **Patent No.:** **US 12,404,227 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **METHODS OF MAKING BEMPEDOIC ACID AND COMPOSITIONS OF THE SAME**

(71) Applicant: **Esperion Therapeutics, Inc.**, Ann Arbor, MI (US)

(72) Inventors: **Richard Copp**, Pinckney, MI (US); **Mohamed Abdelnasser**, New City, NY (US); **Christopher M. Cimarusti**, Clementon, NJ (US); **Jonathan Lane**, Longmont, CO (US); **Michael Barkman**, Louisville, CO (US); **Rasidul Amin**, Cary, NC (US); **Arthur John Cooper**, Mentor, OH (US); **Damodaragounder Gopal**, Highland Heights, OH (US); **Philipp Selig**, Linz (AT)

(73) Assignee: **Esperion Therapeutics, Inc.**, Ann Arbor, MI (US)

(\*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 134 days.

(21) Appl. No.: **18/450,591**

(22) Filed: **Aug. 16, 2023**

(65) **Prior Publication Data**

US 2024/0182396 A1 Jun. 6, 2024

**Related U.S. Application Data**

(63) Continuation of application No. 17/577,829, filed on Jan. 18, 2022, now Pat. No. 11,760,714, which is a continuation of application No. 17/150,321, filed on Jan. 15, 2021, now abandoned, which is a continuation of application No. PCT/US2020/038622, filed on Jun. 19, 2020.

(60) Provisional application No. 62/864,873, filed on Jun. 21, 2019.

(51) **Int. Cl.**  
**C07C 51/04** (2006.01)  
**C07C 59/285** (2006.01)

(52) **U.S. Cl.**  
CPC ..... **C07C 51/04** (2013.01); **C07C 59/285** (2013.01); **C07B 2200/13** (2013.01)

(58) **Field of Classification Search**  
CPC .... **C07C 59/285**; **C07C 51/42**; **C07B 2200/13**  
USPC ..... 514/574  
See application file for complete search history.

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Primary Examiner — Ana Z Muresan

(74) Attorney, Agent, or Firm — Goodwin Procter LLP

(57) **ABSTRACT**

The invention provides methods of preparing 8-hydroxy-2, 2,14,14-tetramethylpentadecanedioic acid and methods of making a pharmaceutical material comprising a purified amount of 8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid. Also provided are compositions and pharmaceutical materials including a purified amount of 8-hydroxy-2, 2,14,14-tetramethylpentadecanedioic acid as well as methods of treating various diseases and conditions using the compositions and pharmaceutical materials.

**20 Claims, 6 Drawing Sheets**

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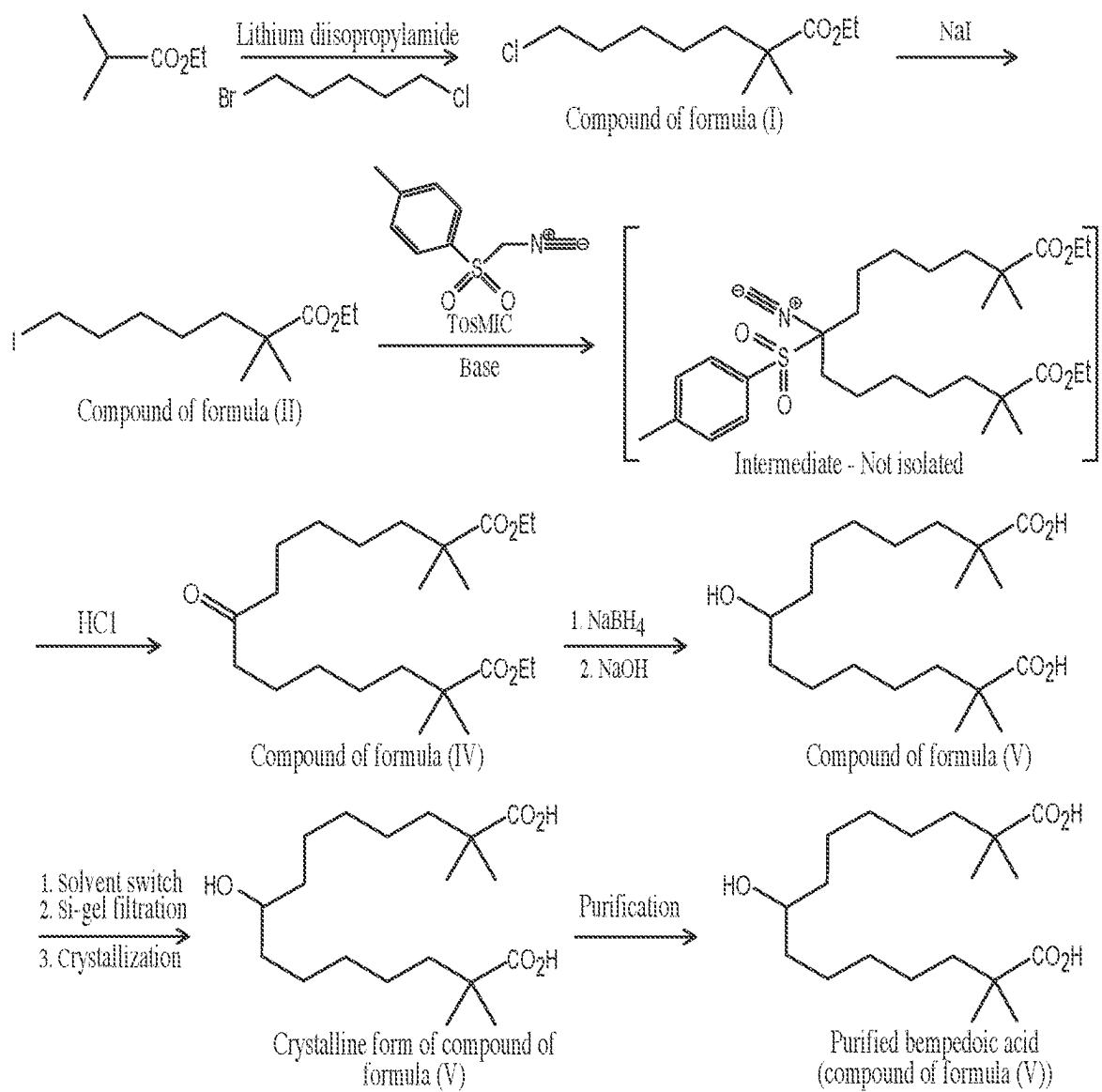


FIG. 1



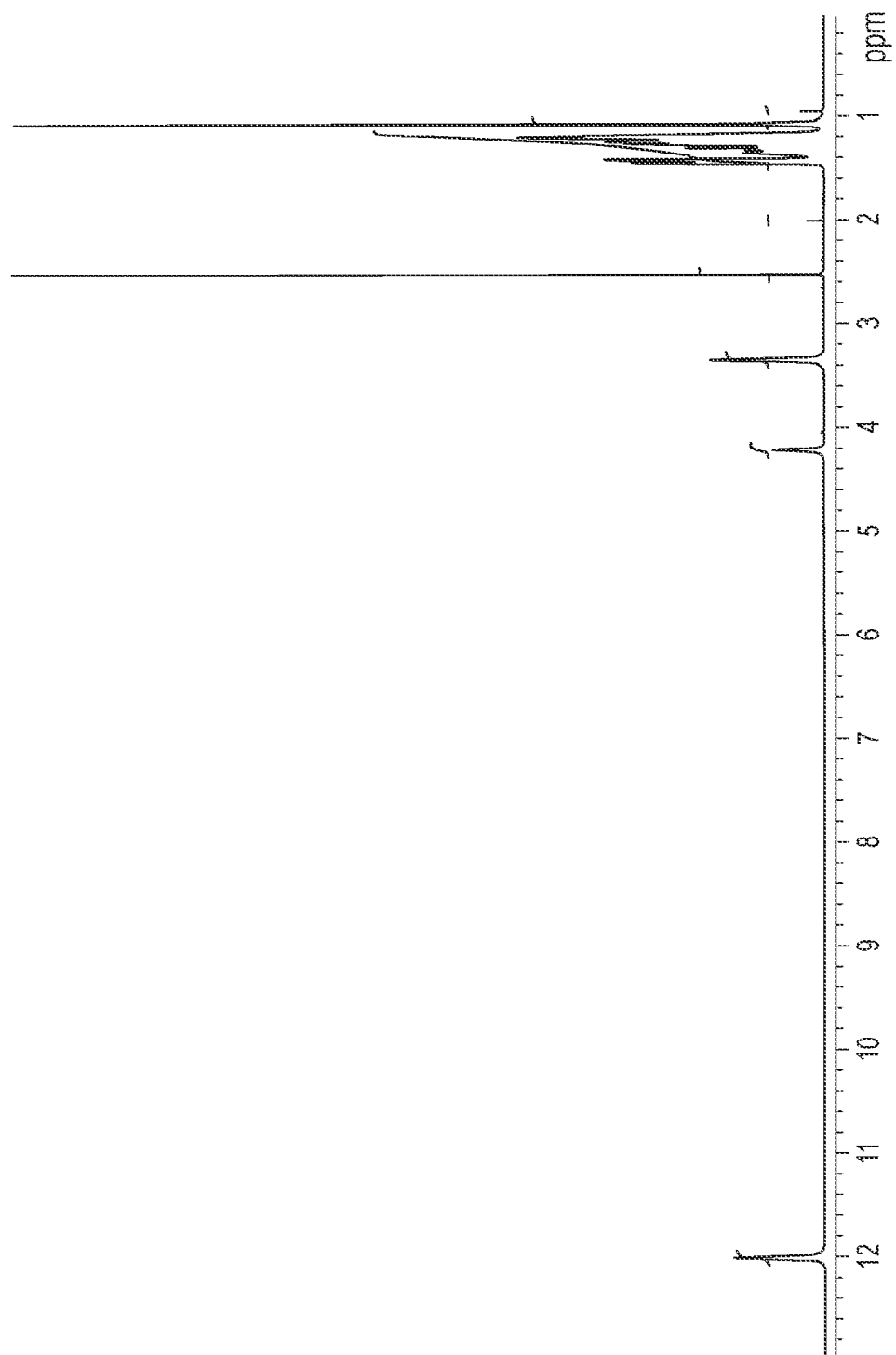


FIG. 2

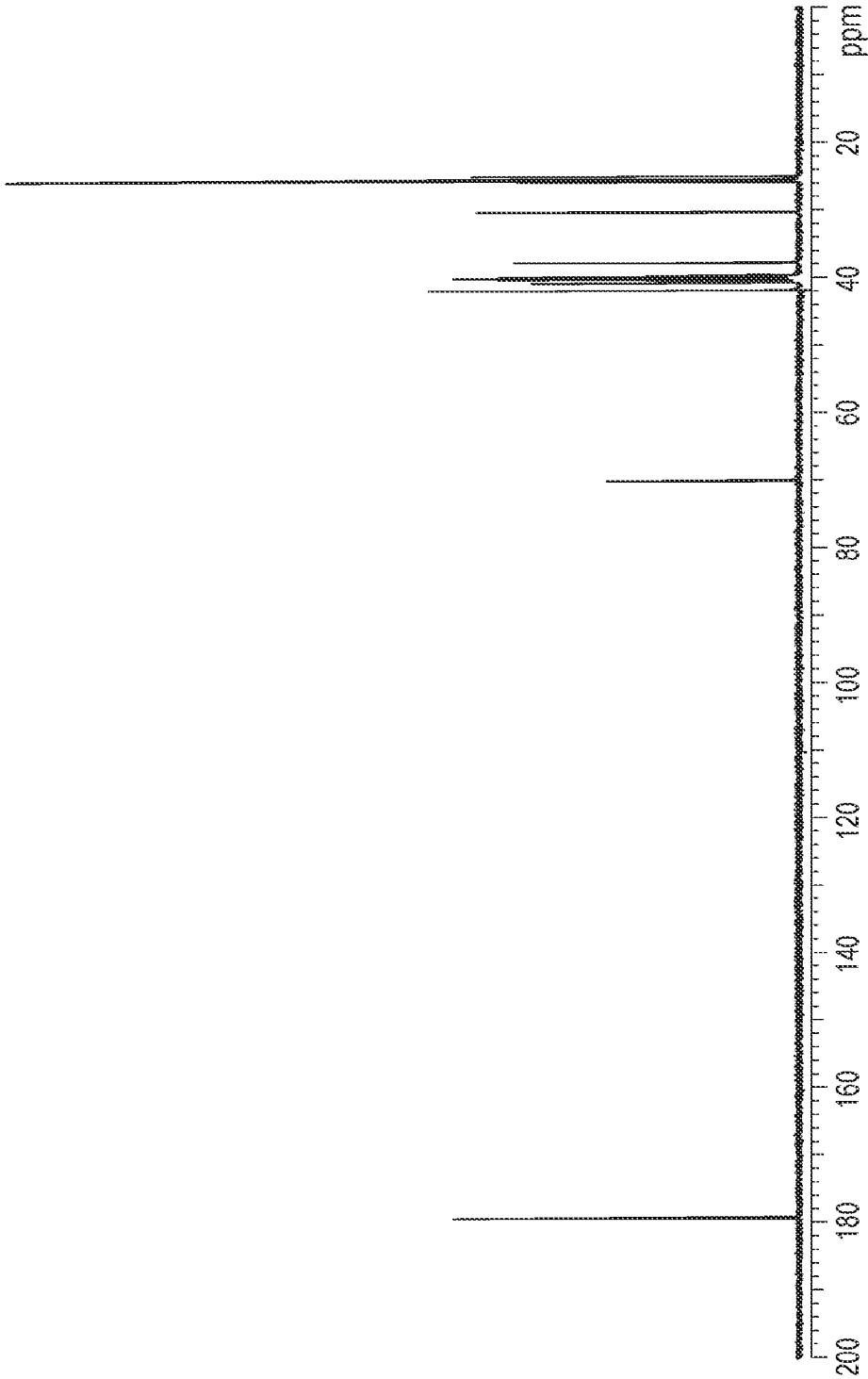


FIG. 3

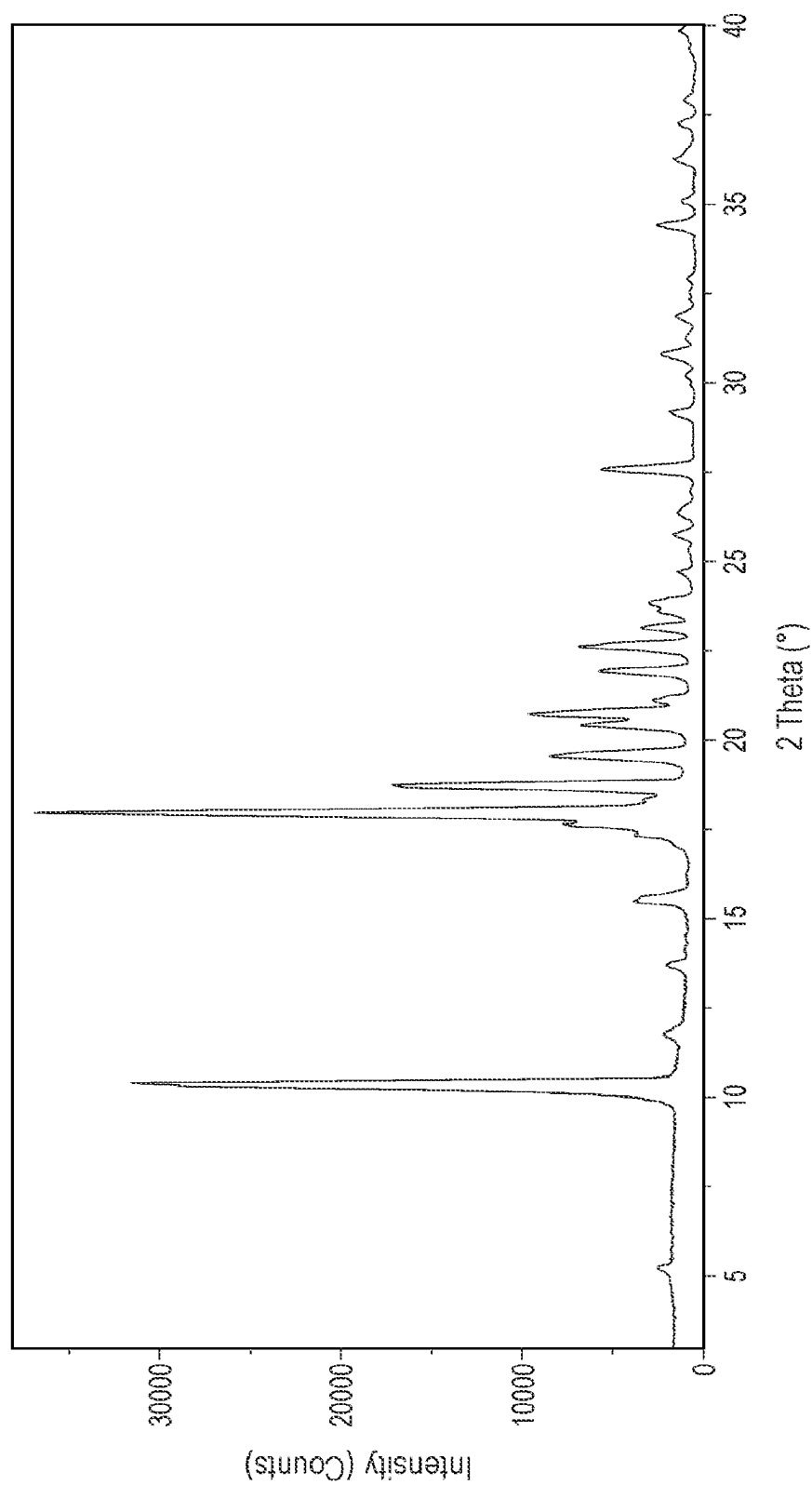


FIG. 4

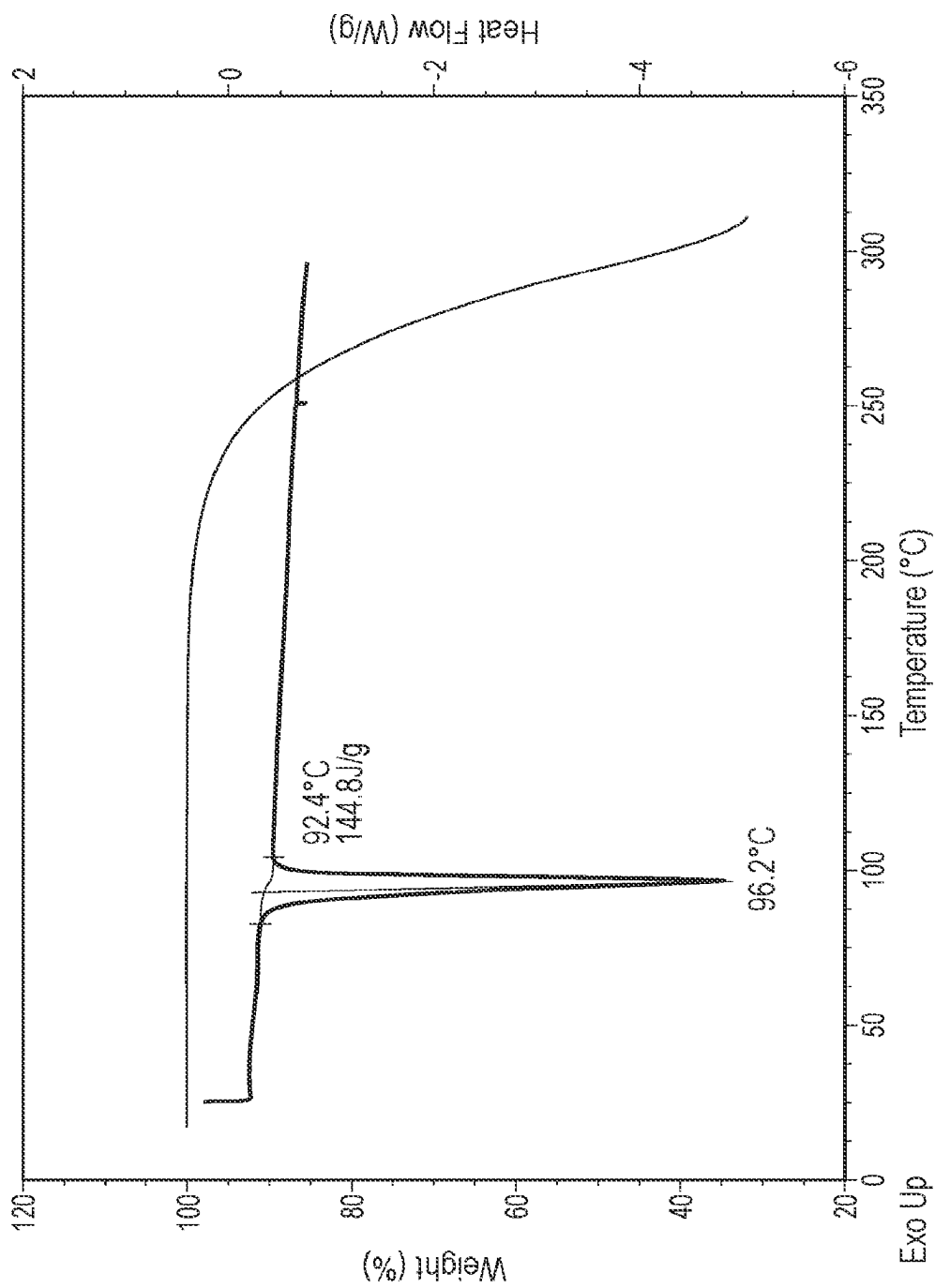


FIG. 5

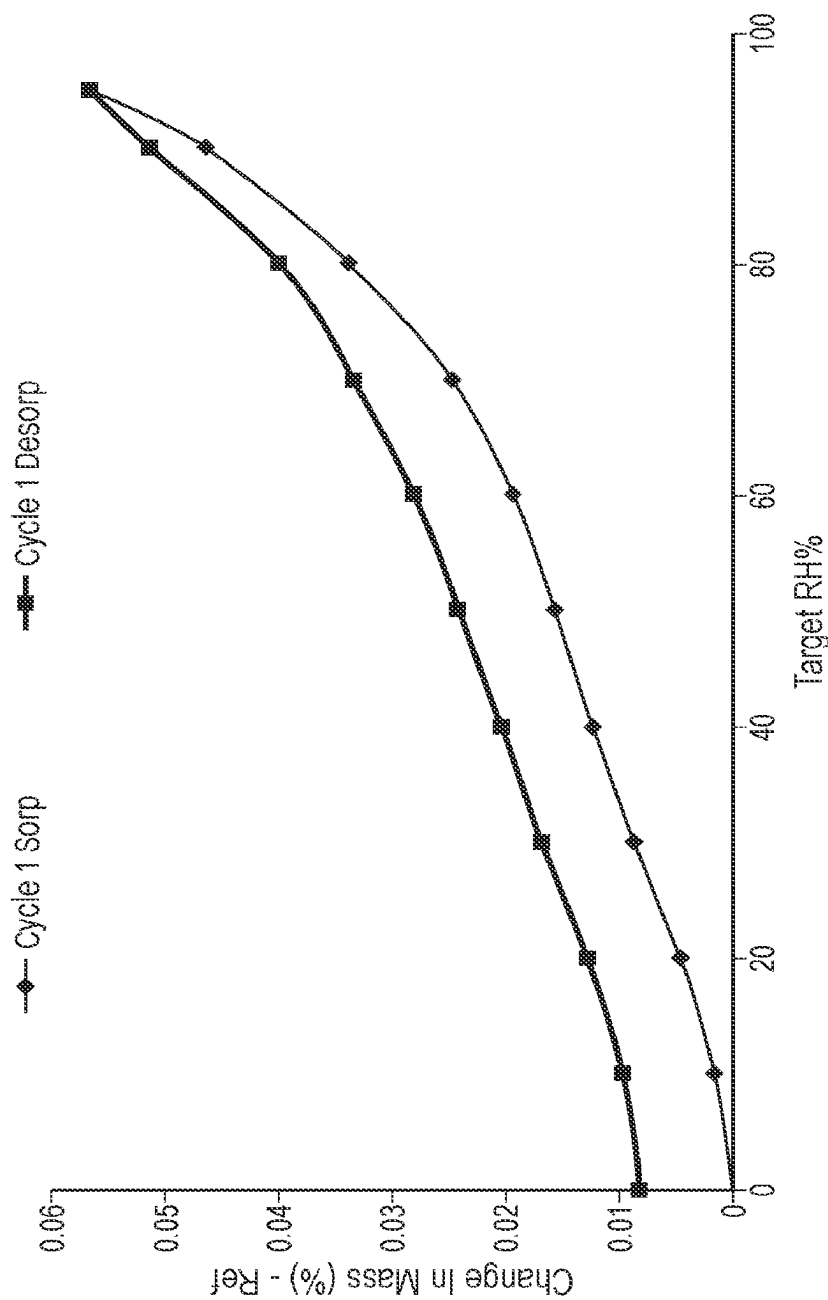


FIG. 6

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# METHODS OF MAKING BEMPEDOIC ACID AND COMPOSITIONS OF THE SAME

## CROSS-REFERENCE

This application is a continuation of U.S. patent application Ser. No. 17/577,829, filed on Jan. 18, 2022, which is a continuation of U.S. patent application Ser. No. 17/150,321, filed on Jan. 15, 2021, which is a continuation of International Application No. PCT/US2020/038622, with an international filing date of Jun. 19, 2020, which international application claims the benefit of and priority to U.S. Patent Application No. 62/864,873, filed on Jun. 21, 2019, the entire contents of which are incorporated by reference herein.

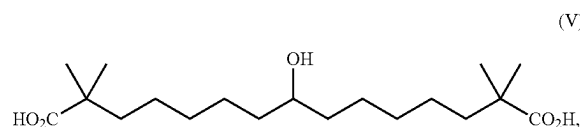
## BACKGROUND

The development of robust, cost-effective and efficient manufacturing methods for the production of pharmaceutically active compounds with desired yield and purity remains a significant challenge. Bempedoic acid (8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid) is a compound under development for the treatment of a wide variety of diseases including liver disorders and cardiovascular disease. Accordingly, a process for synthesizing bempedoic acid (8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid) is desired, whereby the product has purity and impurity profiles required by regulatory agencies for the production of a commercializable drug product.

## SUMMARY

The inventors have discovered an efficient process for producing high purity bempedoic acid, as well as highly pure, stable forms of bempedoic acid suitable for use as an active pharmaceutical ingredient.

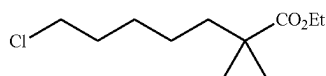
In one aspect, the invention provides methods of preparing a compound of formula (V):



or a pharmaceutically acceptable salt thereof. The compound of formula (V), or a pharmaceutically acceptable salt thereof, can be of high purity. Accordingly, in certain embodiments, the invention provides methods of preparing a pharmaceutical material comprising a compound of formula (V), or a pharmaceutically acceptable salt thereof, where the pharmaceutical material includes the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material.

In various embodiments of the invention, the method generally comprises:

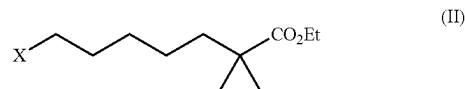
- (a) contacting ethyl isobutyrate with a substituted 5-chloropentane in the presence of a first base to form a compound of formula (I):



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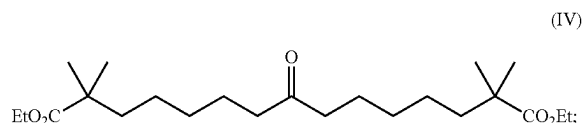
wherein the substituted 5-chloropentane is selected from the group consisting of 1-bromo-5-chloropentane and 1-iodo-5-chloropentane;

- (b) contacting the compound of formula (I) with a salt of formula  $[M]^+[X]^-$  to form a compound of formula (II):



wherein  $[M]^+$  is selected from the group consisting of  $Li^+$ ,  $Na^+$  and  $K^+$ , wherein  $[X]^-$  is selected from the group consisting of  $Br^-$  and  $I^-$ ;

- (c) contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of a second base to form a first intermediate, and contacting the first intermediate with an acid to form a compound of formula (IV):

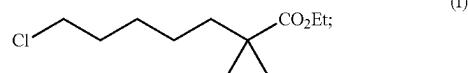


and

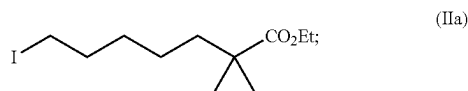
- (d) contacting the compound of formula (IV) with a reducing agent to form a second intermediate, and contacting the second intermediate with a hydrolyzing base to form a compound of formula (V), or a pharmaceutically acceptable salt thereof.

In certain embodiments of the invention, the method comprises:

- (a) contacting 1-bromo-5-chloropentane with about 1 to about 1.21 molar equivalents of ethyl isobutyrate in the presence of lithium diisopropylamide at a temperature in the range of about  $-20^\circ\text{C}$ . to about  $0^\circ\text{C}$ . to form a compound of formula (I):

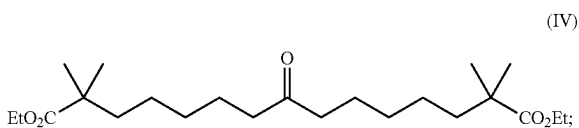


- (b) contacting the compound of formula (I) with about 1.1 molar equivalents of sodium iodide in 2-butanone at a temperature in the range of about  $78^\circ\text{C}$ . to about  $82^\circ\text{C}$ . to form a compound of formula (IIa):



- (c) contacting the compound of formula (IIa) with toluenesulfonylmethyl isocyanide in the presence of sodium tert-pentoxide in dimethylacetamide at a temperature in the range of about  $-20^\circ\text{C}$ . to about  $10^\circ\text{C}$ . to form an intermediate, and contacting the intermediate with an acid at a temperature in the range of about  $-10^\circ\text{C}$ . to about  $35^\circ\text{C}$ . to form a compound of formula (IV):

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and

- (d) contacting the compound of formula (IV) with about 0.35 molar equivalents of sodium borohydride to form a second intermediate, and contacting the second intermediate with sodium hydroxide in a solution to form a compound of formula (V).

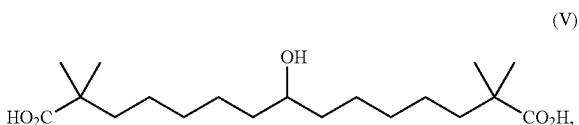
In certain embodiments of the invention, the method further comprises:

- (e) purifying the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In certain embodiments of the invention, purifying the compound of formula (V) comprises:

- (f) adjusting the pH of the solution comprising the compound of formula (V) to about 6;  
 (g) extracting the compound of formula (V) from the solution using methyl tert-butyl ether to provide a methyl tert-butyl ether solution comprising the compound of formula (V);  
 (h) exchanging the methyl tert-butyl ether of the methyl tert-butyl ether solution with ethyl acetate to provide an ethyl acetate solution comprising the compound of formula (V);  
 (i) filtering the ethyl acetate solution comprising the compound of formula (V) through silica gel;  
 (j) crystallizing the compound of formula (V) using ethyl acetate and water to provide a crystalline form of the compound of formula (V); and  
 (k) recrystallizing the crystalline form of the compound of formula (V) using ethyl acetate and water to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In another aspect, the invention provides high purity or purified bempedoic acid, or a pharmaceutically acceptable salt thereof. For example, described herein are pharmaceutical materials comprising a compound of formula (V):



or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical material comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material.

In various embodiments, the pharmaceutical material comprises a crystalline form of the compound of formula (V), or a pharmaceutically acceptable salt thereof. In some embodiments, the pharmaceutical material comprises the compound of formula (V) in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material.

In another aspect, the invention provides pharmaceutical compositions or formulations including high purity bempedoic acid, or a pharmaceutically acceptable salt thereof, such

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as the pharmaceutical materials described herein. For example, a pharmaceutical composition can include a pharmaceutical material of the invention (e.g., a pharmaceutical material comprising the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material); and a pharmaceutically acceptable excipient. In some embodiments, a pharmaceutical composition can include a therapeutically effective amount of a pharmaceutical material of the invention; and a pharmaceutically acceptable excipient. In certain embodiments, the pharmaceutical composition comprises a crystalline form of the compound of formula (V), or a pharmaceutically acceptable salt thereof; and a pharmaceutically acceptable excipient.

Purified bempedoic acid, or a pharmaceutically acceptable salt thereof; a crystalline form of bempedoic acid, or a pharmaceutically acceptable salt thereof; a pharmaceutical material of the invention (e.g., a pharmaceutical material comprising the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material); or a pharmaceutical composition of the invention can be used in treating the various conditions and diseases described herein. For example, the methods of treatment can include inhibiting adenosine triphosphate citrate lyase (ACL), inhibiting cholesterol synthesis, and/or suppressing fatty acid biosynthesis. In some embodiments, the condition or disease can be hyperlipidemia such as primary hyperlipidemia and the methods include treating hyperlipidemia such as primary hyperlipidemia. In some embodiments, the disease can be cardiovascular disease and the methods include treating cardiovascular disease. In various embodiments, the methods of treatment can include improving or lowering low density lipid cholesterol (LDL-C), non-high density lipid cholesterol (non-HDL-C), total serum cholesterol (TC), apolipoprotein B (apoB), and/or high sensitivity C-reactive protein (hsCRP).

#### BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an exemplary reaction scheme of the invention for the synthesis of bempedoic acid (i.e., a compound of formula (V)) as described in Example 1, which reaction scheme includes the synthesis of a pharmaceutical material comprising the compound of formula (V) in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material.

FIG. 2 is an exemplary <sup>1</sup>H-NMR spectrum of the compound of formula (V).

FIG. 3 is an exemplary <sup>13</sup>C-NMR spectrum of the compound of formula (V).

FIG. 4 is an X-ray powder diffraction pattern of the crystalline form of the compound of formula (V), as further described in Example 1.

FIG. 5 is an overlay of differential scanning calorimetry (DSC) and thermogravimetric analysis (TGA) curves of the crystalline form of the compound of formula (V), as further described in Example 1.

FIG. 6 is a water sorption isotherm of the crystalline form of the compound of formula (V).

#### DETAILED DESCRIPTION OF THE INVENTION

It has now been discovered that bempedoic acid, including pharmaceutically acceptable salts thereof, can be pre-

pared with high purity and/or in bulk quantities. In various embodiments, a crystalline form of bempedoic acid, or a pharmaceutically acceptable salt thereof, is provided.

The methods for preparing bempedoic acid described herein can provide a pharmaceutical material containing a high level or amount of bempedoic acid, or a pharmaceutically acceptable salt thereof, in part, due to the control of the formation of hard-to-remove impurities during the synthetic process.

In addition, a pharmaceutical material with a high purity crystalline form of bempedoic acid, or a pharmaceutically acceptable salt thereof, is provided, for example, where the pharmaceutical material comprises the crystalline form of bempedoic acid, or a pharmaceutically acceptable salt thereof, and wherein the pharmaceutical material comprises bempedoic acid, or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material.

### I. Definitions

To facilitate an understanding of the present invention, a number of terms and phrases are defined below.

Unless defined otherwise, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this invention belongs. The abbreviations used herein have their conventional meaning within the chemical and biological arts. The chemical structures and formulae set forth herein are constructed according to the standard rules of chemical valency known in the chemical arts.

Throughout the description, where compositions and kits are described as having, including, or comprising specific components, or where processes and methods are described as having, including, or comprising specific steps, it is contemplated that, additionally, there are compositions and kits of the present invention that consist essentially of, or consist of, the recited components, and that there are processes and methods according to the present invention that consist essentially of, or consist of, the recited processing steps.

In the application, where an element or component is said to be included in and/or selected from a list of recited elements or components, it should be understood that the element or component can be any one of the recited elements or components, or the element or component can be selected from a group consisting of two or more of the recited elements or components.

Further, it should be understood that elements and/or features of a composition or a method described herein can be combined in a variety of ways without departing from the spirit and scope of the present invention, whether explicit or implicit herein. For example, where reference is made to a particular compound, that compound can be used in various embodiments of compositions of the present invention and/or in methods of the present invention, unless otherwise understood from the context. In other words, within this application, embodiments have been described and depicted in a way that enables a clear and concise application to be written and drawn, but it is intended and will be appreciated that embodiments may be variously combined or separated without parting from the present teachings and invention(s). For example, it will be appreciated that all features described and depicted herein can be applicable to all aspects of the invention(s) described and depicted herein.

The articles "a" and "an" are used in this disclosure to refer to one or more than one (i.e., to at least one) of the

grammatical object of the article, unless the context is inappropriate. By way of example, "an element" means one element or more than one element.

The term "and/or" is used in this disclosure to mean either "and" or "or" unless indicated otherwise.

It should be understood that the expression "at least one of" includes individually each of the recited objects after the expression and the various combinations of two or more of the recited objects unless otherwise understood from the context and use. The expression "and/or" in connection with three or more recited objects should be understood to have the same meaning unless otherwise understood from the context.

The use of the term "include," "includes," "including," "have," "has," "having," "contain," "contains," or "containing," including grammatical equivalents thereof, should be understood generally as open-ended and non-limiting, for example, not excluding additional unrecited elements or steps, unless otherwise specifically stated or understood from the context.

Where the use of the term "about" is before a quantitative value, the present invention also includes the specific quantitative value itself, unless specifically stated otherwise. As used herein, the term "about" refers to a  $\pm 10\%$  variation from the nominal value unless otherwise indicated or inferred from the context.

Where a molecular weight is provided and not an absolute value, for example, of a polymer, then the molecular weight should be understood to be an average molecule weight, unless otherwise stated or understood from the context.

It should be understood that the order of steps or order for performing certain actions is immaterial so long as the present invention remain operable. Moreover, two or more steps or actions may be conducted simultaneously.

At various places in the present specification, variable or parameters are disclosed in groups or in ranges. It is specifically intended that the description include each and every individual subcombination of the members of such groups and ranges. For example, an integer in the range of 0 to 40 is specifically intended to individually disclose 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, and 40, and an integer in the range of 1 to 20 is specifically intended to individually disclose 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, and 20.

The use of any and all examples, or exemplary language herein, for example, "such as" or "including," is intended merely to illustrate better the present invention and does not pose a limitation on the scope of the invention unless claimed. No language in the specification should be construed as indicating any non-claimed element as essential to the practice of the present invention.

As a general matter, compositions specifying a percentage are by weight unless otherwise specified. Further, if a variable is not accompanied by a definition, then the previous definition of the variable controls.

As used herein, "pharmaceutically acceptable salt" refers to any salt of an acidic or a basic group that may be present in a compound of the present invention, which salt is compatible with pharmaceutical administration. For example, one or both of the carboxylic acid groups of bempedoic acid can be transformed to pharmaceutically acceptable salt(s).

As is known to those of skill in the art, "salts" of compounds may be derived from inorganic or organic acids and bases. Examples of acids include, but are not limited to, hydrochloric, hydrobromic, sulfuric, nitric, perchloric,



fumaric, maleic, phosphoric, glycolic, lactic, salicylic, succinic, toluene-p-sulfonic, tartaric, acetic, citric, methanesulfonic, ethanesulfonic, formic, benzoic, malonic, naphthalene-2-sulfonic and benzenesulfonic acid.

Other acids, such as oxalic, while not in themselves pharmaceutically acceptable, may be employed in the preparation of salts useful as intermediates in obtaining the compounds described herein and their pharmaceutically acceptable acid addition salts.

Examples of bases include, but are not limited to, alkali metal (e.g., sodium and potassium) hydroxides, alkaline earth metal (e.g., magnesium and calcium) hydroxides, ammonia, and compounds of formula  $NW_4^+$ , wherein W is  $C_{1-4}$  alkyl, and the like.

Examples of salts include, but are not limited, to acetate, adipate, alginate, aspartate, benzoate, benzenesulfonate, bisulfate, butyrate, citrate, camphorate, camphorsulfonate, cyclopentanepropionate, digluconate, dodecylsulfate, ethanesulfonate, fumarate, flucoheptanoate, glycerophosphate, hemisulfate, heptanoate, hexanoate, hydrochloride, hydrobromide, hydriodide, 2-hydroxyethanesulfonate, lactate, maleate, methanesulfonate, 2-naphthalenesulfonate, nicotine, oxalate, palmoate, pectinate, persulfate, phenylpropionate, picrate, pivalate, propionate, succinate, tartrate, thiocyanate, tosylate, undecanoate, and the like.

Other examples of salts include anions of the compounds of the present invention compounded with a suitable cation such as  $Na^+$ ,  $K^+$ ,  $Ca^{2+}$ ,  $NH_4^+$ , and  $NW_4^+$  (where W can be a  $C_{1-4}$  alkyl group), and the like.

For therapeutic use, salts of the compounds of the present invention are contemplated as being pharmaceutically acceptable. However, salts of acids and bases that are non-pharmaceutically acceptable may also find use, for example, in the preparation or purification of a pharmaceutically acceptable compound.

As used herein, "pharmaceutical composition" or "pharmaceutical formulation" refers to the combination of an active agent with a carrier, inert or active, making the composition especially suitable for diagnostic or therapeutic use in vivo or ex vivo.

The phrases "pharmaceutically acceptable" and "pharmacologically acceptable," as used herein, refer to compounds, molecular entities, compositions, materials, and/or dosage forms that do not produce an adverse, allergic or other untoward reaction when administered to an animal, or a human, as appropriate. For human administration, preparations should meet sterility, pyrogenicity, and general safety and purity standards as required by FDA Office of Biologics standards. "Pharmaceutically acceptable" and "pharmacologically acceptable" can mean approved or approvable by a regulatory agency of the federal or a state government or the corresponding agency in countries other than the United States, or that is listed in the U.S. Pharmacopoeia or other generally recognized pharmacopoeia for use in animals, and more particularly, in humans.

As used herein, "carrier" refers to a material, composition or vehicle, such as a liquid or solid filler, diluent, excipient, solvent or encapsulating material, involved in carrying or transporting a pharmaceutical agent such as bempedoic acid, or a pharmaceutically acceptable salt thereof, from one organ, or portion of the body, to another organ, or portion of the body.

As used herein, "pharmaceutically acceptable excipient" refers to a substance that aids the administration of an active agent to and/or absorption by a subject and can be included in the compositions of the present invention without causing a significant adverse toxicological effect on the patient.

Non-limiting examples of pharmaceutically acceptable excipients include water, NaCl, normal saline solutions, such as a phosphate buffered saline solution, emulsions (e.g., such as an oil/water or water/oil emulsions), lactated Ringer's, normal sucrose, normal glucose, binders, fillers, disintegrants, lubricants, coatings, sweeteners, flavors, salt solutions (such as Ringer's solution), alcohols, oils, gelatins, carbohydrates such as lactose, amylose or starch, fatty acid esters, hydroxymethylcellulose, polyvinyl pyrrolidone, and colors, and the like. Such preparations can be sterilized and, if desired, mixed with auxiliary agents such as lubricants, preservatives, stabilizers, wetting agents, emulsifiers, salts for influencing osmotic pressure, buffers, coloring, and/or aromatic substances and the like that do not deleteriously react with the compounds of the invention. For examples of excipients, see Martin, Remington's Pharmaceutical Sciences, 15th Ed., Mack Publ. Co., Easton, PA (1975).

As used herein, "treating" or "treatment" includes any effect, for example, lessening, reducing, modulating, ameliorating or eliminating, that results in the improvement of the condition, disease, disorder, and the like, or ameliorating a symptom thereof. Treating can be curing, improving, or at least partially ameliorating the disorder. In certain embodiments, treating is curing the disease.

As used herein, "reducing" or "reduction" of a symptom or symptoms (and grammatical equivalents of this phrase) means decreasing of the severity or frequency of the symptom(s), or elimination of the symptom(s).

As used herein, "effective amount" or "therapeutically effective amount" refers to the amount of a compound (e.g., a compound of the present invention) sufficient to effect beneficial or desired results. An effective amount can be administered in one or more administrations, applications or dosages and is not intended to be limited to a particular formulation or administration route. As used herein, the term "treating" includes any effect, e.g., lessening, reducing, modulating, ameliorating or eliminating, that results in the improvement of the condition, disease, disorder, and the like, or ameliorating a symptom thereof.

As used herein, "subject" and "patient" are used interchangeably and refer to an organism to be treated by the methods and compositions of the present invention. Such organisms are preferably a mammal (e.g., human, mouse, rat, guinea pig, dog, cat, horse, cow, pig, or non-human primate, such as a monkey, chimpanzee, baboon, and rhesus), and more preferably, a human.

As used herein, "disease," "disorder," "condition," or "illness," can be used interchangeably unless otherwise underacted or understood from the context, refers to a state of being or health status of a patient or subject capable of being treated with a compound, pharmaceutical materials, pharmaceutical composition, or method provided herein. In some embodiments, the compounds and methods described herein comprise reduction or elimination of one or more symptoms of the disease, disorder, or condition, or illness e.g., through administration of the compound of formula (V), or a pharmaceutically acceptable salt thereof.

As used herein, "administering" means oral administration, administration as a suppository, topical contact, intravenous, parenteral, intraperitoneal, intramuscular, intral-lesional, intrathecal, intracranial, intranasal or subcutaneous administration, or the implantation of a slow-release device, e.g., a mini-osmotic pump, to a subject. Administration is by any route, including parenteral and transmucosal (e.g., buccal, sublingual, palatal, gingival, nasal, vaginal, rectal, or transdermal). Parenteral administration includes, e.g., intravenous, intramuscular, intra-arterial, intradermal, subcuta-

neous, intraperitoneal, intraventricular, and intracranial. Other modes of delivery include, but are not limited to, the use of liposomal formulations, intravenous infusion, transdermal patches, etc. By “co-administer” it is meant that a composition described herein is administered at the same time, just prior to, or just after the administration of one or more additional therapies (e.g., anti-cancer agent, chemotherapeutic, or treatment for a neurodegenerative disease). The compound of the invention can be administered alone or can be co-administered to the patient. Co-administration is meant to include simultaneous or sequential administration of the compound individually or in combination (more than one compound or agent). Thus, the preparations can also be combined, when desired, with other active substances (e.g., to reduce metabolic degradation).

As used herein, “liver disorder” refers generally to a disease, a disorder, and/or a condition affecting the liver, and may have a wide range of severity encompassing, for example, simple accumulation of fat in the hepatocytes (steatosis), macrovesicular steatosis, periportal and lobular inflammation (steatohepatitis), cirrhosis, fibrosis, liver cancers, and liver failure.

As used herein, “fatty liver disease” (“FLD”), which is also called “fatty liver,” refers to a disease leading to liver injury caused by abnormal fat accumulation in liver cells. FLD may arise from a number of sources, including excessive alcohol consumption and metabolic disorders, such as those associated with insulin resistance, obesity, and hypertension.

As used herein, “non-alcoholic fatty liver disease” (“NAFLD”) refers to the spectrum of disorders resulting from an accumulation of fat in liver cells in individuals with no history of excessive alcohol consumption. In the mildest form, NAFLD refers to hepatic steatosis.

As used herein, “drug-induced liver disease” or “toxic liver injury” refers to a disease or a condition in which an active agent has caused injury to the liver.

As used herein, “alcoholic liver disease,” also called “alcoholic liver injury,” refers to a disease caused by fat accumulation in liver cells, caused at least in part by alcohol ingestion. Examples include, but are not limited to, diseases such as alcoholic simple fatty liver, alcoholic steatohepatitis (“ASH”), alcoholic hepatic fibrosis, alcoholic cirrhosis, alcoholic fatty liver disease, and the like. It should be noted that alcoholic steatohepatitis is also called alcoholic fatty hepatitis and includes alcoholic hepatic fibrosis.

As used herein, “fatty liver of pregnancy” refers to acute fatty liver conditions that can arise during pregnancy and can be life-threatening.

As used herein, “altering lipid metabolism” refers to an observable (measurable) change in at least one aspect of lipid metabolism, including but not limited to total blood lipid content, blood HDL cholesterol, blood LDL cholesterol, blood VLDL cholesterol, blood triglyceride, blood Lp(a), blood apo A-I, blood apo E and blood non-esterified fatty acids.

As used herein, “altering glucose metabolism” refers to an observable (measurable) change in at least one aspect of glucose metabolism, including but not limited to total blood glucose content, blood insulin, the blood insulin to blood glucose ratio, insulin sensitivity, and oxygen consumption.

As used herein, “purified bempedoic acid” means that, when isolated as a solid, a pharmaceutical material contains at least 95% by weight of 8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid based on the total weight of the pharmaceutical material. In certain embodiments, purified bempedoic acid means that, when isolated as a solid, a

pharmaceutical material contains at least 99.0% by weight of 8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid based on the total weight of the pharmaceutical material. In addition, purified bempedoic acid can include a pharmaceutically acceptable salt thereof, unless stated otherwise or understood from the context.

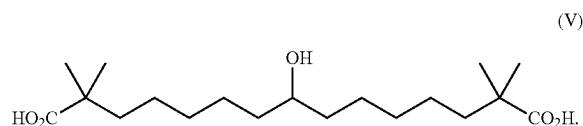
As used herein, a reaction that is “substantially complete” means that the reaction contains more than about 80% by weight of the desired product. In certain embodiments, a substantially complete reaction contains more than about 90% by weight of the desired product. In certain embodiments, a substantially complete reaction contains more than about 95% by weight of the desired product. In certain embodiments, a substantially complete reaction contains more than about 97% by weight of the desired product.

Unless stated otherwise, all X-ray powder diffraction (XRPD) patterns described herein correspond to XRPD patterns measured using a Cu K $\alpha$  radiation source, and the crystalline forms of bempedoic acid are analyzed by XRPD at ambient temperature.

## II. Crystalline Forms of Bempedoic Acid

### A. Crystalline Bempedoic Acid

In one aspect, the invention provides a crystalline form of 8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid, which is also known as and referred to herein as “bempedoic acid” and/or a compound of formula (V):



In certain embodiments, the crystalline form of the compound of formula (V) may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $10.3\pm 0.2$ ,  $10.4\pm 0.2$ ,  $17.9\pm 0.2$ ,  $18.8\pm 0.2$ ,  $19.5\pm 0.2$ , and  $20.7\pm 0.2$ . In certain embodiments, the crystalline form of the compound of formula (V) may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $10.3\pm 0.2$ ,  $10.4\pm 0.2$ ,  $17.6\pm 0.2$ ,  $17.9\pm 0.2$ ,  $18.8\pm 0.2$ ,  $19.5\pm 0.2$ ,  $19.7\pm 0.2$ ,  $20.4\pm 0.2$ ,  $20.7\pm 0.2$  and  $22.6\pm 0.2$ .

In certain embodiments, the crystalline form of the compound of formula (V) is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 1.

TABLE 1

| X-ray Powder Diffraction Data of the Crystalline Form of the Compound of Formula (V) |                            |                        |
|--------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                  | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 5.2                                                                                  | 16.84                      | 2.33                   |
| 10.3                                                                                 | 8.61                       | 70.75                  |
| 10.4                                                                                 | 8.48                       | 78.65                  |
| 11.8                                                                                 | 7.51                       | 2.88                   |
| 13.7                                                                                 | 6.44                       | 2.72                   |
| 15.5                                                                                 | 5.73                       | 8.08                   |
| 15.6                                                                                 | 5.67                       | 7.16                   |
| 17.3                                                                                 | 5.12                       | 8.20                   |

TABLE 1-continued

| X-ray Powder Diffraction Data of the Crystalline Form of the Compound of Formula (V) |               |                        |
|--------------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                                  | d-spacing [Å] | Relative Intensity [%] |
| 17.6                                                                                 | 5.04          | 18.72                  |
| 17.9                                                                                 | 4.95          | 100.00                 |
| 18.8                                                                                 | 4.73          | 42.30                  |
| 19.5                                                                                 | 4.55          | 21.42                  |
| 19.7                                                                                 | 4.51          | 15.07                  |
| 20.4                                                                                 | 4.35          | 16.93                  |
| 20.7                                                                                 | 4.29          | 23.95                  |
| 21.1                                                                                 | 4.21          | 5.78                   |
| 22.0                                                                                 | 4.05          | 13.87                  |
| 22.6                                                                                 | 3.94          | 17.54                  |
| 23.1                                                                                 | 3.84          | 7.78                   |
| 23.6                                                                                 | 3.78          | 4.97                   |
| 23.9                                                                                 | 3.73          | 6.19                   |
| 24.7                                                                                 | 3.60          | 1.98                   |
| 25.8                                                                                 | 3.46          | 3.04                   |
| 26.3                                                                                 | 3.39          | 2.10                   |
| 27.5                                                                                 | 3.24          | 13.36                  |
| 29.2                                                                                 | 3.06          | 3.86                   |
| 30.2                                                                                 | 2.96          | 1.27                   |
| 30.8                                                                                 | 2.90          | 5.34                   |
| 31.3                                                                                 | 2.86          | 1.40                   |
| 31.9                                                                                 | 2.81          | 2.95                   |
| 32.9                                                                                 | 2.72          | 1.27                   |
| 34.4                                                                                 | 2.61          | 5.98                   |
| 35.1                                                                                 | 2.56          | 2.07                   |
| 36.2                                                                                 | 2.48          | 3.16                   |
| 37.2                                                                                 | 2.42          | 2.37                   |
| 37.9                                                                                 | 2.37          | 1.79                   |

In certain embodiments, the crystalline form of the compound of formula (V) is characterized by an X-ray powder diffraction pattern substantially the same as shown in FIG. 4.

In certain embodiments, the crystalline form of the compound of formula (V) exists in a monoclinic crystal system and has a  $P2_1/c$  space group. In certain embodiments, the crystalline form of the compound of formula (V) is characterized by the crystallographic unit cell parameters as set forth in Table 2.

TABLE 2

| Unit Cell Parameters of the Crystalline Form of Compound of Formula (V) |                                                       |                                                                           |
|-------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------------------------------------|
| Unit cell dimensions                                                    | a = 17.9209(8) Å<br>b = 9.8547(5) Å<br>c = 12.2775(6) | $\alpha = 90^\circ$<br>$\beta = 106.834(10)^\circ$<br>$\gamma = 90^\circ$ |
| Volume                                                                  | 2075.35(17) Å <sup>3</sup>                            |                                                                           |
| Z                                                                       | 4                                                     |                                                                           |
| Density (calculated)                                                    | 1.102 Mg/m <sup>3</sup>                               |                                                                           |

The crystalline form of the compound of formula (V) may also be characterized according to the temperature of melting point onset. Accordingly, in certain embodiments, the crystalline form of the compound of formula (V) has a melting point onset as determined by differential scanning calorimetry in the range of from about 82° C. to about 94° C. In certain embodiments, the crystalline form of the compound of formula (V) has a melting point onset as determined by differential scanning calorimetry in the range of about 90° C. to about 94° C. In certain embodiments, the crystalline form of the compound of formula (V) has a melting point onset as determined by differential scanning calorimetry at about 92° C. In certain embodiments, the crystalline form of the compound of formula (V) has a differential scanning calorimetry curve substantially the same as shown in FIG. 5.

The crystalline form of the compound of formula (V) may also be characterized according to its mass gain/mass loss as a function of temperature. Accordingly, in certain embodiments, the crystalline form of the compound of formula (V) exhibits a reduction in mass, as determined by thermogravimetric analysis, of from about 0.1% to about 0.7% upon heating to about 200° C. In certain embodiments, the crystalline form of the compound of formula (V) exhibits a reduction in mass, as determined by thermogravimetric analysis, of less than or equal to about 0.7% upon heating to about 200° C. In certain embodiments, the crystalline form of the compound of formula (V) has a thermogravimetric analysis curve substantially the same as shown in FIG. 5.

The crystalline form of the compound of formula (V) may also be characterized according to its water sorption properties. Accordingly, in certain embodiments, the crystalline form of the compound of formula (V) exhibits a change in mass, as determined by dynamic vapor sorption, of from about 0.01% to about 0.05% at a relative humidity of 80% and a temperature of 25° C. In certain embodiments, the crystalline form of the compound of formula (V) exhibits a change in mass, as determined by dynamic vapor sorption, of about 0.03% at a relative humidity of 80% and a temperature of 25° C. In certain embodiments, the crystalline form of the compound of formula (V) has a water sorption isotherm, when measured at 25° C., substantially the same as shown in FIG. 6.

It should be understood that reference herein to bempedoic acid or a purified bempedoic acid includes the crystalline form of bempedoic acid, unless otherwise stated or understood from the context.

#### B. Crystalline Salt Forms of Bempedoic Acid

In addition, it has been discovered that various crystalline salt forms of bempedoic acid can be prepared. In particular, the following counter ions produced crystalline salt forms of bempedoic acid: ammonium, sodium, potassium, calcium (two crystal forms), lysine, diethylamine, ethylenediamine, piperazine, betaine, tromethamine, and isonicotinamide.

##### (i) Crystalline Betaine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline betaine salt of bempedoic acid. In certain embodiments, the crystalline betaine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.2±0.2, 13.5±0.2, 17.5±0.2, 19.3±0.2, and 25.6±0.2. In certain embodiments, the crystalline betaine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.2±0.2, 13.5±0.2, 16.1±0.2, 17.5±0.2, 19.3±0.2, 19.9±0.2, 25.6±0.2, and 27.2±0.2.

In certain embodiments, the crystalline betaine salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 3.

TABLE 3

| X-ray Powder Diffraction Data of the Crystalline Betaine Salt of Bempedoic Acid |               |                        |
|---------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                             | d-spacing [Å] | Relative Intensity [%] |
| 6.20                                                                            | 14.25         | 95.15                  |
| 10.33                                                                           | 8.56          | 1.75                   |

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TABLE 3-continued

| X-ray Powder Diffraction Data of the Crystalline Betaine Salt of Bempedoic Acid |               |                        |
|---------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                             | d-spacing [Å] | Relative Intensity [%] |
| 11.68                                                                           | 7.58          | 6.08                   |
| 12.38                                                                           | 7.15          | 3.16                   |
| 13.52                                                                           | 6.55          | 35.24                  |
| 15.25                                                                           | 5.81          | 4.53                   |
| 16.10                                                                           | 5.51          | 10.15                  |
| 17.48                                                                           | 5.07          | 20.77                  |
| 19.33                                                                           | 4.59          | 20.56                  |
| 19.94                                                                           | 4.45          | 14.49                  |
| 21.48                                                                           | 4.14          | 5.58                   |
| 25.64                                                                           | 3.47          | 29.74                  |
| 27.24                                                                           | 3.27          | 13.51                  |
| 31.47                                                                           | 2.84          | 7.33                   |
| 39.02                                                                           | 2.31          | 2.27                   |

## (ii) Crystalline Calcium Salt Forms of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline calcium salt of bempedoic acid. In certain embodiments, the crystalline calcium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $4.9\pm0.2$ ,  $9.1\pm0.2$ , and  $19.7\pm0.2$ . In certain embodiments, the crystalline calcium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $4.9\pm0.2$ ,  $6.4\pm0.2$ ,  $9.1\pm0.2$ ,  $14.8\pm0.2$ ,  $19.7\pm0.2$ , and  $37.1\pm0.2$ .

In certain embodiments, the crystalline calcium salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 4.

TABLE 4

| X-ray Powder Diffraction Data of the Crystalline Calcium Salt of Bempedoic Acid |               |                        |
|---------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                             | d-spacing [Å] | Relative Intensity [%] |
| 4.86                                                                            | 18.16         | 21.41                  |
| 6.44                                                                            | 13.72         | 5.99                   |
| 7.51                                                                            | 11.77         | 1.16                   |
| 9.15                                                                            | 9.67          | 59.65                  |
| 12.25                                                                           | 7.22          | 1.94                   |
| 14.79                                                                           | 5.99          | 6.51                   |
| 16.11                                                                           | 5.50          | 4.16                   |
| 19.71                                                                           | 4.50          | 10.32                  |
| 27.26                                                                           | 3.27          | 2.07                   |
| 32.66                                                                           | 2.74          | 1.12                   |
| 37.10                                                                           | 2.42          | 5.22                   |
| 38.55                                                                           | 2.34          | 1.40                   |

In certain embodiments, the crystalline calcium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $6.0\pm0.2$ ,  $6.8\pm0.2$ ,  $8.5\pm0.2$ , and  $9.8\pm0.2$ . In certain embodiments, the crystalline calcium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $6.0\pm0.2$ ,  $6.8\pm0.2$ ,  $8.5\pm0.2$ ,  $9.8\pm0.2$ ,  $17.1\pm0.2$ , and  $19.0\pm0.2$ .

In certain embodiments, the crystalline calcium salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ ,

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and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 5.

TABLE 5

| X-ray Powder Diffraction Data of the Crystalline Calcium Salt of Bempedoic Acid |               |                        |
|---------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                             | d-spacing [Å] | Relative Intensity [%] |
| 6.03                                                                            | 14.65         | 49.46                  |
| 6.79                                                                            | 13.01         | 28.89                  |
| 8.53                                                                            | 10.37         | 100.00                 |
| 9.80                                                                            | 9.02          | 77.92                  |
| 12.05                                                                           | 7.35          | 2.31                   |
| 14.09                                                                           | 6.29          | 4.03                   |
| 17.10                                                                           | 5.18          | 13.68                  |
| 19.03                                                                           | 4.66          | 5.37                   |
| 33.15                                                                           | 2.70          | 0.90                   |
| 35.89                                                                           | 2.50          | 1.92                   |

## (iii) Crystalline Diethylamine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline diethylamine salt of bempedoic acid. In certain embodiments, the crystalline diethylamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $9.6\pm0.2$ ,  $14.1\pm0.2$ , and  $19.8\pm0.2$ . In certain embodiments, the crystalline diethylamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $9.6\pm0.2$ ,  $14.1\pm0.2$ ,  $17.8\pm0.2$ ,  $19.8\pm0.2$ ,  $22.6\pm0.2$ , and  $38.7\pm0.2$ .

In certain embodiments, the crystalline diethylamine salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 6.

TABLE 6

| X-ray Powder Diffraction Data of the Crystalline Diethylamine Salt of Bempedoic Acid |               |                        |
|--------------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                                  | d-spacing [Å] | Relative Intensity [%] |
| 9.55                                                                                 | 9.26          | 29.82                  |
| 14.08                                                                                | 6.29          | 69.49                  |
| 17.79                                                                                | 4.99          | 11.34                  |
| 19.77                                                                                | 4.49          | 37.13                  |
| 22.60                                                                                | 3.93          | 3.08                   |
| 38.70                                                                                | 2.33          | 6.24                   |

## (iv) Crystalline Ethylenediamine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline ethylenediamine salt of bempedoic acid. In certain embodiments, the crystalline ethylenediamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $6.8\pm0.2$ ,  $10.8\pm0.2$ ,  $16.2\pm0.2$ ,  $18.3\pm0.2$ , and  $18.8\pm0.2$ . In certain embodiments, the crystalline ethylenediamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ):  $6.8\pm0.2$ ,  $7.7\pm0.2$ ,  $10.8\pm0.2$ ,  $13.9\pm0.2$ ,  $15.2\pm0.2$ ,  $16.2\pm0.2$ ,  $18.3\pm0.2$ ,  $18.8\pm0.2$ ,  $21.4\pm0.2$ , and  $22.3\pm0.2$ .

In certain embodiments, the crystalline ethylenediamine salt of bempedoic acid is characterized by the X-ray powder

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diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 7.

TABLE 7

| X-ray Powder Diffraction Data of the Crystalline Ethylenediamine Salt of Bempedoic Acid |                            |                        |
|-----------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                     | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 6.76                                                                                    | 13.07                      | 66.63                  |
| 7.73                                                                                    | 11.44                      | 52.39                  |
| 10.83                                                                                   | 8.17                       | 85.3                   |
| 13.53                                                                                   | 6.54                       | 19.73                  |
| 13.92                                                                                   | 6.36                       | 24.35                  |
| 15.23                                                                                   | 5.82                       | 29.21                  |
| 16.23                                                                                   | 5.46                       | 100.00                 |
| 16.71                                                                                   | 5.31                       | 10.30                  |
| 17.39                                                                                   | 5.10                       | 10.74                  |
| 18.28                                                                                   | 4.85                       | 71.04                  |
| 18.84                                                                                   | 4.71                       | 53.54                  |
| 19.74                                                                                   | 4.50                       | 11.79                  |
| 20.96                                                                                   | 4.24                       | 12.84                  |
| 21.37                                                                                   | 4.16                       | 24.85                  |
| 21.65                                                                                   | 4.10                       | 22.22                  |
| 22.25                                                                                   | 4.00                       | 28.05                  |
| 22.85                                                                                   | 3.89                       | 21.63                  |
| 24.86                                                                                   | 3.58                       | 17.81                  |
| 26.03                                                                                   | 3.42                       | 3.96                   |
| 27.02                                                                                   | 3.30                       | 7.08                   |
| 28.10                                                                                   | 3.18                       | 10.24                  |
| 28.33                                                                                   | 3.15                       | 12.26                  |
| 31.17                                                                                   | 2.87                       | 9.27                   |
| 32.07                                                                                   | 2.79                       | 8.88                   |
| 33.13                                                                                   | 2.70                       | 6.27                   |
| 34.60                                                                                   | 2.59                       | 3.21                   |
| 37.45                                                                                   | 2.40                       | 3.41                   |

#### (v) Crystalline Isonicotinamide Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline isonicotinamide salt of bempedoic acid. In certain embodiments, the crystalline isonicotinamide salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.4\pm 0.2$ ,  $18.8\pm 0.2$ ,  $20.1\pm 0.2$ , and  $24.5\pm 0.2$ . In certain embodiments, the crystalline isonicotinamide salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.4\pm 0.2$ ,  $14.5\pm 0.2$ ,  $18.8\pm 0.2$ ,  $20.1\pm 0.2$ ,  $24.5\pm 0.2$ ,  $26.2\pm 0.2$ , and  $29.5\pm 0.2$ .

In certain embodiments, the crystalline isonicotinamide salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 8.

TABLE 8

| X-ray Powder Diffraction Data of the Crystalline Isonicotinamide Salt of Bempedoic Acid |                            |                        |
|-----------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                     | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 4.36                                                                                    | 20.26                      | 31.75                  |
| 8.89                                                                                    | 9.95                       | 0.44                   |
| 11.49                                                                                   | 7.70                       | 1.86                   |
| 13.05                                                                                   | 6.78                       | 1.75                   |
| 14.47                                                                                   | 6.12                       | 4.77                   |
| 18.36                                                                                   | 4.83                       | 3.21                   |

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TABLE 8-continued

| X-ray Powder Diffraction Data of the Crystalline Isonicotinamide Salt of Bempedoic Acid |                            |                        |
|-----------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                     | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 18.84                                                                                   | 4.71                       | 15.30                  |
| 20.09                                                                                   | 4.42                       | 13.54                  |
| 24.51                                                                                   | 3.63                       | 8.22                   |
| 25.90                                                                                   | 3.44                       | 2.54                   |
| 26.24                                                                                   | 3.40                       | 4.32                   |
| 26.76                                                                                   | 3.33                       | 2.22                   |
| 27.69                                                                                   | 3.22                       | 1.44                   |
| 28.69                                                                                   | 3.11                       | 1.38                   |
| 29.49                                                                                   | 3.03                       | 3.67                   |
| 30.08                                                                                   | 2.97                       | 1.49                   |
| 30.77                                                                                   | 2.91                       | 1.16                   |
| 32.56                                                                                   | 2.75                       | 1.14                   |
| 34.83                                                                                   | 2.58                       | 1.16                   |
| 36.79                                                                                   | 2.44                       | 0.69                   |

#### (vi) Crystalline Potassium Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline potassium salt of bempedoic acid. In certain embodiments, the crystalline potassium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $5.7\pm 0.2$ ,  $7.3\pm 0.2$ ,  $9.6\pm 0.2$ , and  $22.1\pm 0.2$ . In certain embodiments, the crystalline potassium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $5.7\pm 0.2$ ,  $7.3\pm 0.2$ ,  $9.6\pm 0.2$ ,  $16.0\pm 0.2$ ,  $22.1\pm 0.2$ , and  $23.0\pm 0.2$ .

In certain embodiments, the crystalline potassium salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 9.

TABLE 9

| X-ray Powder Diffraction Data of the Crystalline Potassium Salt of Bempedoic Acid |                            |                        |
|-----------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                               | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 5.71                                                                              | 15.48                      | 95.20                  |
| 7.33                                                                              | 12.06                      | 100.00                 |
| 9.58                                                                              | 9.23                       | 17.10                  |
| 15.99                                                                             | 5.54                       | 9.16                   |
| 22.10                                                                             | 4.02                       | 18.51                  |
| 22.97                                                                             | 3.87                       | 3.21                   |
| 24.83                                                                             | 3.59                       | 2.48                   |
| 29.94                                                                             | 2.98                       | 0.86                   |
| 37.72                                                                             | 2.39                       | 1.40                   |

#### (vii) Crystalline Lysine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline lysine salt of bempedoic acid. In certain embodiments, the crystalline lysine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.2\pm 0.2$ ,  $10.2\pm 0.2$ ,  $19.1\pm 0.2$ ,  $19.7\pm 0.2$ , and  $21.9\pm 0.2$ . In certain embodiments, the crystalline lysine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.2\pm 0.2$ ,  $10.2\pm 0.2$ ,  $13.5\pm 0.2$ ,  $14.2\pm 0.2$ ,  $16.0\pm 0.2$ ,  $19.1\pm 0.2$ ,  $19.7\pm 0.2$ , and  $21.9\pm 0.2$ .

In certain embodiments, the crystalline lysine salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ ,

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and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 10.

TABLE 10

| X-ray Powder Diffraction Data of the Crystalline Lysine Salt of Bempedoic Acid |               |                        |
|--------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                            | d-spacing [Å] | Relative Intensity [%] |
| 4.22                                                                           | 20.95         | 79.22                  |
| 10.23                                                                          | 8.65          | 24.52                  |
| 13.53                                                                          | 6.55          | 18.67                  |
| 14.22                                                                          | 6.23          | 20.50                  |
| 15.96                                                                          | 5.55          | 16.26                  |
| 19.12                                                                          | 4.64          | 100.00                 |
| 19.68                                                                          | 4.51          | 30.60                  |
| 21.91                                                                          | 4.06          | 36.00                  |
| 23.09                                                                          | 3.85          | 11.64                  |
| 25.45                                                                          | 3.50          | 13.58                  |
| 33.18                                                                          | 2.70          | 4.36                   |

## (viii) Crystalline Sodium Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline sodium salt of bempedoic acid. In certain embodiments, the crystalline sodium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.1 $\pm$ 0.2, 14.2 $\pm$ 0.2, 18.3 $\pm$ 0.2, and 24.5 $\pm$ 0.2. In certain embodiments, the crystalline sodium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.1 $\pm$ 0.2, 13.4 $\pm$ 0.2, 14.2 $\pm$ 0.2, 16.6 $\pm$ 0.2, 18.3 $\pm$ 0.2, 19.1 $\pm$ 0.2, and 24.5 $\pm$ 0.2.

In certain embodiments, the crystalline sodium salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 11.

TABLE 11

| X-ray Powder Diffraction Data of the Crystalline Sodium Salt of Bempedoic Acid |               |                        |
|--------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                            | d-spacing [Å] | Relative Intensity [%] |
| 6.10                                                                           | 14.48         | 100.00                 |
| 8.16                                                                           | 10.83         | 1.15                   |
| 10.89                                                                          | 8.13          | 2.68                   |
| 12.19                                                                          | 7.28          | 3.98                   |
| 13.36                                                                          | 6.63          | 6.65                   |
| 14.22                                                                          | 6.23          | 11.60                  |
| 16.63                                                                          | 5.33          | 8.50                   |
| 16.86                                                                          | 5.26          | 1.01                   |
| 18.32                                                                          | 4.84          | 24.10                  |
| 19.12                                                                          | 4.64          | 9.38                   |
| 21.36                                                                          | 4.16          | 1.36                   |
| 21.83                                                                          | 4.07          | 1.51                   |
| 22.07                                                                          | 4.03          | 2.91                   |
| 22.42                                                                          | 3.97          | 1.34                   |
| 22.67                                                                          | 3.92          | 1.91                   |
| 24.08                                                                          | 3.70          | 3.19                   |
| 24.50                                                                          | 3.63          | 11.00                  |
| 25.11                                                                          | 3.55          | 3.04                   |
| 28.92                                                                          | 3.09          | 0.79                   |
| 29.79                                                                          | 3.00          | 1.62                   |
| 30.77                                                                          | 2.91          | 1.87                   |
| 32.32                                                                          | 2.77          | 0.59                   |
| 33.02                                                                          | 2.71          | 1.13                   |
| 34.12                                                                          | 2.63          | 0.32                   |
| 37.11                                                                          | 2.42          | 0.34                   |

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TABLE 11-continued

| X-ray Powder Diffraction Data of the Crystalline Sodium Salt of Bempedoic Acid |               |                        |
|--------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                            | d-spacing [Å] | Relative Intensity [%] |
| 37.89                                                                          | 2.37          | 0.41                   |
| 38.77                                                                          | 2.32          | 0.40                   |

- (ix) Crystalline Ammonium Salt Form of Bempedoic Acid
- In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline ammonium salt of bempedoic acid. In certain embodiments, the crystalline ammonium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.9 $\pm$ 0.2, 7.1 $\pm$ 0.2, 14.3 $\pm$ 0.2, 16.0 $\pm$ 0.2, and 21.4 $\pm$ 0.2. In certain embodiments, the crystalline ammonium salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.9 $\pm$ 0.2, 7.1 $\pm$ 0.2, 9.3 $\pm$ 0.2, 14.3 $\pm$ 0.2, 16.0 $\pm$ 0.2, 18.2 $\pm$ 0.2, 19.2 $\pm$ 0.2, 21.4 $\pm$ 0.2, and 22.3 $\pm$ 0.2.

In certain embodiments, the crystalline ammonium salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 12.

TABLE 12

| X-ray Powder Diffraction Data of the Crystalline Ammonium Salt of Bempedoic Acid |               |                        |
|----------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                              | d-spacing [Å] | Relative Intensity [%] |
| 6.92                                                                             | 12.78         | 32.02                  |
| 7.12                                                                             | 12.41         | 52.37                  |
| 9.27                                                                             | 9.54          | 25.96                  |
| 12.37                                                                            | 7.15          | 11.85                  |
| 14.26                                                                            | 6.21          | 45.64                  |
| 15.96                                                                            | 5.55          | 47.86                  |
| 16.72                                                                            | 5.30          | 22.51                  |
| 17.08                                                                            | 5.19          | 24.62                  |
| 18.16                                                                            | 4.89          | 33.61                  |
| 19.17                                                                            | 4.63          | 31.17                  |
| 21.43                                                                            | 4.15          | 44.99                  |
| 22.26                                                                            | 3.99          | 31.44                  |
| 24.05                                                                            | 3.70          | 16.43                  |
| 24.56                                                                            | 3.62          | 10.08                  |
| 27.32                                                                            | 3.26          | 12.47                  |
| 27.79                                                                            | 3.21          | 9.98                   |
| 27.98                                                                            | 3.19          | 10.01                  |
| 29.36                                                                            | 3.04          | 3.14                   |
| 29.83                                                                            | 3.00          | 3.47                   |
| 30.30                                                                            | 2.95          | 4.46                   |
| 30.94                                                                            | 2.89          | 4.92                   |
| 35.56                                                                            | 2.52          | 14.29                  |
| 36.67                                                                            | 2.45          | 5.59                   |
| 37.62                                                                            | 2.39          | 2.72                   |
| 38.66                                                                            | 2.33          | 2.81                   |

## (x) Crystalline Piperazine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline piperazine salt of bempedoic acid. In certain embodiments, the crystalline piperazine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.7 $\pm$ 0.2, 8.7 $\pm$ 0.2, 10.7 $\pm$ 0.2, 15.7 $\pm$ 0.2, and 16.0 $\pm$ 0.2. In certain embodiments, the crystalline piperazine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 6.7 $\pm$ 0.2, 8.7 $\pm$ 0.2, 10.7 $\pm$ 0.2, 15.7 $\pm$ 0.2, 16.0 $\pm$ 0.2, 19.4 $\pm$ 0.2, 20.1 $\pm$ 0.2, and 21.4 $\pm$ 0.2.

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In certain embodiments, the crystalline piperazine salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 13.

TABLE 13

| X-ray Powder Diffraction Data of the Crystalline Piperazine Salt of Bempedoic Acid |                            |                        |
|------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 6.66                                                                               | 13.27                      | 74.02                  |
| 8.69                                                                               | 10.18                      | 40.75                  |
| 10.70                                                                              | 8.27                       | 33.84                  |
| 13.35                                                                              | 6.63                       | 11.79                  |
| 15.68                                                                              | 5.65                       | 48.82                  |
| 15.99                                                                              | 5.54                       | 100.00                 |
| 19.38                                                                              | 4.58                       | 23.99                  |
| 20.10                                                                              | 4.42                       | 18.23                  |
| 21.35                                                                              | 4.16                       | 18.56                  |
| 27.52                                                                              | 3.24                       | 3.45                   |
| 28.61                                                                              | 3.12                       | 6.01                   |
| 34.01                                                                              | 2.64                       | 5.19                   |

#### (xi) Crystalline Tromethamine Salt Form of Bempedoic Acid

In certain embodiments, the crystalline salt form of bempedoic acid is a crystalline tromethamine salt of bempedoic acid. In certain embodiments, the crystalline tromethamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $6.6\pm 0.2$ ,  $18.2\pm 0.2$ ,  $18.6\pm 0.2$ , and  $19.8\pm 0.2$ . In certain embodiments, the crystalline tromethamine salt of bempedoic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $6.6\pm 0.2$ ,  $13.6\pm 0.2$ ,  $18.2\pm 0.2$ ,  $18.6\pm 0.2$ ,  $19.8\pm 0.2$ , and  $26.5\pm 0.2$ .

In certain embodiments, the crystalline tromethamine salt of bempedoic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 14.

TABLE 14

| X-ray Powder Diffraction Data of the Crystalline Tromethamine Salt Bempedoic Acid |                            |                        |
|-----------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                               | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 6.60                                                                              | 13.39                      | 100.00                 |
| 9.10                                                                              | 9.72                       | 1.10                   |
| 13.57                                                                             | 6.53                       | 1.90                   |
| 13.94                                                                             | 6.35                       | 1.47                   |
| 17.08                                                                             | 5.19                       | 1.34                   |
| 18.19                                                                             | 4.88                       | 2.33                   |
| 18.62                                                                             | 4.77                       | 2.76                   |
| 19.31                                                                             | 4.60                       | 1.40                   |
| 19.79                                                                             | 4.49                       | 10.06                  |
| 21.68                                                                             | 4.10                       | 0.33                   |
| 26.55                                                                             | 3.36                       | 2.17                   |
| 28.21                                                                             | 3.16                       | 0.61                   |
| 30.67                                                                             | 2.92                       | 0.27                   |
| 33.69                                                                             | 2.66                       | 0.24                   |

#### C. Co-Crystal Forms of Bempedoic Acid

Moreover, it was discovered that certain co-crystal forms of bempedoic acid could be prepared. In particular, the following co-formers produced co-crystals with bempedoic acid: palmitic acid and aspartame (two crystal forms).

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#### (i) Co-Crystal of Bempedoic Acid and Aspartame

In certain embodiments, the co-crystal form of bempedoic acid is a co-crystal form of bempedoic acid and aspartame. In certain embodiments, the co-crystal form of bempedoic acid and aspartame may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $7.6\pm 0.2$ ,  $8.6\pm 0.2$ ,  $17.3\pm 0.2$ ,  $18.4\pm 0.2$ , and  $25.1\pm 0.2$ . In certain embodiments, the co-crystal form of bempedoic acid and aspartame may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $7.6\pm 0.2$ ,  $8.6\pm 0.2$ ,  $14.4\pm 0.2$ ,  $17.3\pm 0.2$ ,  $18.4\pm 0.2$ ,  $25.1\pm 0.2$ ,  $25.2\pm 0.2$ ,  $26.1$ .

In certain embodiments, the co-crystal form of bempedoic acid and aspartame is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 15.

TABLE 15

| X-ray Powder Diffraction Data of the Co-crystal Form of Bempedoic Acid and Aspartame |                            |                        |
|--------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                  | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 7.6                                                                                  | 11.60                      | 82.14                  |
| 8.6                                                                                  | 10.23                      | 22.88                  |
| 13.3                                                                                 | 6.67                       | 3.69                   |
| 14.4                                                                                 | 6.14                       | 19.99                  |
| 15.3                                                                                 | 5.80                       | 7.52                   |
| 17.3                                                                                 | 5.12                       | 32.09                  |
| 18.4                                                                                 | 4.82                       | 100.00                 |
| 20.9                                                                                 | 4.26                       | 4.07                   |
| 23.0                                                                                 | 3.87                       | 6.98                   |
| 25.1                                                                                 | 3.54                       | 23.14                  |
| 25.2                                                                                 | 3.53                       | 17.65                  |
| 26.1                                                                                 | 3.42                       | 19.40                  |
| 29.0                                                                                 | 3.08                       | 1.60                   |
| 31.2                                                                                 | 2.87                       | 8.23                   |
| 32.4                                                                                 | 2.76                       | 2.29                   |
| 35.6                                                                                 | 2.52                       | 2.32                   |
| 36.6                                                                                 | 2.45                       | 3.48                   |

In certain embodiments, the co-crystal form of bempedoic acid and aspartame may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.4\pm 0.2$ ,  $6.8\pm 0.2$ ,  $10.6\pm 0.2$ ,  $13.2\pm 0.2$ , and  $18.4\pm 0.2$ . In certain embodiments, the co-crystal form of bempedoic acid and aspartame may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles ( $2\theta$ ):  $4.4\pm 0.2$ ,  $5.6\pm 0.2$ ,  $6.8\pm 0.2$ ,  $10.6\pm 0.2$ ,  $12.3\pm 0.2$ ,  $13.2\pm 0.2$ ,  $13.6\pm 0.2$ ,  $16.2\pm 0.2$ ,  $17.6\pm 0.2$ , and  $18.4\pm 0.2$ .

In certain embodiments, the co-crystal form of bempedoic acid and aspartame is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle  $2\theta$ , and optionally inter-planar distances  $d$ , and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 16.

TABLE 16

| X-ray Powder Diffraction Data of the Co-crystal Form of Bempedoic Acid and Aspartame |                            |                        |
|--------------------------------------------------------------------------------------|----------------------------|------------------------|
| Angle [ $2\theta$ ]                                                                  | d-spacing [ $\text{\AA}$ ] | Relative Intensity [%] |
| 4.4                                                                                  | 20.07                      | 100.00                 |
| 5.6                                                                                  | 15.71                      | 18.95                  |
| 6.8                                                                                  | 13.01                      | 23.46                  |

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TABLE 16-continued

| X-ray Powder Diffraction Data of the<br>Co-crystal Form of Bempedoic Acid and Aspartame |               |                        |
|-----------------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                                     | d-spacing [Å] | Relative Intensity [%] |
| 8.6                                                                                     | 10.27         | 3.51                   |
| 10.6                                                                                    | 8.38          | 26.31                  |
| 12.3                                                                                    | 7.22          | 14.92                  |
| 13.2                                                                                    | 6.70          | 23.91                  |
| 13.6                                                                                    | 6.53          | 18.83                  |
| 16.2                                                                                    | 5.47          | 15.84                  |
| 16.8                                                                                    | 5.28          | 13.01                  |
| 17.6                                                                                    | 5.03          | 19.15                  |
| 18.4                                                                                    | 4.83          | 25.66                  |
| 19.0                                                                                    | 4.68          | 12.76                  |
| 22.9                                                                                    | 3.88          | 8.04                   |
| 25.1                                                                                    | 3.54          | 5.95                   |
| 29.2                                                                                    | 3.05          | 6.26                   |
| 31.0                                                                                    | 2.89          | 5.42                   |
| 31.5                                                                                    | 2.83          | 5.95                   |
| 32.9                                                                                    | 2.72          | 4.77                   |

## (ii) Co-Crystal of Bempedoic Acid and Palmitic Acid

In certain embodiments, the co-crystal form of bempedoic acid is a co-crystal form of bempedoic acid and palmitic acid. In certain embodiments, the co-crystal form of bempedoic acid and palmitic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 4.3 $\pm$ 0.2, 6.3 $\pm$ 0.2, 8.5 $\pm$ 0.2, and 17.0 $\pm$ 0.2. In certain embodiments, the co-crystal form of bempedoic acid and palmitic acid may be characterized by an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2 $\theta$ ): 4.3 $\pm$ 0.2, 6.3 $\pm$ 0.2, 8.5 $\pm$ 0.2, 10.5 $\pm$ 0.2, 17.0 $\pm$ 0.2, and 25.5 $\pm$ 0.2.

In certain embodiments, the co-crystal form of bempedoic acid and palmitic acid is characterized by the X-ray powder diffraction pattern expressed in terms of diffraction angle 2 $\theta$ , and optionally inter-planar distances d, and relative intensity (expressed as a percentage with respect to the most intense peak) as set forth in Table 17.

TABLE 17

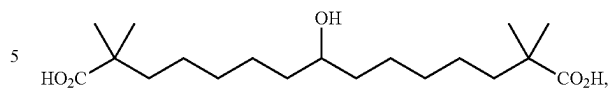
| X-ray Powder Diffraction Data of the Co-crystal<br>Form of Bempedoic Acid and Palmitic Acid |               |                        |
|---------------------------------------------------------------------------------------------|---------------|------------------------|
| Angle [2 $\theta$ ]                                                                         | d-spacing [Å] | Relative Intensity [%] |
| 4.3                                                                                         | 20.74         | 100.00                 |
| 6.3                                                                                         | 14.00         | 27.40                  |
| 8.5                                                                                         | 10.42         | 11.40                  |
| 10.5                                                                                        | 8.41          | 6.34                   |
| 17.0                                                                                        | 5.22          | 9.46                   |
| 21.2                                                                                        | 4.19          | 4.72                   |
| 24.6                                                                                        | 3.61          | 4.33                   |
| 25.5                                                                                        | 3.49          | 4.99                   |
| 29.9                                                                                        | 2.99          | 2.02                   |
| 34.4                                                                                        | 2.61          | 2.54                   |

### III. Methods of Preparing Bempedoic Acid Including Purified Bempedoic Acid

As described herein, in one aspect, the invention provides methods of preparing 8-hydroxy-2,2,14,14-tetramethylpentadecanedioic acid, which is a compound of formula (V):

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(V)

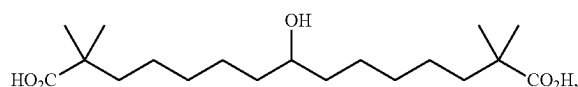


which methods also include making a pharmaceutically acceptable salt thereof.

It should be understood that methods of the invention include preparing bempedoic acid. In certain embodiments, the methods of preparing bempedoic acid result in purified bempedoic acid, which also can be described herein with respect to a pharmaceutical material, i.e., a pharmaceutical material comprising an amount of bempedoic acid or an amount of a compound of formula (V), or a pharmaceutically acceptable salt thereof. These terms and phrases can be used interchangeably herein, unless otherwise stated or understood from the context.

Accordingly, in various embodiments, methods are provided for preparing a pharmaceutical material comprising a compound of formula (V):

(V)

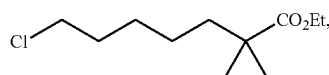


or a pharmaceutically acceptable salt thereof.

In various embodiments, the methods generally include:

- (a) contacting ethyl isobutyrate with a substituted 5-chloropentane in the presence of a first base to form a compound of formula (I):

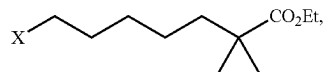
(I)



wherein the substituted 5-chloropentane is selected from the group consisting of 1-bromo-5-chloropentane and 1-iodo-5-chloropentane;

- (b) contacting the compound of formula (I) with a salt of formula [M]<sup>+</sup>[X]<sup>-</sup> to form a compound of formula (II):

(II)

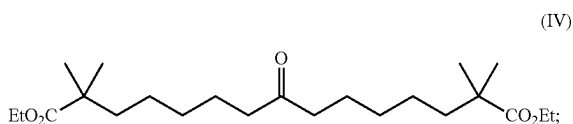


wherein [M]<sup>+</sup> is selected from the group consisting of Li<sup>+</sup>, Na<sup>+</sup> and K<sup>+</sup>, wherein [X]<sup>-</sup> is selected from the group consisting of Br<sup>-</sup> and I<sup>-</sup>;

- (c) contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of a second base to form a first intermediate, and contacting the first intermediate with an acid to form a compound of formula (IV):



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and

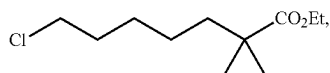
- (d) contacting the compound of formula (IV) with a reducing agent to form a second intermediate, and contacting the second intermediate with a hydrolyzing base to form a compound of formula (V).

In certain embodiments of the invention, the method further comprises:

- (e) purifying the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

#### Synthesis of a Compound of Formula (I)—Step (a)

In various embodiments, synthesis of the compound of formula (I):



generally comprises contacting ethyl isobutyrate with a substituted 5-chloropentane in the presence of a first base.

In certain embodiments, in step (a), contacting ethyl isobutyrate with the substituted 5-chloropentane in the presence of a first base is conducted at a temperature in the range of from about  $-30^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $5^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-15^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-18^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-20^{\circ}\text{C}$ ., from about  $-30^{\circ}\text{C}$ . to about  $-25^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $-15^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $-18^{\circ}\text{C}$ ., from about  $-25^{\circ}\text{C}$ . to about  $-20^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-15^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-18^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-18^{\circ}\text{C}$ . to about  $-15^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., or from about  $-5^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ . In certain embodiments, in step (a), contacting ethyl isobutyrate with the substituted 5-chloropentane in the presence of a first base is conducted at a temperature in the range of from about  $-20^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ . In certain embodiments, in step (a), contacting ethyl isobutyrate with the

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substituted 5-chloropentane in the presence of a first base is conducted at a temperature in the range of from about  $-18^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ .

In certain embodiments, in step (a), less than about 0.5% by weight, about 0.6% by weight, about 0.7% by weight, about 0.8% by weight, about 0.9% by weight, about 1% by weight, about 1.1% by weight, about 1.2% by weight, about 1.3% by weight, about 1.4% by weight, or about 1.5% by weight of the substituted 5-chloropentane remains after forming the compound of formula (I). In some embodiments, in step (a), less than about 1% by weight of the substituted 5-chloropentane remains after forming the compound of formula (I).

In certain embodiments, in step (a), the molar ratio of ethyl isobutyrate to the substituted 5-chloropentane is about 1:1, about 1.01:1, about 1.02:1, about 1.03:1, about 1.04:1, about 1.05:1, about 1.06:1, about 1.07:1, about 1.08:1, about 1.09:1, about 1.1:1, about 1.11:1, about 1.12:1, about 1.13:1, about 1.14:1, about 1.15:1, about 1.16:1, about 1.17:1, about 1.18:1, about 1.19:1, about 1.2:1, or about 1.21:1, including the ranges between each of these ratios. In some embodiments, in step (a), the molar ratio of ethyl isobutyrate to the substituted 5-chloropentane is about 1.1:1. In some embodiments, the molar ratio of ethyl isobutyrate to the substituted 5-chloropentane is from about 1.1:1 to about 1.21:1.

In certain embodiments, in step (a), the substituted 5-chloropentane is contacted ethyl isobutyrate, which is present in an amount of about 1 molar equivalent, about 1.01 molar equivalents, about 1.02 molar equivalents, about 1.03 molar equivalents, about 1.04 molar equivalents, about 1.05 molar equivalents, about 1.06 molar equivalents, about 1.07 molar equivalents, about 1.08 molar equivalents, about 1.09 molar equivalents, about 1.1 molar equivalents, about 1.11 molar equivalents, about 1.12 molar equivalents, about 1.13 molar equivalents, about 1.14 molar equivalents, about 1.15 molar equivalents, about 1.16 molar equivalents, about 1.17 molar equivalents, about 1.18 molar equivalents, about 1.19 molar equivalents, about 1.2 molar equivalents, or about 1.21 molar equivalents. In some embodiments, in step (a), the substituted 5-chloropentane is contacted with about 1.1 molar equivalents of ethyl isobutyrate.

In certain embodiments, in step (a), contacting ethyl isobutyrate and the substituted 5-chloropentane occurs by adding ethyl isobutyrate and the substituted 5-chloropentane to a reactor. In some embodiments, in step (a), adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs at a temperature of less than about  $10^{\circ}\text{C}$ ., less than about  $5^{\circ}\text{C}$ ., less than about  $0^{\circ}\text{C}$ ., less than about  $-5^{\circ}\text{C}$ ., less than about  $-10^{\circ}\text{C}$ ., less than about  $-15^{\circ}\text{C}$ ., less than about  $-20^{\circ}\text{C}$ ., less than about  $-25^{\circ}\text{C}$ ., or less than about  $-30^{\circ}\text{C}$ . In some embodiments, in step (a), adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs at a temperature of about  $10^{\circ}\text{C}$ ., about  $5^{\circ}\text{C}$ ., about  $0^{\circ}\text{C}$ ., about  $-5^{\circ}\text{C}$ ., about  $-7^{\circ}\text{C}$ ., about  $-10^{\circ}\text{C}$ ., about  $-12^{\circ}\text{C}$ ., about  $-14^{\circ}\text{C}$ ., about  $-16^{\circ}\text{C}$ ., about  $-18^{\circ}\text{C}$ ., about  $-20^{\circ}\text{C}$ ., about  $-22^{\circ}\text{C}$ ., about  $-24^{\circ}\text{C}$ ., about  $-26^{\circ}\text{C}$ ., about  $-28^{\circ}\text{C}$ ., or about  $-30^{\circ}\text{C}$ . In some embodiments, in step (a), adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs at a temperature of about  $-5^{\circ}\text{C}$ . In some embodiments, in step (a), adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs at a temperature of about  $-12^{\circ}\text{C}$ . In some embodiments, in step (a), adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs at a temperature of about  $-18^{\circ}\text{C}$ .

In certain embodiments, in step (a), the time of adding ethyl isobutyrate and the substituted 5-chloropentane to the

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reactor is about 5 mins, about 10 mins, about 15 mins, about 20 mins, about 30 mins, about 40 mins, about 50 mins, about 1 hour, about 2 hours, about 3 hours, about 4 hours, about 5 hours, about 6 hours, about 7 hours, about 8 hours, about 9 hours, about 10 hours, about 11 hours, or about 12 hours.

In certain embodiments, in step (a), the time of adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor is from about 10 mins to about 60 mins, from about 20 mins to about 60 mins, from about 30 mins to about 60 mins, from about 40 mins to about 60 mins, from about 50 mins to about 60 mins, from about 10 mins to about 50 mins, from about 10 mins to about 40 mins, from about 10 mins to about 30 mins, from about 10 mins to about 20 mins, from about 20 mins to about 50 mins, from about 20 mins to about 40 mins, from about 20 mins to about 30 mins, from about 30 mins to about 50 mins, from about 30 mins to about 40 mins, or from about 40 mins to about 50 mins.

In certain embodiments, in step (a), the time of adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor is from about 1 hour to about 12 hours, from about 2 hours to about 12 hours, from about 3 hours to about 12 hours, from about 4 hours to about 12 hours, from about 5 hours to about 12 hours, from about 6 hours to about 12 hours, from about 7 hours to about 12 hours, from about 8 hours to about 12 hours, from about 9 hours to about 12 hours, from about 10 hours to about 12 hours, from about 11 hours to about 12 hours, from about 1 hours to about 11 hours, from about 1 hours to about 10 hours, from about 1 hours to about 9 hours, from about 1 hours to about 8 hours, from about 1 hours to about 7 hours, from about 1 hours to about 6 hours, from about 1 hours to about 5 hours, from about 1 hours to about 4 hours, from about 1 hours to about 3 hours, from about 1 hours to about 2 hours, from about 2 hours to about 11 hours, from about 2 hours to about 10 hours, from about 2 hours to about 9 hours, from about 2 hours to about 8 hours, from about 2 hours to about 7 hours, from about 2 hours to about 6 hours, from about 2 hours to about 5 hours, from about 2 hours to about 4 hours, from about 2 hours to about 3 hours, from about 3 hours to about 11 hours, from about 3 hours to about 10 hours, from about 3 hours to about 9 hours, from about 3 hours to about 8 hours, from about 3 hours to about 7 hours, from about 3 hours to about 6 hours, from about 3 hours to about 5 hours, from about 3 hours to about 4 hours, from about 4 hours to about 11 hours, from about 4 hours to about 10 hours, from about 4 hours to about 9 hours, from about 4 hours to about 8 hours, from about 4 hours to about 7 hours, from about 4 hours to about 6 hours, from about 4 hours to about 5 hours, from about 5 hours to about 11 hours, from about 5 hours to about 10 hours, from about 5 hours to about 9 hours, from about 5 hours to about 8 hours, from about 5 hours to about 7 hours, from about 5 hours to about 6 hours, from about 6 hours to about 11 hours, from about 6 hours to about 10 hours, from about 6 hours to about 9 hours, from about 6 hours to about 8 hours, from about 6 hours to about 7 hours, from about 7 hours to about 11 hours, from about 7 hours to about 10 hours, from about 7 hours to about 9 hours, from about 7 hours to about 8 hours, from about 7 hours to about 11 hours, from about 8 hours to about 10 hours, from about 8 hours to about 9 hours, from about 9 hours to about 11 hours, from about 9 hours to about 10 hours, or from about 10 hours to about 11 hours.

In some embodiments, adding ethyl isobutyrate and the substituted 5-chloropentane to the reactor occurs simultaneously. In some embodiments, adding ethyl isobutyrate to the reactor occurs prior to adding the substituted 5-chloropentane to the reactor. In some embodiments, adding ethyl

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isobutyrate to the reactor occurs after adding the substituted 5-chloropentane to the reactor.

In certain embodiments, contacting ethyl isobutyrate with the substituted 5-chloropentane in the presence of a first base forms a reaction mixture. In certain embodiments, in step (a), at the end of the reaction, the methods include quenching the reaction mixture with an acid. In some embodiments, the acid is hydrochloric acid.

In certain embodiments, ethyl isobutyrate and the substituted 5-chloropentane are starting materials used in the production of the compound of formula (I). In certain embodiments, the purity of the substituted 5-chloropentane is  $\geq 99\%$ ,  $\geq 99.1\%$ ,  $\geq 99.2\%$ ,  $\geq 99.3\%$ ,  $\geq 99.4\%$ ,  $\geq 99.5\%$ ,  $\geq 99.6\%$ ,  $\geq 99.7\%$ ,  $\geq 99.8\%$ , or  $\geq 99.9\%$ , as measured by gas chromatography (GC). In some embodiments, the purity of the substituted 5-chloropentane is  $\geq 99\%$ , as measured by GC.

In certain embodiments, the purity of ethyl isobutyrate is  $\geq 99\%$ ,  $\geq 99.1\%$ ,  $\geq 99.2\%$ ,  $\geq 99.3\%$ ,  $\geq 99.4\%$ ,  $\geq 99.5\%$ ,  $\geq 99.6\%$ ,  $\geq 99.7\%$ ,  $\geq 99.8\%$ , or  $\geq 99.9\%$ , as measured by gas chromatography (GC). In some embodiments, the purity of ethyl isobutyrate is  $\geq 99.5\%$ , as measured by GC.

In certain embodiments, the concentration of ethanol present in ethyl isobutyrate is  $\leq 0.05\%$ ,  $\leq 0.06\%$ ,  $\leq 0.07\%$ ,  $0.08\%$ ,  $\leq 0.09\%$ ,  $0.1\%$ ,  $\leq 0.11\%$ ,  $\leq 0.12\%$ ,  $\leq 0.13\%$ ,  $\leq 0.14\%$ , or  $\leq 0.15\%$ , as measured by GC. In some embodiments, the concentration of ethanol present in ethyl isobutyrate is  $0.1\%$ , as measured by GC.

In certain embodiments, the substituted 5-chloropentane is 1-iodo-5-chloropentane. In certain embodiments, the substituted 5-chloropentane is 1-bromo-5-chloropentane.

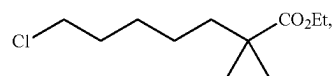
In certain embodiments, the purity of 1-iodo-5-chloropentane or 1-bromo-5-chloropentane is  $\geq 99\%$ ,  $\geq 99.1\%$ ,  $\geq 99.2\%$ ,  $\geq 99.3\%$ ,  $\geq 99.4\%$ ,  $\geq 99.5\%$ ,  $\geq 99.6\%$ ,  $\geq 99.7\%$ ,  $99.8\%$ , or  $\geq 99.9\%$ , as measured by gas chromatography (GC).

In certain embodiments, in step (a), the first base is selected from the group consisting of lithium diisopropylamide, lithium bis(trimethylsilyl)amide, sodium hydride, sodium amide, lithium amide, and lithium tetramethylpiperidide. In some embodiments, in step (a), the first base is lithium diisopropylamide.

In certain embodiments, the amount of unreacted substituted 5-chloropentane remaining upon completion of step (a) is  $\leq 0.05\%$ ,  $\leq 0.06\%$ ,  $\leq 0.07\%$ ,  $\leq 0.08\%$ ,  $\leq 0.09\%$ ,  $\leq 0.1\%$ ,  $\leq 0.11\%$ ,  $\leq 0.12\%$ ,  $\leq 0.12\%$ ,  $\leq 0.13\%$ ,  $\leq 0.14\%$ ,  $\leq 0.15\%$ ,  $\leq 0.16\%$ ,  $\leq 0.17\%$ ,  $\leq 0.18\%$ ,  $\leq 0.19\%$ , or  $\leq 0.2\%$ , as measured by GC. In some embodiments, the amount of unreacted substituted 5-chloropentane remaining upon completion of step (a) is  $\leq 0.21\%$ ,  $\leq 0.22\%$ ,  $\leq 0.23\%$ ,  $\leq 0.24\%$ ,  $\leq 0.25\%$ ,  $\leq 0.25\%$ ,  $\leq 0.26\%$ ,  $\leq 0.27\%$ ,  $\leq 0.28\%$ ,  $\leq 0.29\%$ ,  $\leq 0.3\%$ ,  $\leq 0.31\%$ ,  $\leq 0.32\%$ ,  $\leq 0.33\%$ ,  $\leq 0.34\%$ ,  $\leq 0.35\%$ ,  $\leq 0.36\%$ ,  $\leq 0.37\%$ ,  $\leq 0.38\%$ ,  $\leq 0.39\%$ , or  $\leq 0.4\%$ , as measured by GC.

#### Synthesis of Lithium Diisopropylamide (a First Base for Making the Compound of Formula (I))

In various embodiments, the synthesis of lithium diisopropylamide, which is a base used for making the compound of formula (I):



(I)

generally comprises contacting diisopropylamine with butyllithium.

In certain embodiments, the molar ratio of butyllithium to diisopropylamine is about 1:1.04, about 1:1.05, about 1:1.06, about 1:1.07, about 1:1.08, about 1:1.09, about 1:1.1, about 1:1.2, about 1:1.3, about 1:1.4, about 1:1.5, or about 1:1.6. In some embodiments, the molar ratio of butyllithium to diisopropylamine is about 1:1.07. In some embodiments, the molar ratio of butyllithium to diisopropylamine is about 1:1.5.

In certain embodiments, the molar ratio of butyllithium to diisopropylamine is from about 1:1.04 to about 1:1.1, from about 1:1.05 to about 1:1.1, from about 1:1.06 to about 1:1.1, from about 1:1.07 to about 1:1.1, from about 1:1.08 to about 1:1.1, from about 1:1.09 to about 1:1, from about 1:1.04 to about 1:1.09, from about 1:1.04 to about 1:1.08, from about 1:1.04 to about 1:1.07, from about 1:1.04 to about 1:1.06, from about 1:1.04 to about 1:1.05, from about 1:1.05 to about 1:1.09, from about 1:1.05 to about 1:1.08, from about 1:1.05 to about 1:1.07, from about 1:1.05 to about 1:1.06, from about 1:1.06 to about 1:1.09, from about 1:1.06 to about 1:1.08, from about 1:1.06 to about 1:1.07, from about 1:1.07 to about 1:1.09, from about 1:1.07 to about 1:1.08, or from about 1:1.08 to about 1:1.09. In some embodiments, the molar ratio of butyllithium to diisopropylamine is from about 1:1.06 to about 1:1.07.

In certain embodiments, contacting diisopropylamine with butyllithium is conducted at a temperature of  $\leq 0^\circ \text{C}$ .,  $\leq -5^\circ \text{C}$ .,  $\leq -10^\circ \text{C}$ .,  $\leq -15^\circ \text{C}$ ., or  $\leq -20^\circ \text{C}$ . In some embodiments, contacting diisopropylamine with butyllithium is conducted at a temperature of  $\leq -5^\circ \text{C}$ .

In certain embodiments, contacting diisopropylamine with butyllithium is conducted in tetrahydrofuran (THF).

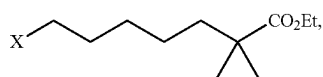
In certain embodiments, lithium diisopropylamide is prepared before step (a), for example, before contacting ethyl isobutyrate with 1-bromo-5-chloropentane.

In certain embodiments, lithium diisopropylamide is prepared in situ during step (a), for example, while contacting ethyl isobutyrate with 1-bromo-5-chloropentane. In some embodiments, when lithium diisopropylamide is prepared in situ during step (a), the molar ratio of the substituted 5-chloropentane to ethyl isobutyrate to butyllithium to diisopropylamine is about 1:1.1:1.2:1.26, about 1:1.1:1.15:1.75, about 1:1.1:1.24:1.3, about 1:1.1:1.2:1.29, about 1:1.1:1.2:1.28, or about 1:1.1:1.25:1.15:1.2:1.25-1.75. In some embodiments, when lithium diisopropylamide is prepared in situ during step (a), the molar ratio of the substituted 5-chloropentane to ethyl isobutyrate to butyllithium to diisopropylamine is about 1:1.1:1.2:1.28.

In some embodiments, the substituted 5-chloropentane is 1-bromo-5-chloropentane.

#### Synthesis of a Compound of Formula (II)—Step (b)

In various embodiments, the synthesis of a compound of formula (II):



wherein X is Br or I, generally comprises contacting the compound of formula (I) with a salt of formula  $[M]^+[X]^-$ .

In certain embodiments, in step (b), contacting the compound of formula (I) with a salt of formula  $[M]^+[X]^-$ , is conducted in a solvent comprising one or more of acetone, 2-butanone, methyl isobutyl ketone, THE and 3-pentanone, wherein M is selected from the group consisting of Li, Na, and K, and X is selected from the group consisting of Br and I.

In certain embodiments, in step (b), the solvent comprises less than about 3.5% by weight water, less than about 3% by weight water, less than about 2.5% by weight water, less than about 2% by weight water, less than about 1.5% by weight water, less than about 1% by weight water, or less than about 0.5% by weight water. In some embodiments, in step (b), the solvent comprises less than less than about 3% by weight water.

In certain embodiments, in step (b), contacting the compound of formula (I) with the salt of formula  $[M]^+[X]^-$  comprises contacting the compound of formula (I) with about 1 molar equivalent, about 1.05 molar equivalents, about 1.1 molar equivalents, about 1.15 molar equivalents, about 1.2 molar equivalents, or about 1.25 molar equivalents of the salt of formula  $[M]^+[X]^-$  based on the molar amount of the compound of formula (I). In certain embodiments, in step (b), contacting the compound of formula (I) with the salt of formula  $[M]^+[X]^-$  comprises contacting the compound of formula (I) with about 1.1 molar equivalents of the salt of formula  $[M]^+[X]^-$  based on the molar amount of the compound of formula (I).

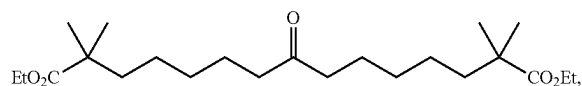
In certain embodiments, in step (b), contacting the compound of formula (I) with the salt of formula  $[M]^+[X]^-$  is conducted at a temperature in the range of from about  $75^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $81^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $82^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $83^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $84^\circ \text{C}$ . to about  $85^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $79^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $78^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $77^\circ \text{C}$ ., from about  $75^\circ \text{C}$ . to about  $76^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $79^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $78^\circ \text{C}$ ., from about  $76^\circ \text{C}$ . to about  $77^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $79^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $78^\circ \text{C}$ ., from about  $77^\circ \text{C}$ . to about  $77^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $78^\circ \text{C}$ . to about  $79^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $79^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $81^\circ \text{C}$ ., from about  $80^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $81^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $81^\circ \text{C}$ . to about  $82^\circ \text{C}$ ., from about  $81^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., from about  $81^\circ \text{C}$ . to about  $80^\circ \text{C}$ ., from about  $82^\circ \text{C}$ . to about  $84^\circ \text{C}$ ., from about  $82^\circ \text{C}$ . to about  $83^\circ \text{C}$ ., or from about  $83^\circ \text{C}$ . to about  $84^\circ \text{C}$ . In some embodiments, in step (b), contacting the compound of formula (I) with the salt of formula  $[M]^+[X]^-$  is conducted at a temperature in the range of from about  $78^\circ \text{C}$ . to about  $82^\circ \text{C}$ .

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In certain embodiments, in step (b), the salt of formula  $[M]^+[X]^-$  is selected from the group consisting of lithium bromide (LiBr), lithium iodide (LiI), potassium bromide (KBr), potassium iodide (KI), sodium bromide (NaBr) and sodium iodide (NaI). In some embodiments, in step (b), the salt of formula  $[M]^+[X]^-$  is sodium iodide.

#### Synthesis of a Compound of Formula (IV)—Step (c)

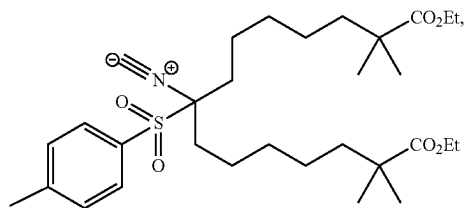
In various embodiments, the synthesis of a compound of formula (IV):



generally comprises contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of a second base to form a first intermediate, and contacting the first intermediate with an acid.

#### Synthesis of the First Intermediate

In various embodiments, the synthesis of the first intermediate:



generally comprises contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of a second base.

In certain embodiments, in step (c), the second base is selected from sodium hydride, potassium tert-butoxide, and sodium tert-pentoxide. In some embodiments, in step (c), the second base is sodium tert-pentoxide.

In certain embodiments, in step (c), contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of sodium tert-pentoxide to form the first intermediate is conducted at a temperature in a range of from about  $-20^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., or from about  $-10^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ . In certain embodiments, in step (c), contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide in the presence of sodium tert-pentoxide to form the first intermediate is conducted at a temperature in a range of from about  $-15^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ .

In certain embodiments, in step (c), contacting the compound of formula (II) with toluenesulfonylmethyl isocya-

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nide in the presence of a second base to form the first intermediate is conducted at a temperature of about  $-20^{\circ}\text{C}$ ., about  $-15^{\circ}\text{C}$ ., about  $-10^{\circ}\text{C}$ ., about  $-5^{\circ}\text{C}$ ., or about  $0^{\circ}\text{C}$ .

In certain embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide is about 1.7:1, about 1.8:1, about 1.9:1, about 2:1, about 2.1:1, or about 2.2:1. In certain embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide is about 1.9:1.

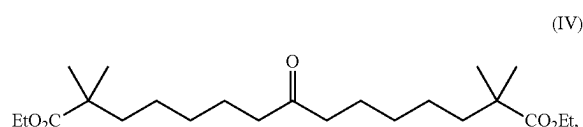
In some embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide to the second base is about 1.9:1.0:2.1. In some embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide to the second base is about 1.9:1.0:2.2.

In some embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide to sodium tert-pentoxide is about 1.9:1.0:2.1. In some embodiments, in step (c), the molar ratio of the compound of formula (II) to toluenesulfonylmethyl isocyanide to sodium tert-pentoxide is about 1.9:1.0:2.2.

In certain embodiments, in step (c), contacting the compound of formula (II) with toluenesulfonylmethyl isocyanide is conducted in a solvent comprising dimethylacetamide or that is dimethylacetamide.

#### Synthesis of a Compound of Formula (IV)

In various embodiments, the synthesis of a compound of formula (IV):



generally comprises contacting the first intermediate with an acid.

In certain embodiments, in step (c), contacting the first intermediate with an acid is conducted at a temperature in a range of from about  $-15^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $5^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $10^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $15^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $20^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $25^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $30^{\circ}\text{C}$ . to about  $35^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $30^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $25^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $20^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $15^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $30^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $25^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $20^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $15^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $30^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $25^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $20^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $15^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $30^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $25^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $20^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $15^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., from about  $0^{\circ}\text{C}$ . to about  $30^{\circ}\text{C}$ ., from about  $5^{\circ}\text{C}$ . to about  $25^{\circ}\text{C}$ ., from

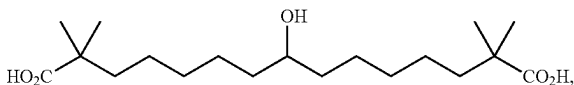
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about 5° C. to about 20° C., from about 5° C. to about 15° C., from about 5° C. to about 10° C., from about 10° C. to about 30° C., from about 10° C. to about 25° C., from about 10° C. to about 20° C., from about 10° C. to about 15° C., from about 15° C. to about 30° C., from about 15° C. to about 25° C., from about 15° C. to about 20° C., from about 20° C. to about 30° C., from about 20° C. to about 25° C., or from about 25° C. to about 30° C. In some embodiments, in step (c), contacting the first intermediate with an acid is conducted at a temperature in a range of from about -10° C. to about 35° C. In some embodiments, in step (c), contacting the first intermediate with an acid is conducted at a temperature in a range of from about -15° C. to about 25° C. In some embodiments, in step (c), contacting the first intermediate with an acid is conducted at a temperature in a range of from about 10° C. to about 25° C.

In certain embodiments, in step (c), the acid is hydrochloric acid.

#### Synthesis of a Compound of Formula (V)—Step (d)

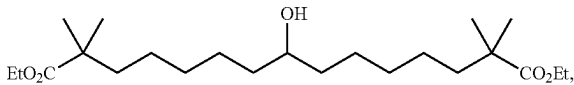
In various embodiments the synthesis of a compound of formula (V):



generally comprises contacting the compound of formula (IV) with a reducing agent to form a second intermediate, and contacting the second intermediate with a hydrolyzing base.

#### Synthesis of the Second Intermediate

In various embodiments, the synthesis of the second intermediate:



generally comprises contacting the compound of formula (IV) with a reducing agent.

In certain embodiments, in step (d), the reducing agent is selected from the group consisting of sodium borohydride, sodium cyanoborohydride, cerium borohydride, zinc borohydride and diisobutylaluminum hydride. In some embodiments, the reducing agent is sodium borohydride.

In certain embodiments, in step (d), contacting the compound of formula (IV) with a reducing agent comprises contacting the compound of formula (IV) with about 0.25 molar equivalents, about 0.3 molar equivalents, about 0.35 molar equivalents, about 0.4 molar equivalents, about 0.45, about 0.5 molar equivalents, about 0.6 molar equivalents, about 0.7 molar equivalents, about 0.8 molar equivalents, about 0.9 molar equivalents, about 1.0 molar equivalents, about 1.1 molar equivalents, about 1.2 molar equivalents, about 1.3 molar equivalents, about 1.4 molar equivalents, or about 1.5 molar equivalents of the reducing agent based on the molar amount of the compound of formula (IV). In some embodiments, in step (d), contacting the compound of formula (IV) with a reducing agent comprises contacting the

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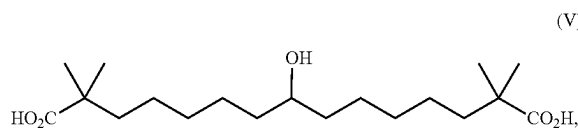
compound of formula (IV) with about 0.35 molar equivalents of the reducing agent based on the molar amount of the compound of formula (IV).

In certain embodiments, in step (c), contacting the compound of formula (IV) with a reducing agent is conducted at a temperature in a range of from about 5° C. to about 30° C., from about 10° C. to about 30° C., from about 15° C. to about 30° C., from about 20° C. to about 30° C., from about 25° C. to about 30° C., from about 5° C. to about 25° C., from about 5° C. to about 20° C., from about 5° C. to about 15° C., from about 5° C. to about 10° C., from about 10° C. to about 25° C., from about 10° C. to about 20° C., from about 10° C. to about 15° C., from about 15° C. to about 25° C., from about 15° C. to about 20° C., or from about 20° C. to about 25° C.

In certain embodiments, in step (c), contacting the compound of formula (IV) with a reducing agent is conducted at a temperature of about 5° C., about 10° C., about 15° C., about 20° C., about 25° C., or about 30° C.

#### Synthesis of a Compound of Formula (V)

In various embodiments, the synthesis of a compound of formula (V):



generally comprises contacting the second intermediate with a hydrolyzing base.

In certain embodiments, in step (d), the concentration of the hydrolyzing base is about 20% w/w, about 25% w/w, about 30% w/w, about 35% w/w, about 40% w/w, about 45% w/w, about 50% w/w, about 55% w/w, or about 60% w/w. In some embodiments, in step (d), the concentration of the hydrolyzing base is about 50% w/w.

In certain embodiments, in step (d), contacting the second intermediate with a hydrolyzing base to form a compound of formula (V) is conducted in a solution. In some embodiments, in step (d), the method further comprises adjusting the pH of the solution comprising the compound of formula (V) to between about 3 to about 7.

In certain embodiments, in step (d), contacting the second intermediate with a hydrolyzing base is conducted at a temperature in a range of from about 30° C. to about 60° C., from about 35° C. to about 60° C., from about 40° C. to about 60° C., from about 45° C. to about 60° C., from about 50° C. to about 60° C., from about 55° C. to about 60° C., from about 30° C. to about 55° C., from about 30° C. to about 50° C., from about 30° C. to about 45° C., from about 30° C. to about 40° C., from about 30° C. to about 35° C., from about 35° C. to about 55° C., from about 35° C. to about 50° C., from about 35° C. to about 45° C., from about 35° C. to about 40° C., from about 40° C. to about 55° C., from about 40° C. to about 50° C., from about 40° C. to about 45° C., from about 45° C. to about 55° C., from about 45° C. to about 50° C., or from about 50° C. to about 60° C.

In certain embodiments, in step (d), contacting the second intermediate with a hydrolyzing base is conducted at a temperature of about 30° C., about 35° C., about 40° C., about 45° C., about 50° C., about 55° C., or about 60° C. In

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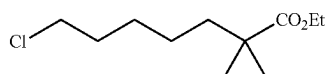
certain embodiments, in step (d), contacting the second intermediate with a hydrolyzing base is conducted at a temperature of about 50° C.

In certain embodiments, in step (d), the hydrolyzing base is sodium hydroxide.

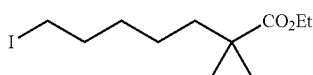
In certain embodiments, in step (d), contacting the compound of formula (IV) with a reducing agent to form a second intermediate, and contacting the second intermediate with a hydrolyzing base to form a compound of formula (V) is conducted in a single reaction vessel.

In some embodiments of the present aspect, the methods can include:

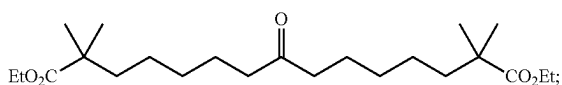
- (a) contacting ethyl isobutyrate with 1-bromo-5-chloropentane in the presence of a first base to form a compound of formula (I):



- (b) contacting the compound of formula (I) with sodium iodide to form a compound of formula (IIa):



- (c) contacting the compound of formula (IIa) with toluenesulfonylmethyl isocyanide in the presence of a second base to form a first intermediate, and contacting the first intermediate with an acid to form a compound of formula (IV):



and

- (d) contacting the compound of formula (IV) with a reducing agent to form a second intermediate, and contacting the second intermediate with a hydrolyzing base to form a compound of formula (V).

In some embodiments of the invention, the method further comprises:

- (e) purifying the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In some embodiments as described above, the same conditions (e.g., temperatures) amounts, ratios, equivalents, times, purities, and other parameters or variables that were previously described can be equally applicable, for example, where “1-bromo-5-chloropentane” is substituted for “substituted-5-chloropentane;” “sodium iodide” is substituted for “a salt of formula [M]<sup>+</sup>[X]<sup>-</sup>;” “Na” is substituted for “M;” “I” is substituted for “X;” and a “compound of formula (IIa)” is substituted for a “compound of formula II.”

In addition, in certain embodiments, the concentration of 1,5-dichloropentane in 1-bromo-5-chloropentane is ≤0.1%, ≤0.2%, ≤0.3%, ≤0.4%, ≤0.5%, ≤0.6%, ≤0.7%, ≤0.8%,

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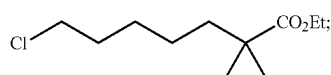
≤0.9%, or ≤1%, as measured by GC. In some embodiments, the concentration of 1,5-dichloropentane in 1-bromo-5-chloropentane is ≤0.5%, as measured by GC.

- In certain embodiments, the concentration of 1,5-dibromopentane in 1-bromo-5-chloropentane is ≤0.05%, ≤0.1%, ≤0.15%, ≤0.2%, ≤0.25%, ≤0.3%, ≤0.35%, ≤0.4%, ≤0.45%, ≤0.5%, ≤0.6%, ≤0.7%, ≤0.8%, ≤0.9%, or ≤1.0%, as measured by GC. In some embodiments, the concentration of 1,5-dibromopentane in 1-bromo-5-chloropentane is ≤0.2%, as measured by GC. In some embodiments, the concentration of 1,5-dibromopentane in 1-bromo-5-chloropentane is ≤1.0%, as measured by GC.

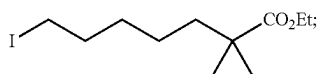
- In certain embodiments, the amount of unreacted 1-bromo-5-chloropentane remaining upon completion of step (a) is ≤0.05%, ≤0.06%, ≤0.07%, ≤0.08%, ≤0.09%, ≤0.1%, ≤0.11%, ≤0.12%, ≤0.12%, ≤0.13%, ≤0.14%, ≤0.15%, ≤0.16%, ≤0.17%, ≤0.18%, ≤0.19%, or ≤0.2%, as measured by GC. In some embodiments, the amount of unreacted 1-bromo-5-chloropentane remaining upon completion of step (a) is ≤0.21%, ≤0.22%, ≤0.23%, ≤0.24%, ≤0.25%, ≤0.26%, ≤0.27%, ≤0.28%, ≤0.29%, ≤0.3%, ≤0.31%, ≤0.32%, ≤0.33%, ≤0.34%, ≤0.35%, ≤0.36%, ≤0.37%, ≤0.38%, ≤0.39%, or ≤0.4%, as measured by GC.

In certain embodiments, a method of preparing a compound of formula (V) comprises:

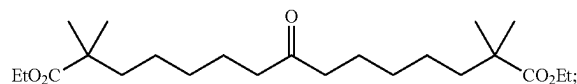
- (a) contacting 1-bromo-5-chloropentane with about 1.1 molar equivalents of ethyl isobutyrate in the presence of lithium diisopropylamide at a temperature in the range of from about -20° C. to about 0° C. to form a compound of formula (I):



- (b) contacting the compound of formula (I) with about 1.1 molar equivalents of sodium iodide in 2-butanone at a temperature in the range of from about 78° C. to about 82° C. to form a compound of formula (IIa):



- (c) contacting the compound of formula (IIa) with toluenesulfonylmethyl isocyanide in the presence of sodium tert-pentoxide in dimethylacetamide at a temperature in the range of from about -20° C. to about 10° C. to form a first intermediate, and contacting the first intermediate with an acid at a temperature in the range of from about -10° C. to about 35° C. to form a compound of formula (IV):



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and

- (d) contacting the compound of formula (IV) with about 0.35 molar equivalents of sodium borohydride to form a second intermediate, and contacting the second intermediate with sodium hydroxide in a solution to form a compound of formula (V).

In certain embodiments of the invention, the method further comprises:

- (e) purifying the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In certain embodiments as described above, the same conditions (e.g., temperatures) amounts, ratios, equivalents, times, purities, and other parameters or variables that were previously described can be equally applicable, for example, where "lithium diisopropylamide" is substituted for "a first base;" "sodium tert-pentoxide" is substituted for "a second base;" "sodium borohydride" is substituted for "a reducing agent;" and "sodium hydroxide" is substituted for a hydrolyzing base."

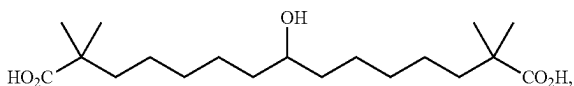
In addition, in certain embodiments, in step (a), contacting ethyl isobutyrate with 1-bromo-5-chloropentane in the presence of lithium diisopropylamide is conducted at a temperature in the range of from about  $-20^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-10^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-5^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., from about  $-20^{\circ}\text{C}$ . to about  $-15^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., from about  $-15^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., or from about  $-10^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ .

In certain embodiments, in step (b), contacting the compound of formula (I) with sodium iodide is conducted at a temperature in the range of from about  $78^{\circ}\text{C}$ . to about  $82^{\circ}\text{C}$ ., from about  $78^{\circ}\text{C}$ . to about  $80^{\circ}\text{C}$ ., or from about  $80^{\circ}\text{C}$ . to about  $82^{\circ}\text{C}$ .

In certain embodiments, in step (c), the molar ratio of the compound of formula (IIa) to toluenesulfonylmethyl isocyanide to sodium tert-pentoxide is about 1.9:1:2.1.

Purification of a Compound of Formula (V)—Step (e)

Various embodiments of the invention include methods for producing a pharmaceutical material comprising a purified amount of a compound of formula (V):



i.e., the methods can include purifying the compound of formula (V).

In various embodiments, purifying the compound of formula (V) comprises filtering the compound of formula (V) in a solvent through silica gel. In some embodiments, the solvent comprises ethyl acetate. In some embodiments, the solvent is ethyl acetate.

In certain embodiments, purifying the compound of formula (V) comprises crystallizing the compound of formula (V) to provide a crystalline form of the compound of formula (V).

In certain embodiments, purifying the compound of formula (V) comprises contacting the compound of formula (V) with charcoal and then filtering the charcoal. In some embodiments, contacting the compound of formula (V) with charcoal comprises contacting the compound of formula (V) with a solution, wherein the solution comprises acetonitrile and activated charcoal (e.g., 5% (w/w) activated charcoal).

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In certain embodiments, purifying the compound of formula (V) comprises recrystallizing the crystalline form of the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In various embodiments, purifying the compound of formula (V) comprises:

- (f) adjusting the pH of the solution comprising the compound of formula (V) to about 5 to about 6;
- (g) extracting the compound of formula (V) from the solution using methyl tert-butyl ether to provide a methyl tert-butyl ether solution comprising the compound of formula (V);
- (h) exchanging the methyl tert-butyl ether of the methyl tert-butyl ether solution with ethyl acetate to provide an ethyl acetate solution comprising the compound of formula (V);
- (i) filtering the ethyl acetate solution comprising the compound of formula (V) through silica gel;
- (j) crystallizing the compound of formula (V) using ethyl acetate and water to provide a crystalline form of the compound of formula (V); and
- (k) recrystallizing the crystalline form of the compound of formula (V) using ethyl acetate and water to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In certain embodiments, in step (g), extracting the compound of formula (V) from the solution using methyl tert-butyl ether is conducted at a temperature less than or equal to about  $5^{\circ}\text{C}$ ., about  $10^{\circ}\text{C}$ ., about  $15^{\circ}\text{C}$ ., about  $20^{\circ}\text{C}$ ., about  $25^{\circ}\text{C}$ ., about  $30^{\circ}\text{C}$ ., about  $35^{\circ}\text{C}$ ., about  $40^{\circ}\text{C}$ ., about  $45^{\circ}\text{C}$ ., about  $50^{\circ}\text{C}$ ., or about  $55^{\circ}\text{C}$ . In certain embodiments, in step (g), extracting the compound of formula (V) from the solution using methyl tert-butyl ether is conducted at a temperature less than or equal to about  $15^{\circ}\text{C}$ . In certain embodiments, in step (g), extracting the compound of formula (V) from the solution using methyl tert-butyl ether is conducted at a temperature less than or equal to about  $50^{\circ}\text{C}$ .

In certain embodiments, in step (j), crystallizing the compound of formula (V) using ethyl acetate and water is conducted over a temperature range of about  $60^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $55^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $50^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $45^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $40^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $35^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $30^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $25^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $20^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $15^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $10^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $5^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $0^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $-5^{\circ}\text{C}$ . to about  $-10^{\circ}\text{C}$ ., about  $60^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $55^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $50^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $45^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $40^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $35^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $30^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $25^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $20^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $15^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $10^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $5^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $0^{\circ}\text{C}$ . to about  $-5^{\circ}\text{C}$ ., about  $60^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $55^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $50^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $45^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $40^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $35^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $30^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $25^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $20^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $15^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $10^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $5^{\circ}\text{C}$ . to about  $0^{\circ}\text{C}$ ., about  $60^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $55^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $50^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $45^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $40^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $35^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $30^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $25^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $20^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $15^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $10^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $5^{\circ}\text{C}$ . to about  $5^{\circ}\text{C}$ ., about  $60^{\circ}\text{C}$ . to about  $10^{\circ}\text{C}$ ., about  $55^{\circ}\text{C}$ . to

about 10° C., about 50° C. to about 10° C., about 45° C. to about 10° C., about 40° C. to about 10° C., about 35° C. to about 10° C., about 30° C. to about 10° C., about 25° C. to about 10° C., about 20° C. to about 10° C., about 15° C. to about 10° C., about 60° C. to about 15° C., about 55° C. to about 15° C., about 50° C. to about 15° C., about 45° C. to about 15° C., about 40° C. to about 15° C., about 35° C. to about 15° C., about 30° C. to about 15° C., about 25° C. to about 15° C., about 20° C. to about 15° C., about 60° C. to about 20° C., about 55° C. to about 20° C., about 50° C. to about 20° C., about 45° C. to about 20° C., about 40° C. to about 20° C., about 35° C. to about 20° C., about 30° C. to about 20° C., about 25° C. to about 20° C., about 60° C. to about 25° C., about 55° C. to about 25° C., about 50° C. to about 25° C., about 45° C. to about 25° C., about 40° C. to about 25° C., about 35° C. to about 25° C., about 30° C. to about 25° C., about 60° C. to about 30° C., about 55° C. to about 30° C., about 50° C. to about 30° C., about 45° C. to about 30° C., about 40° C. to about 30° C., about 35° C., about 55° C. to about 35° C., about 50° C. to about 35° C., about 45° C. to about 35° C., about 40° C. to about 35° C., about 60° C. to about 40° C., about 55° C. to about 40° C., about 50° C. to about 40° C., about 45° C. to about 40° C., about 60° C. to about 45° C., about 55° C. to about 45° C., about 50° C. to about 45° C., about 60° C. to about 50° C., about 55° C. to about 50° C., or about 60° C. to about 55° C. In certain embodiments, in step (j), crystallizing the compound of formula (V) using ethyl acetate and water is conducted over a temperature range of about 50° C. to about -5° C.

In certain embodiments, in step (k), recrystallizing the crystalline form of the compound of formula (V) using ethyl acetate and water is conducted over a temperature range of about 70° C. to about 5° C., about 65° C. to about 5° C., about 60° C. to about 5° C., about 55° C. to about 5° C., about 50° C. to about 5° C., about 45° C. to about 5° C., about 40° C. to about 5° C., about 35° C. to about 5° C., about 30° C. to about 5° C., about 25° C. to about 5° C., about 20° C. to about 5° C., about 15° C. to about 5° C., about 10° C. to about 5° C., about 70° C. to about 10° C., about 65° C. to about 10° C., about 60° C. to about 10° C., about 55° C. to about 10° C., about 50° C. to about 10° C., about 45° C. to about 10° C., about 40° C. to about 10° C., about 35° C. to about 10° C., about 30° C. to about 10° C., about 25° C. to about 10° C., about 20° C. to about 10° C., about 15° C. to about 10° C., about 70° C. to about 15° C., about 65° C. to about 15° C., about 60° C. to about 15° C., about 55° C. to about 15° C., about 50° C. to about 15° C., about 45° C. to about 15° C., about 40° C. to about 15° C., about 35° C. to about 15° C., about 30° C. to about 15° C., about 25° C. to about 15° C., about 20° C. to about 15° C., about 70° C. to about 20° C., about 65° C. to about 20° C., about 60° C. to about 20° C., about 55° C. to about 20° C., about 50° C. to about 20° C., about 45° C. to about 20° C., about 40° C. to about 20° C., about 35° C. to about 20° C., about 30° C. to about 20° C., about 25° C. to about 20° C., about 70° C. to about 25° C., about 65° C. to about 25° C., about 60° C. to about 25° C., about 55° C. to about 25° C., about 50° C. to about 25° C., about 45° C. to about 25° C., about 40° C. to about 25° C., about 35° C. to about 25° C., about 30° C. to about 25° C., about 70° C. to about 30° C., about 65° C. to about 30° C., about 60° C. to about 30° C., about 55° C. to about 30° C., about 50° C. to about 30° C., about 45° C. to about 30° C., about 40° C. to about 30° C., about 35° C. to about 30° C., about 70° C. to about 35° C., about 65° C. to about 35° C., about 60° C. to about 35° C., about 55° C. to about 35° C., about 50° C. to about 35° C., about 45° C. to about 35° C., about 40° C. to about 35° C.,

about 70° C. to about 40° C., about 65° C. to about 40° C., about 60° C. to about 40° C., about 55° C. to about 40° C., about 50° C. to about 40° C., about 45° C. to about 40° C., about 70° C. to about 45° C., about 65° C. to about 45° C., about 60° C. to about 45° C., about 55° C. to about 45° C., about 50° C. to about 45° C., about 70° C. to about 50° C., about 65° C. to about 50° C., about 60° C. to about 50° C., about 55° C. to about 50° C., about 70° C. to about 55° C., about 65° C. to about 55° C., about 60° C. to about 55° C., about 70° C. to about 60° C., about 65° C. to about 60° C., or about 70° C. to about 65° C. In certain embodiments, in step (k), recrystallizing the crystalline form of the compound of formula (V) using ethyl acetate and water is conducted over a temperature range of about 70° C. to about 5° C.

In certain embodiments, purifying the compound of formula (V) comprises:

- (l) dissolving the crystalline form of the compound of formula (V) in acetonitrile, thereby forming a solution;  
(m) contacting the solution with charcoal;  
(n) filtering the charcoal to provide a purified solution comprising the compound of formula (V); and  
(o) crystallizing the compound of formula (V) from the purified solution to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

It should be understood that, in various embodiments, the above steps (1)-(o) can be conducted after or without conducting steps (f)-(k).

### Crystallization of the Compound of Formula (V)

In various embodiments, purifying the compound of formula (V) comprises crystallizing the compound of formula (V) from a solvent or a mixture of solvents, for example, ethyl acetate and water.

In certain embodiments, the concentration of water in the mixture of solvents comprising ethyl acetate and water is about 0.5% (w/w), about 0.6% (w/w), about 0.75% (w/w), about 0.9% (w/w), about 1.05% (w/w), about 1.2% (w/w), about 1.35% (w/w), about 1.4% (w/w), or about 1.5% (w/w). In some embodiments, the concentration of water in the mixture of solvents comprising ethyl acetate and water is about 1.05% (w/w).

In certain embodiments, the concentration of water in the mixture of solvents comprising ethyl acetate and water is from about 0.5% (w/w) to about 1.5% (w/w), from about 0.5% (w/w) to about 1.4% (w/w), from about 0.5% (w/w) to about 1.35% (w/w), from about 0.5% (w/w) to about 1.2% (w/w), from about 0.5% (w/w) to about 1.05% (w/w), from about 0.5% (w/w) to about 0.9% (w/w), from about 0.5% (w/w) to about 0.75% (w/w), from about 0.5% (w/w) to about 0.6% (w/w), from about 0.6% (w/w) to about 1.5% (w/w), from about 0.6% (w/w) to about 1.4% (w/w), from about 0.6% (w/w) to about 1.35% (w/w), from about 0.6% (w/w) to about 1.2% (w/w), from about 0.6% (w/w) to about 1.05% (w/w), from about 0.6% (w/w) to about 0.9% (w/w), from about 0.6% (w/w) to about 0.75% (w/w), from about 0.75% (w/w) to about 1.5% (w/w), from about 0.75% (w/w) to about 1.4% (w/w), from about 0.75% (w/w) to about 1.35% (w/w), from about 0.75% (w/w) to about 1.2% (w/w), from about 0.75% (w/w) to about 1.05% (w/w), from about 0.75% (w/w) to about 0.9% (w/w), from about 0.9% (w/w) to about 1.35% (w/w), from about 0.9% (w/w) to about 1.2% (w/w), from about 0.9% (w/w) to about 1.05% (w/w), from about 1.05% (w/w) to about 1.5% (w/w), from about 1.05% (w/w) to about 1.35% (w/w), from about 1.05% (w/w) to about 1.2% (w/w), from about 1.2% (w/w) to about 1.5% (w/w), from about 1.2% (w/w) to about 1.35% (w/w), from about 1.2% (w/w) to about 1.05% (w/w), from about 1.2% (w/w) to about 0.9% (w/w), from about 1.2% (w/w) to about 0.75% (w/w), from about 1.2% (w/w) to about 0.6% (w/w), from about 1.2% (w/w) to about 0.5% (w/w), from about 1.35% (w/w) to about 1.5% (w/w), from about 1.35% (w/w) to about 1.4% (w/w), from about 1.35% (w/w) to about 1.3% (w/w), from about 1.35% (w/w) to about 1.2% (w/w), from about 1.35% (w/w) to about 1.1% (w/w), from about 1.35% (w/w) to about 1.0% (w/w), from about 1.35% (w/w) to about 0.9% (w/w), from about 1.35% (w/w) to about 0.8% (w/w), from about 1.35% (w/w) to about 0.7% (w/w), from about 1.35% (w/w) to about 0.6% (w/w), from about 1.35% (w/w) to about 0.5% (w/w), from about 1.4% (w/w) to about 1.5% (w/w), from about 1.4% (w/w) to about 1.3% (w/w), from about 1.4% (w/w) to about 1.2% (w/w), from about 1.4% (w/w) to about 1.1% (w/w), from about 1.4% (w/w) to about 1.0% (w/w), from about 1.4% (w/w) to about 0.9% (w/w), from about 1.4% (w/w) to about 0.8% (w/w), from about 1.4% (w/w) to about 0.7% (w/w), from about 1.4% (w/w) to about 0.6% (w/w), from about 1.4% (w/w) to about 0.5% (w/w), from about 1.5% (w/w) to about 1.4% (w/w), from about 1.5% (w/w) to about 1.3% (w/w), from about 1.5% (w/w) to about 1.2% (w/w), from about 1.5% (w/w) to about 1.1% (w/w), from about 1.5% (w/w) to about 1.0% (w/w), from about 1.5% (w/w) to about 0.9% (w/w), from about 1.5% (w/w) to about 0.8% (w/w), from about 1.5% (w/w) to about 0.7% (w/w), from about 1.5% (w/w) to about 0.6% (w/w), from about 1.5% (w/w) to about 0.5% (w/w).



(w/w) to about 1.35% (w/w), or from about 1.35% (w/w) to about 1.5% (w/w). In some embodiments, the concentration of water in the mixture of solvents comprising ethyl acetate and water is from about 0.6% (w/w) to about 1.4% (w/w). In some embodiments, the concentration of water in the mixture of solvents comprising ethyl acetate and water is from about 0.75% (w/w) to about 1.35% (w/w).

In various embodiments, crystallizing the compound of formula (V) from a mixture comprising the compound of formula (V), ethyl acetate and water comprises:

- (1) cooling the mixture from a first temperature ( $T^1$ ) to a second temperature ( $T^2$ ), wherein  $T^1$  is from about 40° C. to about 60° C.,  $T^2$  is from about 15° C. to about 30° C., and the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 10° C./hour to about 20° C./hour;
- (2) holding the mixture at  $T^2$  for at least 3 hours;
- (3) cooling the mixture from  $T^2$  to a third temperature ( $T^3$ ), wherein  $T^3$  is from about -5° C. to about 10° C. and the mixture is cooled from  $T^2$  to  $T^3$  at a rate of from about 5° C./hour to about 15° C./hour; and
- (4) holding the mixture at  $T^3$  for at least 3 hours, thereby producing a crystalline form of the compound of formula (V).

In certain embodiments, in crystallization step (1),  $T^1$  is from about 40° C. to about 60° C., from about 45° C. to about 60° C., from about 50° C. to about 60° C., from about 55° C. to about 60° C., from about 40° C. to about 55° C., from about 40° C. to about 50° C., from about 40° C. to about 45° C., from about 45° C. to about 55° C., from about 45° C. to about 50° C., or from about 50° C. to about 55° C. In some embodiments, in crystallization step (1),  $T^1$  is from about 40° C. to about 60° C. In some embodiments, in crystallization step (1),  $T^1$  is from about 45° C. to about 55° C.

In certain embodiments, in crystallization step (1),  $T^1$  is about 40° C., about 45° C., about 50° C., about 55° C., or about 60° C. In some embodiments, in crystallization step (1),  $T^1$  is about 50° C.

In certain embodiments, in crystallization step (1),  $T^2$  is from about 15° C. to about 30° C., from about 20° C. to about 30° C., from about 25° C. to about 30° C., from about 15° C. to about 25° C., from about 15° C. to about 20° C., or from about 20° C. to about 25° C. In some embodiments, in crystallization step (i),  $T^2$  is from about 15° C. to about 30° C. In some embodiments, in crystallization step (1),  $T^2$  is from about 20° C. to about 25° C.

In certain embodiments, in crystallization step (1),  $T^2$  is about 15° C., about 16° C., about 17° C., about 18° C., about 19° C., about 20° C., about 21° C., about 22° C., about 23° C., about 24° C., about 25° C., about 26° C., about 27° C., about 28° C., about 29° C., or about 30° C. In some embodiments, in crystallization step (1),  $T^2$  is about 22° C.

In certain embodiments, in crystallization step (1), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 10° C./hour to about 20° C./hour, from about 12° C./hour to about 20° C./hour, from about 14° C./hour to about 20° C./hour, from about 16° C./hour to about 20° C./hour, from about 18° C./hour to about 20° C./hour, from about 10° C./hour to about 18° C./hour, from about 10° C./hour to about 16° C./hour, from about 10° C./hour to about 14° C./hour, from about 10° C./hour to about 12° C./hour, from about 12° C./hour to about 18° C./hour, from about 12° C./hour to about 16° C./hour, from about 12° C./hour to about 14° C./hour, from about 14° C./hour to about 18° C./hour, from about 14° C./hour to about 16° C./hour, or from about 16° C./hour to about 18° C./hour. In some embodiments, the mixture is cooled from  $T^1$  to  $T^2$  at a rate

of from about 10° C./hour to about 20° C./hour. In some embodiments, the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 10° C./hour to about 12° C./hour.

In certain embodiments, in crystallization step (1), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of about 10° C./hour, about 11° C./hour, about 12° C./hour, about 13° C./hour, about 14° C./hour, about 15° C./hour, about 16° C./hour, about 17° C./hour, about 18° C./hour, about 19° C./hour, or about 20° C./hour. In certain embodiments, in crystallization step (1), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of about 11° C./hour.

In certain embodiments, in crystallization step (2), the mixture is held at  $T^2$  for at least 3 hours, at least 3.5 hours, at least 4 hours, at least 4.5 hours, at least 5 hours, at least 5.5 hours, at least 6 hours, at least 6.5 hours, at least 7 hours, at least 7.5 hours, at least 8 hours, at least 8.5 hours, at least 9 hours, at least 9.5 hours, or at least 10 hours. In some embodiments, in crystallization step (2), the mixture is held at  $T^2$  for at least 6 hours.

In certain embodiments, in crystallization step (2), the mixture is held at  $T^2$  for no greater than 3 hours, no greater than 3.5 hours, no greater than 4 hours, no greater than 4.5 hours, no greater than 5 hours, no greater than 5.5 hours, no greater than 6 hours, no greater than 6.5 hours, no greater than 7 hours, no greater than 7.5 hours, no greater than 8 hours, no greater than 8.5 hours, no greater than 9 hours, no greater than 9.5 hours, no greater than 10 hours, no greater than 10.5 hours, no greater than 11 hours, no greater than 11.5 hours, no greater than 12 hours, no greater than 16 hours, no greater than 20 hours, or no greater than 24 hours. In some embodiments, in crystallization step (2), the mixture is held at  $T^2$  for no greater than 6 hours.

In certain embodiments, in crystallization step (3),  $T^3$  is from about -5° C. to about 10° C., from about 0° C. to about 10° C., from about 5° C. to about 10° C., from about -5° C. to about 5° C., from about -5° C. to about 0° C., or from about 0° C. to about 5° C. In some embodiments,  $T^3$  is from about -5° C. to about 10° C. In some embodiments,  $T^3$  is from about -5° C. to about 5° C.

In certain embodiments, in crystallization step (3),  $T^3$  is about -5° C., about 0° C., about 5° C., or about 10° C. In some embodiments, in crystallization step (3),  $T^3$  is about 0° C.

In certain embodiments, in crystallization step (3), the mixture is cooled from  $T^2$  to  $T^3$  at a rate of from about 5° C./hour to about 15° C./hour, from about 7° C./hour to about 15° C./hour, from about 9° C./hour to about 15° C./hour, from about 11° C./hour to about 15° C./hour, from about 13° C./hour to about 15° C./hour, from about 5° C./hour to about 13° C./hour, from about 5° C./hour to about 11° C./hour, from about 5° C./hour to about 9° C./hour, from about 5° C./hour to about 7° C./hour, from about 7° C./hour to about 13° C./hour, from about 7° C./hour to about 11° C./hour, from about 7° C./hour to about 9° C./hour, from about 9° C./hour to about 13° C./hour, from about 9° C./hour to about 11° C./hour, or from about 11° C./hour to about 13° C./hour. In some embodiments, in crystallization step (3), the mixture is cooled from  $T^2$  to  $T^3$  at a rate of from about 5° C./hour to about 15° C./hour. In some embodiments, in crystallization step (3), the mixture is cooled from  $T^2$  to  $T^3$  at a rate of from about 7° C./hour to about 13° C./hour.

In certain embodiments, in crystallization step (3), the mixture is cooled from  $T^2$  to  $T^3$  at a rate of about 5° C./hour, about 6° C./hour, about 7° C./hour, about 8° C./hour, about 9° C./hour, about 10° C./hour, about 11° C./hour, about 12° C./hour, about 13° C./hour, about 14° C./hour, or about 15° C./hour.

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C./hour. In some embodiments, in crystallization step (3), the mixture is cooled from  $T^2$  to  $T^3$  at a rate of about 11° C./hour.

In certain embodiments, in crystallization step (4), the mixture is held at  $T^3$  for at least 3 hours, at least 3.5 hours, at least 4 hours, at least 4.5 hours, at least 5 hours, at least 5.5 hours, at least 6 hours, at least 6.5 hours, at least 7 hours, at least 7.5 hours, at least 8 hours, at least 8.5 hours, at least 9 hours, at least 9.5 hours, or at least 10 hours. In some embodiments, in crystallization step (4), the mixture is held at  $T^3$  for at least 6 hours.

In certain embodiments, in crystallization step (4), the mixture is held at  $T^1$  for no greater than 3 hours, no greater than 3.5 hours, no greater than 4 hours, no greater than 4.5 hours, no greater than 5 hours, no greater than 5.5 hours, no greater than 6 hours, no greater than 6.5 hours, no greater than 7 hours, no greater than 7.5 hours, no greater than 8 hours, no greater than 8.5 hours, no greater than 9 hours, no greater than 9.5 hours, no greater than 10 hours, no greater than 10.5 hours, no greater than 11 hours, no greater than 11.5 hours, no greater than 12 hours, no greater than 16 hours, no greater than 20 hours, or no greater than 24 hours. In some embodiments, in crystallization step (4), the mixture is held at  $T^3$  for no greater than 10 hours.

In certain embodiments, crystallizing the compound of formula (V) from a mixture comprising the compound of formula (V), ethyl acetate and water further comprises seeding the mixture with an amount of a crystalline form of the compound of formula (V) prior to crystallization step (1), during crystallization step (1), during crystallization step (2), during crystallization step (3), during crystallization step (4), or any combination thereof, in order to facilitate the crystallization of the compound of formula (V).

In certain embodiments, the amount of seed added to the mixture is about 0.001 kg/kg, about 0.0012 kg/kg, about 0.0014 kg/kg, about 0.0016 kg/kg, about 0.0018 kg/kg, or about 0.002 kg/kg, based on the weight of seed per kilogram of the compound of formula (IV) produced in step (c). In some embodiments, the amount of seed added to the mixture is about 0.0014 kg/kg, based on the weight of seed per kilogram of the compound of formula (IV) produced in step (c).

In certain embodiments, the crystalline form of the compound of formula (V) added as seed to the mixture may be a crystalline form of the compound of formula (V) as characterized herein, for example, by an X-ray powder diffraction pattern or peak(s), and/or other characteristic properties of the crystalline form of bempedoic acid.

In certain embodiments, crystallizing the compound of formula (V) from a mixture comprising the compound of formula (V), ethyl acetate and water comprises filtration of the compound of formula (V).

In certain embodiments, filtering the crystalline form of the compound of formula (V) occurs at a temperature of from about -20° C. to about 5° C., from about -15° C. to about 5° C., from about -10° C. to about 5° C., from about -5° C. to about 5° C., from about 0° C. to about 5° C., from about -20° C. to about 0° C., from about -20° C. to about -5° C., from about -20° C. to about -10° C., from about -20° C. to about -15° C., from about -15° C. to about 5° C., from about -15° C. to about 0° C., from about -15° C. to about -5° C., from about -15° C. to about -10° C., from about -10° C. to about 5° C., from about -10° C. to about 0° C., from about 0° C. to about -10° C. to about -5° C., from about -5° C. to about 5° C., from about -5° C. to about 0° C., or from about 0° C. to about 5° C. In some embodiments, filtering the crystalline form of the compound of formula (V) occurs

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at a temperature of from about -20° C. to about -5° C. In some embodiments, filtering the crystalline form of the compound of formula (V) occurs at a temperature of about -5° C. to about 5° C.

In certain embodiments, filtering the crystalline form of the compound of formula (V) occurs at a temperature of about -20° C., about -15° C., about -10° C., about -5° C., about 0° C., or about 5° C. In some embodiments, filtering the crystalline form of the compound of formula (V) occurs at a temperature of about -10° C., about -5° C., or about 0° C.

In certain embodiments, filtration further comprises washing. In certain embodiments, washing comprises washing the crystalline form of the compound of formula (V) with a solvent. In some embodiments, washing comprises washing the crystalline form of the compound of formula (V) with ethyl acetate.

In certain embodiments, the temperature of the solvent, for example, ethyl acetate, is from about -20° C. to about 10° C., from about -10° C. to about 10° C., from about 0° C. to about 10° C., from about -20° C. to about 0° C., from about -20° C. to about -10° C., or from about -10° C. to about 0° C. In some embodiments, the temperature of the solvent is from about -10° C. to about 10° C.

In certain embodiments, the temperature of the solvent, for example, ethyl acetate, is about -20° C., about -15° C., about -10° C., about -5° C., about 0° C., about 5° C., or about 10° C.

In some embodiments, the temperature of the solvent is about 0° C.

In various embodiments, crystallizing the compound of formula (V) from a mixture comprising the compound of formula (V), ethyl acetate and water comprises:

- (1) cooling the mixture from a first temperature ( $T^1$ ) to a second temperature ( $T^2$ ), wherein  $T^1$  is about 50° C.,  $T^2$  is about 22° C., and the mixture is cooled from  $T^1$  to  $T^2$  at a rate of about 11° C./hour;
- (2) holding the compound of formula (V) at  $T^2$  for at least 6 hours;
- (3) cooling the compound of formula (V) from  $T^2$  to a third temperature ( $T^3$ ), wherein  $T^3$  is about 0° C. and the mixture is cooled from  $T^2$  to  $T^3$  at a rate of about 11° C./hour; and
- (4) holding the compound of formula (V) at  $T^3$  for at least 6 hours, thereby producing a crystalline form of the compound of formula (V).

It should be understood that the conditions (e.g., temperatures), times, seeding, amounts, compounds, and other parameters and/or variables for crystallizing the compound of formula (V) as described herein can be equally applicable to the immediately above-described crystallizing process, unless otherwise stated or understood from the context (e.g., the conditions or parameters fall outside the values or ranges in the immediately above-described crystallizing process).

In certain embodiments, the crystalline form of the compound of formula (V) produced by any of the crystallization methods described herein may be a crystalline form of the compound of formula (V) as characterized herein, for example, by an X-ray powder diffraction pattern or peak(s), and/or other characteristic properties of the crystalline form of bempedoic acid.

In certain embodiments, the purity of the crystalline form of the compound of formula (V) produced by any of the crystallization methods described herein is greater than about 85%, greater than about 90%, greater than about 95%, greater than about 96%, greater than about 97%, greater than about 98%, greater than about 99%, greater than about

99.1%, greater than about 99.2%, greater than about 99.3%, greater than about 99.4%, greater than about 99.5%, greater than about 99.6%, greater than about 99.7%, greater than about 99.8%, greater than about 99.85%, greater than about 99.9%, greater than about 99.95%, or greater than about 99.98% by weight of the total weight of the crystalline form of the compound of formula (V).

Recrystallization of the Crystalline Form of the Compound of Formula (V)

In various embodiments, in step (e), purifying the compound of formula (V) comprises one or more recrystallizations of the crystalline form of the compound of formula (V) to provide a pharmaceutical material comprising a purified amount of the compound of formula (V).

In certain embodiments, the one or more recrystallizations of the crystalline form of the compound of formula (V) comprises:

- (1) dissolving the crystalline compound of formula (V) in one or more solvents, thereby forming a mixture;
- (2) cooling the mixture from a first temperature ( $T^1$ ) to a second temperature ( $T^2$ ), wherein  $T^1$  is from about 40° C. to about 65° C.,  $T^2$  is from about 20° C. to about 40° C., and the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 3° C./hour to about 11° C./hour;
- (3) holding the mixture at  $T^2$  for at least 0.5 hours;
- (4) heating the mixture from  $T^2$  to a third temperature ( $T^3$ ), wherein  $T^3$  is from about 30° C. to about 50° C., and the mixture is heated from  $T^2$  to  $T^3$  at a rate of from about 3° C./hour to about 11° C./hour;
- (5) holding the mixture at  $T^3$  for at least 0.5 hours;
- (6) cooling the mixture from  $T^3$  to a fourth temperature ( $T^4$ ), wherein  $T^4$  is from about 25° C. to about 40° C. and the mixture is cooled from  $T^3$  to  $T^4$  at a rate of from about 3° C./hour to about 11° C./hour;
- (7) holding the mixture at  $T^4$  for at least 0.5 hours;
- (8) cooling the mixture from  $T^4$  to a fifth temperature ( $T^5$ ), wherein  $T^5$  is from about -10° C. to about 10° C. and the mixture is cooled from  $T^4$  to  $T^5$  at a rate of from about 3° C./hour to about 11° C./hour; and
- (9) holding the mixture at  $T^5$  for at least 0.5 hours, thereby producing a pharmaceutical material comprising a purified amount of crystalline form of the compound of formula (V).

In certain embodiments, in recrystallization step (1), the one or more solvents comprises ethyl acetate and water. In certain embodiments, the one or more solvents are ethyl acetate and water.

In certain embodiments, in recrystallization step (1), the amount of ethyl acetate in the mixture is from about 2.5 kg/kg to about 3.3 kg/kg, from about 2.7 kg/kg to about 3.3 kg/kg, from about 2.9 kg/kg to about 3.3 kg/kg, from about 3.1 kg/kg to about 3.3 kg/kg, from about 2.5 kg/kg to about 3.1 kg/kg, from about 2.5 kg/kg to about 2.9 kg/kg, from about 2.7 kg/kg to about 3.1 kg/kg, from about 2.7 kg/kg to about 2.9 kg/kg, or from about 2.9 kg/kg to about 3.1 kg/kg, based on the weight of ethyl acetate per kilogram of the crystalline compound of formula (V). In some embodiments, in recrystallization step (1), the amount of ethyl acetate in the mixture is from about 2.5 kg/kg to about 3.3 kg/kg, based on the weight of ethyl acetate per kilogram of the crystalline compound of formula (V). In some embodiments, in recrystallization step (1), the amount of ethyl acetate in the mixture is from about 2.7 kg/kg to about 3.1 kg/kg, based on the weight of ethyl acetate per kilogram of the crystalline compound of formula (V).

In certain embodiments, in recrystallization step (1), the amount of ethyl acetate in the mixture is about 2.5 kg/kg, about 2.7 kg/kg, about 2.9 kg/kg, about 3.1 kg/kg, or about 3.3 kg/kg, based on the weight of ethyl acetate per kilogram of the crystalline compound of formula (V). In some embodiments, in recrystallization step (1), the amount of ethyl acetate in the mixture is about 2.9 kg/kg, based on the weight of ethyl acetate per kilogram of the crystalline compound of formula (V).

In certain embodiments, in recrystallization step (2),  $T^1$  is from about 40° C. to about 65° C., from about 45° C. to about 65° C., from about 50° C. to about 65° C., from about 55° C. to about 65° C., from about 60° C. to about 65° C., from about 40° C. to about 60° C., from about 40° C. to about 55° C., from about 40° C. to about 50° C., from about 40° C. to about 45° C., from about 45° C. to about 60° C., from about 45° C. to about 55° C., from about 45° C. to about 50° C., from about 50° C. to about 60° C., from about 50° C. to about 55° C., or from about 55° C. to about 60° C. In some embodiments, in recrystallization step (2),  $T^1$  is from about 40° C. to about 65° C. In some embodiments, in recrystallization step (2),  $T^1$  is from about 50° C. to about 60° C.

In certain embodiments, in recrystallization step (2),  $T^1$  is about 40° C., about 45° C., about 50° C., about 55° C., about 60° C., or about 65° C. In some embodiments, in recrystallization step (2),  $T^1$  is about 55° C.

In certain embodiments, in recrystallization step (2),  $T^2$  is from about 20° C. to about 40° C., from about 25° C. to about 40° C., from about 30° C. to about 40° C., from about 35° C. to about 40° C., from about 20° C. to about 35° C., from about 20° C. to about 30° C., from about 20° C. to about 25° C., from about 25° C. to about 35° C., from about 25° C. to about 30° C., or from about 30° C. to about 40° C. In certain embodiments, in recrystallization step (2),  $T^2$  is from about 25° C. to about 35° C.

In certain embodiments, in recrystallization step (2),  $T^1$  is about 20° C., about 25° C., about 30° C., about 35° C., or about 40° C. In some embodiments, in recrystallization step (2),  $T^1$  is about 30° C.

In certain embodiments, in recrystallization step (2), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 3° C./hour to about 11° C./hour, from about 5° C./hour to about 11° C./hour, from about 7° C./hour to about 11° C./hour, from about 9° C./hour to about 11° C./hour, from about 3° C./hour to about 9° C./hour, from about 3° C./hour to about 7° C./hour, from about 3° C./hour to about 5° C./hour, from about 5° C./hour to about 9° C./hour, from about 5° C./hour to about 7° C./hour, or from about 7° C./hour to about 9° C./hour. In some embodiments, in recrystallization step (2), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 3° C./hour to about 11° C./hour. In some embodiments, in recrystallization step (2), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 5° C./hour to about 7° C./hour.

In certain embodiments, in recrystallization step (2), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of about 3° C./hour, about 4° C./hour, about 5° C./hour, about 6° C./hour, about 7° C./hour, about 8° C./hour, about 9° C./hour, about 10° C./hour, or about 11° C./hour. In some embodiments, in recrystallization step (2), the mixture is cooled from  $T^1$  to  $T^2$  at a rate of about 5° C./hour, about 6° C./hour, or about 7° C./hour.

In certain embodiments, in recrystallization step (3), the mixture is held at  $T^2$  for at least 1 hour, 1.5 hours, 2 hours, 2.5 hours, 3 hours, 3.5 hours, 4 hours, 4.5 hours, 5 hours, 5.5 hours, 6 hours, 6.5 hours, at least 7 hours, at least 7.5 hours, at least 8 hours, at least 8.5 hours, at least 9 hours, at least



5° C. to about 10° C., from about -10° C. to about 5° C., from about -10° C. to about 0° C., from about -10° C. to about -5° C., from about -5° C. to about 5° C., from about -5° C. to about 0° C., or from about 0° C. to about 5° C. In some embodiments, in recrystallization step (8), T<sup>5</sup> is from about -10° C. to about 10° C. In some embodiments, in recrystallization step (8), T<sup>5</sup> is from about 0° C. to about 10° C.

In certain embodiments, in recrystallization step (8), T<sup>5</sup> is about -10° C., about -5° C., about 0° C., about 5° C., or about 10° C. In some embodiments, in recrystallization step (8), T<sup>5</sup> is about 5° C.

In certain embodiments, in recrystallization step (8), the mixture is cooled from T<sup>4</sup> to T<sup>5</sup> at a rate of from about 3° C./hour to about 11° C./hour, from about 5° C./hour to about 11° C./hour, from about 7° C./hour to about 11° C./hour, from about 9° C./hour to about 11° C./hour, from about 3° C./hour to about 9° C./hour, from about 3° C./hour to about 7° C./hour, from about 3° C./hour to about 5° C./hour, from about 5° C./hour to about 9° C./hour, from about 5° C./hour to about 7° C./hour, or from about 7° C./hour to about 9° C./hour. In some embodiments, in recrystallization step (8), the mixture is cooled from T<sup>4</sup> to T<sup>5</sup> at a rate of from about 3° C./hour to about 11° C./hour. In some embodiments, in recrystallization step (8), the mixture is cooled from T<sup>4</sup> to T<sup>5</sup> at a rate of from about 5° C./hour to about 7° C./hour.

In certain embodiments, in recrystallization step (8), the mixture is cooled from T<sup>4</sup> to T<sup>5</sup> at a rate of about 3° C./hour, about 4° C./hour, about 5° C./hour, about 6° C./hour, about 7° C./hour, about 8° C./hour, about 9° C./hour, about 10° C./hour, or about 11° C./hour. In some embodiments, in recrystallization step (8), the mixture is cooled from T<sup>4</sup> to T<sup>5</sup> at a rate of about 5° C./hour, about 6° C./hour, or about 7° C./hour.

In certain embodiments, in recrystallization step (9), the mixture is held at T<sup>5</sup> for at least 1 hour, 1.5 hours, 2 hours, 2.5 hours, 3 hours, 3.5 hours, 4 hours, 4.5 hours, 5 hours, 5.5 hours, 6 hours, 6.5 hours, at least 7 hours, at least 7.5 hours, at least 8 hours, at least 8.5 hours, at least 9 hours, at least 9.5 hours, or at least 10 hours. In some embodiments, in recrystallization step (9), the mixture is held at T<sup>5</sup> for at least 4 hours.

In certain embodiments, in recrystallization step (9), the mixture is held at T<sup>5</sup> for no greater than 1 hour, no greater than 1.5 hours, no greater than 2 hours, no greater than 2.5 hours, no greater than 3 hours, no greater than 3.5 hours, no greater than 4 hours, no greater than 4.5 hours, no greater than 5 hours, no greater than 5.5 hours, no greater than 6 hours, no greater than 6.5 hours, no greater than 7 hours, no greater than 7.5 hours, no greater than 8 hours, no greater than 8.5 hours, no greater than 9 hours, no greater than 9.5 hours, no greater than 10 hours, no greater than 10.5 hours, no greater than 11 hours, no greater than 11.5 hours, no greater than 12 hours, no greater than 16 hours, no greater than 20 hours, or no greater than 24 hours.

In certain embodiments, recrystallizing the compound of formula (V) further comprises seeding the mixture with an amount of a crystalline form of the compound of formula (V) prior to recrystallization step (2), during recrystallization step (2), during recrystallization step (3), during recrystallization step (4), during recrystallization step (5), during recrystallization step (6), during recrystallization step (7), during recrystallization step (8), during recrystallization step (9), or any combination thereof, in order to facilitate the crystallization of the compound of formula (V).

It should be understood that the conditions (e.g., temperatures), times, seeding, amounts, compounds, and other

parameter and/or variables for crystallizing the compound of formula (V) as described herein can be equally applicable to recrystallization of the compound of formula (V), unless otherwise stated or understood from the context, or as noted below.

For example, in certain embodiments, recrystallizing the compound of formula (V) comprises filtration of the crystalline form of the compound of formula (V). In certain embodiments, filtering the crystalline form of the compound of formula (V) occurs at a temperature of from about -20° C. to about 15° C., from about -15° C. to about 15° C., from about -10° C. to about 15° C., from about -5° C. to about 15° C., from about 0° C. to about 15° C., from about 5° C. to about 15° C., from about 10° C. to about 15° C., from about -20° C. to about 10° C., from about -20° C. to about 5° C., from about -20° C. to about 0° C., from about -20° C. to about -5° C., from about -20° C. to about -10° C., from about -20° C. to about -15° C., from about -15° C. to about 10° C., from about -15° C. to about 5° C., from about -15° C. to about 0° C., from about -15° C. to about -5° C., from about -15° C. to about -10° C., from about -10° C. to about 10° C., from about -10° C. to about 5° C., from about -10° C. to about 0° C., from about -10° C. to about -5° C., from about -5° C. to about 10° C., from about -5° C. to about 5° C., from about -5° C. to about 0° C., from about 0° C. to about 10° C., from about 0° C. to about 5° C., or from about 5° C. to about 10° C. In some embodiments, filtering of the crystalline form of the compound of formula (V) occurs at a temperature of from about 0° C. to about 15° C.

In certain embodiments, filtering of the crystalline form of the compound of formula (V) occurs at a temperature of about -20° C., about -15° C., about -10° C., about -5° C., about 0° C., about 5° C., about 10° C., or about 15° C. In some embodiments, filtering of the crystalline form of the compound of formula (V) occurs at a temperature of about 0° C., about 5° C., about 10° C., or about 15° C.

In certain embodiments, filtration further comprises washing. In certain embodiments, washing comprises washing the crystalline form of the compound of formula (V) with a solvent. In some embodiments, washing comprises washing the crystalline form of the compound of formula (V) with acetonitrile.

In certain embodiments, recrystallizing the compound of formula (V) comprises isolating the crystalline form of the compound of formula (V) from the mixture by centrifugation.

In certain embodiments, isolation of the crystalline form of the compound of formula (V) by centrifugation further comprises washing. In certain embodiments, washing comprises washing the crystalline form of the compound of formula (V) with a solvent. In some embodiments, washing comprises washing the crystalline form of the compound of formula (V) with acetonitrile.

In certain embodiments, the temperature of the solvent, for example, acetonitrile, is from about -20° C. to about 30° C., from about -10° C. to about 30° C., from about 0° C. to about 30° C., from about 10° C. to about 30° C., from about -20° C. to about 20° C., from about -20° C. to about 10° C., from about -20° C. to about 0° C., from about -20° C. to about -10° C., from about -10° C. to about 20° C., from about -10° C. to about 10° C., from about -10° C. to about 0° C., from about 0° C. to about 20° C., from about 0° C. to about 10° C., or from about 10° C. to about 20° C. In some embodiments, the temperature of the solvent is from about 10° C. to about 30° C.

In certain embodiments, the temperature of the solvent, for example, acetonitrile, is about  $-20^{\circ}\text{C.}$ , about  $-10^{\circ}\text{C.}$ , about  $0^{\circ}\text{C.}$ , about  $10^{\circ}\text{C.}$ , about  $20^{\circ}\text{C.}$ , or about  $30^{\circ}\text{C.}$  In some embodiments, the temperature of the solvent is about  $20^{\circ}\text{C.}$

In certain embodiments, recrystallizing the compound of formula (V) comprises drying.

In certain embodiments, drying comprises heating the crystalline form of the compound of formula (V) to a temperature of less than about 85° C., less than about 75° C., less than about 65° C., less than about 55° C., less than about 45° C., less than about 35° C., or less than about 25° C. In some embodiments, the drying step comprises heating the crystalline form of the compound of formula (V) to a temperature of less than about 85° C. In some embodiments, the drying step comprises heating the crystalline form of the compound of formula (V) to a temperature of less than about 45° C.

In certain embodiments, the one or more recrystallizations of the crystalline form of the compound of formula (V) comprises 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 recrystallizations.

In certain embodiments, the one or more recrystallizations of the crystalline form of the compound of formula (V) comprises:

- (1) dissolving the crystalline compound of formula (V) in one or more solvents comprising ethyl acetate and water, thereby forming a mixture;
- (2) cooling the mixture from a first temperature ( $T^1$ ) to a second temperature ( $T^2$ ), wherein  $T^1$  is about 55° C.,  $T^2$  is about 30° C., and the mixture is cooled from  $T^1$  to  $T^2$  at a rate of from about 5° C./hour to about 7° C./hour;
- (3) holding the compound of formula (V) at  $T^2$  for at least 2 hours;
- (4) heating the mixture from  $T^2$  to a third temperature ( $T^3$ ), wherein  $T^3$  is about 40° C. and the mixture is heated from  $T^2$  to  $T^3$  at a rate of from about 5° C./hour to about 7° C./hour;
- (5) holding the mixture at  $T^3$  for at least 1 hour;
- (6) cooling the mixture from  $T^3$  to a fourth temperature ( $T^4$ ), wherein  $T^4$  is about 35° C. and the mixture is cooled from  $T^3$  to  $T^4$  at a rate of from about 5° C./hour to about 7° C./hour;
- (7) holding the mixture at  $T^4$  for at least 2 hours;
- (8) cooling the mixture from  $T^4$  to a fifth temperature ( $T^5$ ), wherein  $T^5$  is about 5° C. and the mixture is cooled from  $T^4$  to  $T^5$  at a rate of from about 5° C./hour to about 7° C./hour; and
- (9) holding the mixture at  $T^5$  for at least 4 hours, thereby producing a pharmaceutical material comprising a purified amount of crystalline form of the compound of formula (V).

It should be understood that the conditions (e.g., temperatures), times, seeding, amounts, compounds, and other parameters and/or variables for crystallizing and/or recrystallizing the compound of formula (V) as described herein can be equally applicable to the immediately above-described recrystallizing process, unless otherwise stated or understood from the context (e.g., the conditions or parameters fall outside the values or ranges in the immediately above-described recrystallizing process).

In certain embodiments, the crystalline form of the compound of formula (V) in the pharmaceutical material produced by any of the recrystallization methods described herein may be a crystalline form of the compound of formula (V) as characterized herein, for example, by an X-ray powder diffraction pattern or peak(s), and/or other characteristic properties of the crystalline form of bempeidoic acid.

In various embodiments, the purified amount of the compound of formula (V) in the pharmaceutical material is greater than 99.0%, greater than about 99.1%, greater than about 99.2%, greater than about 99.3%, greater than about 99.4%, greater than about 99.5%, greater than about 99.6%, greater than about 99.7%, greater than about 99.8%, greater than about 99.85%, greater than about 99.9%, greater than about 99.95%, or greater than about 99.98% by weight of the total weight of the pharmaceutical material. In some embodiments, the purified amount of the compound of formula (V) in the pharmaceutical material is greater than 99.0% by weight of the total weight of the pharmaceutical material. In some embodiments, the purified amount of the compound of formula (V) in the pharmaceutical material is greater than about 99.5% by weight of the total weight of the pharmaceutical material. In some embodiments, the purified amount of the compound of formula (V) in the pharmaceutical material is greater than about 99.7% by weight of the total weight of the pharmaceutical material. In some embodiments, the purified amount of the compound of formula (V) in the pharmaceutical material is greater than about 99.85% by weight of the total weight of the pharmaceutical material.

In certain embodiments, subsequent to recrystallizing the crystalline form of the compound of formula (V), which provides a recrystallized compound of formula (V), the method comprises contacting the recrystallized compound of formula (V) with charcoal, and filtering the charcoal to provide a pharmaceutical material comprising a purified amount of the compound of formula (V). In some embodiments, contacting the compound of formula (V) with charcoal comprises contacting the compound of formula (V) with a solution, wherein the solution comprises acetonitrile and activated charcoal (e.g., 5% (w/w) activated charcoal).

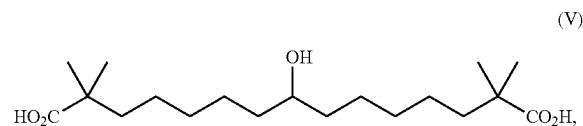
In certain embodiments, the methods described herein can be used to prepare a batch of bempedoic acid. In certain embodiments, the methods described herein can be used to prepare a batch of a pharmaceutical material, wherein the pharmaceutical material comprises a purified amount of the compound of formula (V). In certain embodiments, the purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, is greater than 99.0% by weight of the total weight of the pharmaceutical material.

In certain embodiments, the batch is in an amount of about 1 kg, 2 kg, 3 kg, 4 kg, 5 kg, 10 kg, 20 kg, 30 kg, 40 kg, 50 kg, 60 kg, 70 kg, 80 kg, 90 kg, 100 kg, 200 kg, 300 kg, 400 kg, 500 kg, 600 kg, 700 kg, 800 kg, 900 kg, or 1000 kg.

#### IV. High Purity Compositions of Bempedoic Acid

As described herein, in one aspect, the invention provides pharmaceutical materials comprising bempedoic acid such as a crystalline form of bempedoic acid, or a pharmaceutically acceptable salt thereof.

In various embodiments, a pharmaceutical material generally comprises a crystalline form of the compound of formula (V):



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or a pharmaceutically acceptable salt thereof, wherein the pharmaceutical material comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the compound of formula (V) in the pharmaceutical material is greater than about 99.1%, greater than about 99.2%, greater than about 99.3%, greater than about 99.4%, greater than about 99.5%, greater than about 99.6%, greater than about 99.7%, greater than about 99.8%, greater than about 99.85%, greater than about 99.9%, greater than about 99.95%, or greater than about 99.98% by weight of the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (V) in an amount greater than 99.5% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (V) in an amount greater than 99.7% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (V) in an amount greater than 99.9% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the pharmaceutical material comprises the compound of formula (V) in an amount of from about 98% to about 102% by weight based on the total weight of the pharmaceutical material (anhydrous, solvent-free basis), as determined by a high performance liquid chromatography (HPLC) assay.

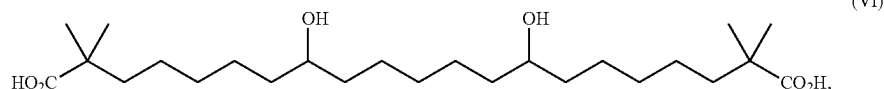
In certain embodiments, the HPLC assay comprises one or more of:

- (i) a Waters XBridge BEH C18 column (4.6 mm i.d.×150 mm, 2.5  $\mu$ m);
- (ii) a column temperature of about 40° C.;
- (iii) a mobile phase comprising about 0.05% phosphoric acid in water/acetonitrile (about 50:50);
- (iv) isocratic elution;
- (v) a flow rate of about 1.2 mL/minute;
- (vi) a sample temperature of ambient temperature;
- (vii) detection at 215 nm; and
- (viii) the retention time of the compound of formula (V) is about 4.6 minutes.

In some embodiments, the HPLC assay comprises each of the above, i.e., (i)-(viii).

In certain embodiments, the crystalline form of the compound of formula (V) may be a crystalline form of the compound of formula (V) as characterized herein, for example, by an X-ray powder diffraction pattern or peak(s), and/or other characteristic properties of the crystalline form of bempedoic acid.

In certain embodiments, a pharmaceutical material described herein can comprise a compound of formula (VI):



or a pharmaceutically acceptable salt thereof. The compound of formula (VI), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “diol impurity.”

In certain embodiments, the amount of the diol impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about

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0.0001% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the diol impurity in the pharmaceutical material is less than about 1.25% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the diol impurity in the pharmaceutical material is less than about 0.15% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the diol impurity in the pharmaceutical material is less than about 0.1% by weight based on the total weight of the pharmaceutical material.

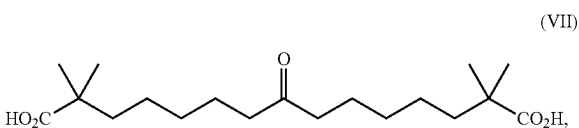
In certain embodiments, the pharmaceutical material comprises the compound of formula (VI), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (VI), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.125% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (VI), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (VI), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.1% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the diol impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.001% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.1% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.25%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the diol impurity in the pharmaceutical material is from about 0.01% to about 0.15% by weight based on the total weight of the pharmaceutical

material. In some embodiments, the amount of the diol impurity in the pharmaceutical material is from about 0.01% to about 0.1% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise the compound of formula (VII):

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or a pharmaceutically acceptable salt thereof. The compound of formula (VII), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “ketone impurity.”

In certain embodiments, the amount of the ketone impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material. In certain embodiments, the amount of the ketone impurity in the pharmaceutical material is less than about 0.15% by weight based on the total weight of the pharmaceutical material. In certain embodiments, the amount of the ketone impurity in the pharmaceutical material is less than about 0.05% by weight based on the total weight of the pharmaceutical material.

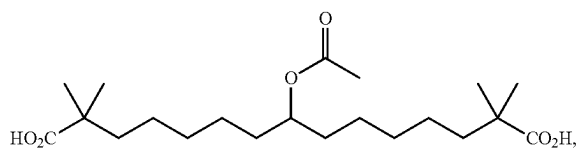
In certain embodiments, the pharmaceutical material comprises the compound of formula (VII), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (VII), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15% by weight based on the total weight of the pharmaceutical material. In some embodiments, the pharmaceutical material comprises the compound of formula (VII), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.05% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the ketone impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.001% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.1% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.025%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the ketone impurity in the pharmaceutical material is from about 0.01% to about 0.15% by weight based on the total weight of the pharmaceutical

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material. In some embodiments, the amount of the ketone impurity in the pharmaceutical material is from about 0.01% to about 0.05% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise the compound of formula (VIII):



or a pharmaceutically acceptable salt thereof. The compound of formula (VIII), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “acetate impurity.”

In certain embodiments, the amount of the acetate impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001% or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the pharmaceutical material comprises the compound of formula (VIII), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material.

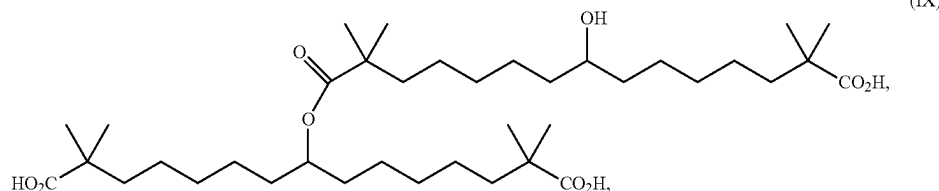
In certain embodiments, the amount of the acetate impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.001% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.1% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.025%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise the compound of formula (IX):



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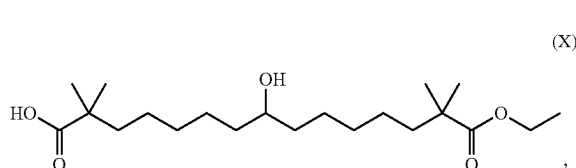
or a pharmaceutically acceptable salt thereof. The compound of formula (IX), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “dimer impurity.”

In certain embodiments, the amount of the dimer impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.010%, about 0.0010% or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the pharmaceutical material comprises the compound of formula (IX), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the dimer impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.0005% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.1% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.025%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise the compound of formula (X):



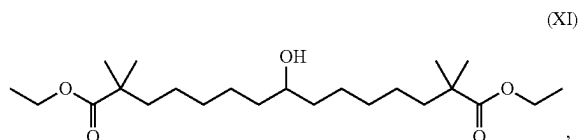
or a pharmaceutically acceptable salt thereof. The compound of formula (X), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “monoethyl ester impurity.”

In certain embodiments, the amount of the monoethyl ester impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001% or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the pharmaceutical material comprises the compound of formula (X), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the monoethyl ester impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.0005% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.1% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.025%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise the compound of formula (XI):



or a pharmaceutically acceptable salt thereof. The compound of formula (XI), or a pharmaceutically acceptable salt thereof, is also referred to herein as the “diethyl ester impurity.”

In certain embodiments, the amount of the diethyl ester impurity in the pharmaceutical material is less than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001% or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the pharmaceutical material comprises the compound of formula (XI), or a pharmaceutically acceptable salt thereof, in an amount no greater than about 0.15%, about 0.125%, about 0.1%, about 0.075%, about 0.05%, about 0.025%, about 0.01%, about 0.001%, or about 0.0001% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the diethyl ester impurity in the pharmaceutical material is from about 0.0001% to about 0.15%, from about 0.0005% to about 0.15%, from about 0.01% to about 0.15%, from about 0.025% to about 0.15%, from about 0.05% to about 0.15%, from about 0.075% to about 0.15%, from about 0.10% to about 0.15%, from about 0.125% to about 0.15%, from about 0.01% to about 0.125%, from about 0.01% to about 0.1%, from about 0.01% to about 0.075%, from about 0.01% to about 0.05%, from about 0.01% to about 0.025%, from about 0.025% to about 0.125%, from about 0.025% to about 0.1%, from about 0.025% to about 0.075%, from about 0.025% to about 0.05%, from about 0.05% to about 0.125%, from about 0.05% to about 0.1%, from about 0.05% to about 0.075%, from about 0.075% to about 0.125%, from about 0.075% to about 0.1%, or from about 0.1% to about 0.125% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a relative retention time (RRT) of about 1.04 to about 1.05, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a relative retention time (RRT) of about 1.06 to about 1.08, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a relative retention time (RRT) of about 1.18 to about 1.20, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a relative retention time (RRT) of about 1.36, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a RRT of about 1.43, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise an impurity with a RRT of about 1.86, as determined by HPLC. In certain embodiments, a pharmaceutical material described herein can comprise a first impurity with a RRT of about 1.36 and a second impurity with a RRT of about 1.86, as determined by HPLC. In certain embodiments, the RRT of the impurity is based on the retention time of bempedoic acid, wherein the RRT of bempedoic acid is about 1.00.

In certain embodiments, a pharmaceutical material described herein can comprise one or more unidentified impurities (e.g., impurities in the pharmaceutical material whose chemical structure cannot be determined but whose RRT is known).

In certain embodiments, the pharmaceutical material comprises one or more of the impurities described herein, as determined by a high performance liquid chromatography (HPLC) assay.

In certain embodiments, the HPLC assay comprises one or more of:

- (i) a Waters XBridge BEH C18 column (4.6 mm i.d.×150 mm, 2.5  $\mu$ m);
- (ii) a column temperature of about 40° C.;
- (iii) a first mobile phase comprising about 0.05% formic acid in water;
- (iv) a second mobile phase comprising about 0.05% formic acid in acetonitrile;
- (v) a flow rate of about 1.2 mL/minute;
- (vi) a sample temperature of ambient temperature; and
- (vii) the retention time of the compound of formula (V) is about 15.2 minutes.

In some embodiments, the HPLC assay comprises each of the above, i.e., (i)-(vii).

In certain embodiments, the amount of the one or more unidentified impurities in the pharmaceutical material is less than about 0.010%, about 0.02%, about 0.03%, about 0.04%, about 0.05%, about 0.06%, about 0.07%, about 0.08%, about 0.09%, or about 0.1% by weight based on the total weight of the pharmaceutical material. In some embodiments, the amount of the one or more unidentified impurities in the pharmaceutical material is less than about 0.05% by weight based on the total weight of the pharmaceutical material.

In certain embodiments, the amount of the one or more unidentified impurities in the pharmaceutical material is from about 0.0001% to about 0.1%, from about 0.0005% to about 0.1%, from about 0.001% to about 0.1%, from about 0.005% to about 0.1%, from about 0.01% to about 0.1%, from about 0.05% to about 0.1%, from about 0.0001% to about 0.05%, from about 0.0001% to about 0.01%, from about 0.0001% to about 0.005%, from about 0.0001% to about 0.001%, from about 0.0001% to about 0.0005%, from about 0.0005% to about 0.05%, from about 0.0005% to about 0.010%, from about 0.0005% to about 0.005%, from about 0.0005% to about 0.001%, from about 0.001% to about 0.05%, from about 0.001% to about 0.01%, from about 0.001% to about 0.005%, from about 0.005% to about 0.05%, from about 0.005% to about 0.01%, or from about 0.01% to about 0.05% by weight based on the total weight of the pharmaceutical material. In certain embodiments, the amount of the one or more unidentified impurities in the pharmaceutical material is from about 0.0001% to about 0.1% by weight based on the total weight of the pharmaceutical material. In certain embodiments, the amount of the one or more unidentified impurities in the pharmaceutical material is from about 0.05% to about 0.1% by weight based on the total weight of the pharmaceutical material.

## V. Pharmaceutical Compositions

In another aspect, provided herein are pharmaceutical compositions comprising bempedoic acid, or a pharmaceutically acceptable salt thereof, as described and/or made herein, including any of the pharmaceutical materials as well as the impurities. In various embodiments, a pharmaceutical composition generally comprises a pharmaceutical material as described herein; and a pharmaceutically acceptable excipient. For example, the pharmaceutical material can comprise greater than 99.0% of the compound of formula (V), or a pharmaceutically acceptable salt thereof.

In certain embodiments, the pharmaceutical compositions may be specially formulated for administration as solid or liquid dosage forms. In some embodiments, the pharmaceutical compositions described herein are formulated for administration as an oral dosage form. Examples of oral dosage forms include, but are not limited to a drench, a tablet, a capsule, a cachet, a pill, an emulsion, a lozenge, a pastille, a mouthwash, a granule, or a paste for application to the tongue. In some embodiments, the pharmaceutical compositions described herein are formulated as a dosage form suitable for parenteral administration. In some embodiments, the pharmaceutical compositions described herein are administered by subcutaneous, intramuscular, intravenous or epidural injection. Examples of dosage forms suitable for parenteral administration include, but are not limited to, a sterile solution or suspension, or a sustained-release formulation. In some embodiments, the pharmaceutical compositions described herein are formulated as a dosage form suitable for topical application.

Examples of dosage forms suitable for topical administration include, but are not limited to, a powder, a spray, an ointment, a paste, a cream, a lotion, a gel, a solution, a patch, an inhalant, or a controlled-release patch or spray applied to the skin. In some embodiments, the pharmaceutical compositions described herein are formulated as a dosage form suitable for intravaginal or intrarectal administration. Examples of dosage forms suitable for intravaginal or intrarectal administration include, but are not limited to, a pessary, a cream, or a foam. In some embodiments, the pharmaceutical compositions described herein are formulated as a dosage form suitable for sublingual, ocular, transdermal or nasal administration.

In certain embodiments, the solid dosage forms described herein to be used for oral administration are prepared by mixing a pharmaceutical material with one or more pharmaceutically acceptable excipients. Pharmaceutical excipients can be selected from the group consisting of a filler or extender, a sweetening agent, a binder, a humectant, a disintegrating agent, a preservative, a perfuming agent, a flavoring agent, an antioxidant, a solution retarding agent, an absorption accelerator, a wetting agent, an absorbent, a lubricant, a coloring agent, and a controlled release agent. In some embodiments, when the solid dosage form is a capsule, a tablet or a pill, the pharmaceutical compositions described herein may also comprise a buffering agent. In some embodiments, when the solid dosage form is a gelatin capsule, the pharmaceutical composition may further comprise one or more excipients selected from lactose, a milk sugar, a high molecular weight polyethylene glycol and combinations thereof.

In certain embodiments, a pharmaceutical composition of the invention may comprise one or more excipients selected from the group consisting of a cyclodextrin, a cellulose, a liposome, a micelle forming agent, and a polymeric carrier. In some embodiments, the pharmaceutical compositions of the present invention comprise an antibacterial agent, an antifungal agent, or combinations thereof. Examples antibacterial and antifungal agents include, but are not limited to, paraben, chlorobutanol, phenol and sorbic acid. In some embodiments, the pharmaceutical compositions of the present invention comprise an isotonic agent.

In certain embodiments, the dosage forms of the present invention may be formulated so as to provide slow or controlled release of the compound of formula (V), or a pharmaceutically acceptable salt thereof. See, e.g., PCT/US2019/018356, which discloses sustained release formulations of bempedoic acid. In some embodiments, the dosage forms of the present invention may be formulated for rapid release.

In certain embodiments, a liquid dosage form of a pharmaceutical composition of the invention comprises one or more of the following; an inert diluent, a solubilizing agent and an emulsifier.

In certain embodiments, oral suspensions of a pharmaceutical composition of the invention comprise one or more suspending agents including an ethoxylated isostearyl alcohol, a polyoxyethylene sorbitol and sorbitan ester, a microcrystalline cellulose, an aluminum metahydroxide, bentonite, agar-agar and tragacanth, and mixtures thereof.

In some embodiments, the ointments, pastes, creams and gels of a pharmaceutical composition of the invention comprise one or more excipients, wherein the one or more excipients can comprise an animal fat, a vegetable fat, an oil, a wax, a paraffin, a starch, a tragacanth, a cellulose derivative, a polyethylene glycol, a silicone, a bentonite, silicic acid, talc, zinc oxide, or mixtures thereof.

In certain embodiments, powders and sprays of a pharmaceutical composition of the invention comprise one or more excipients, wherein the one or more excipients can comprise lactose, talc, silicic acid, aluminum hydroxide, a calcium silicate, a polyamide powder, or mixtures thereof. In some embodiments, a spray of the present invention can comprise a customary propellant, wherein the customary propellant comprises one or more of a chlorofluorohydrocarbon and a volatile unsubstituted hydrocarbon.

In certain embodiments, a transdermal patch of a pharmaceutical composition of the invention provides controlled delivery of the compound of formula (V), or a pharmaceutically acceptable salt thereof, to the body. In some embodiments, ophthalmic formulations, eye ointments, powders, solutions and the like, are also included within the scope of the present invention.

In certain embodiments, the pharmaceutical compositions described herein may be administered in a unit dosage form and may be prepared by any method well known in the art of pharmacy. The amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, present in a single dosage form may vary depending upon the patient being treated and/or the particular mode of administration.

In certain embodiments, the amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, that can be combined with a pharmaceutically acceptable carrier to produce a single dosage form will generally be an amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, that produces a therapeutic effect.

In various embodiments, bempedoic acid, or a pharmaceutically acceptable salt thereof, or a pharmaceutical material of the present invention, can be prepared as a fixed dose formulation (see, e.g., U.S. Patent Application Publication No. 2018/0338922 and International Application No. WO 2018/218147).

## VI. Methods of Treatment and Administration

In various embodiments, bempedoic acid, or a pharmaceutically acceptable salt thereof, as described and/or made herein, including a pharmaceutical material and/or a pharmaceutical composition, may be used for the treatment or prevention of a variety of diseases and disorders. The methods of treating a disease or disorder generally comprise administering to a patient, in need thereof, a therapeutically effective amount of a pharmaceutical material comprising a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, to treat the disease or disorder.

Examples of diseases and disorders include, but are not limited to, cardiovascular disease, atrial fibrillation, blood clotting, coronary heart disease, hypercoagulable states, ischemia, myocardial infarction, myopathy, myositis, pulmonary embolism, stroke, peripheral vascular disease, dyslipidemia, dyslipoproteinemia, a disorder of glucose metabolism, Alzheimer's disease, Parkinson's disease, diabetic nephropathy, diabetic retinopathy, insulin resistance, metabolic syndrome disorders (e.g., Syndrome X), galactosemia, HIV infection, a peroxisome proliferator activated receptor-associated disorder, septicemia, a thrombotic disorder, obesity, pancreatitis, hypertension, renal disease, cancer, inflammation (e.g., liver inflammation), inflammatory muscle diseases (e.g., polymyalgia rheumatica, polymyositis, and fibrositis), impotence, gastrointestinal disease, irritable bowel syndrome, inflammatory bowel disease, inflammatory disorders (e.g., asthma, vasculitis, ulcerative colitis,

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Crohn's disease, Kawasaki disease, Wegener's granulomatosis, (RA), systemic lupus erythematosus (SLE), multiple sclerosis (MS), and autoimmune chronic hepatitis), arthritis (e.g., rheumatoid arthritis, juvenile rheumatoid arthritis, and osteoarthritis), osteoporosis, soft tissue rheumatism (e.g., tendonitis), bursitis, autoimmune disease (e.g., systemic lupus and erythematosus), scleroderma, ankylosing spondylitis, gout, pseudogout, non-insulin dependent diabetes mellitus, diabetes (e.g., type 2), polycystic ovarian disease, hyperlipidemias (e.g., primary hyperlipidemia, familial hypercholesterolemia (FH), Hypercholesterolemia Frederickson Type IIa, Hypercholesterolemia Frederickson Type Jib, familial combined hyperlipidemia (FCH)), lipoprotein lipase deficiencies (e.g., hypertriglyceridemia, hypoalphalipoproteinemia, and hypercholesterolemia), lipoprotein abnormalities associated with diabetes, lipoprotein abnormalities associated with obesity, and lipoprotein abnormalities associated with Alzheimer's disease. In particular embodiments, the methods include treating and/or preventing hyperlipidemia such as primary hyperlipidemia. In some embodiments, the methods include treating and/or preventing cardiovascular disease.

In certain embodiments, bempedoic acid, or a pharmaceutically acceptable salt thereof, as described and/or made herein, including a pharmaceutical material and/or a pharmaceutical composition, may be used for the treatment or prevention of one or more of high levels of low density lipoprotein cholesterol (LDL-C), high levels of apolipoprotein B (apoB), high levels of lipoprotein(a) (Lp(a)), high levels of very low density lipoprotein (VLDL), high levels of non-high density lipid cholesterol (non-HDL-C), high levels of total serum cholesterol (TC), high levels of high sensitivity c-reactive protein (hsCRP), high levels of fibrinogen, high levels of insulin, high levels of glucose, and low levels of high density lipoprotein cholesterol (HDL-C). In other words, methods of the invention can include lowering LDL-C, lowering apoB, lowering Lp(a), lowering VLDL, lowering non-HDL-C, lowering TC, and/or lowering hsCRP. Methods of the invention can include inhibiting adenosine triphosphate citrate lyase (ACL), inhibiting cholesterol synthesis, and/or suppressing fatty acid biosynthesis. In some embodiments, a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, a pharmaceutical composition or a pharmaceutical material of the present invention may be used as an adjunct to diet and maximally tolerated statin therapy to lower LDL-C in adults with heterozygous familial hypercholesterolemia or established atherosclerotic cardiovascular disease. In some embodiments, a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, a pharmaceutical composition or a pharmaceutical material of the present invention may be used for the treatment of non-insulin dependent diabetes mellitus without increasing weight gain.

In certain embodiments, bempedoic acid, or a pharmaceutically acceptable salt thereof, as described and/or made herein, including a pharmaceutical material and/or a pharmaceutical composition, may be used for the treatment or prevention of a variety of diseases and conditions, which include, but are not limited to aging, Alzheimer's disease, cancer, cardiovascular disease, diabetic nephropathy, diabetic retinopathy, a disorder of glucose metabolism, dyslipidemia, dyslipoproteinemia, enhancing bile production, hypertension, impotence, inflammation, insulin resistance, lipid elimination in bile, modulating C reactive protein, obesity, oxysterol elimination in bile, pancreatitis, pancreaticitis, Parkinson's disease, a peroxisome proliferator acti-

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vated receptor-associated disorder, phospholipid elimination in bile, renal disease, rhabdomyolysis, septicemia, sleep apnea, Syndrome X, and a thrombotic disorder.

In certain embodiments, provided herein is a method of treating a liver disorder selected from the group consisting of steatohepatitis, alcoholic liver disease, fatty liver, liver steatosis, liver cirrhosis, liver fibrosis, and acute fatty liver of pregnancy. In some embodiments, the disorder is steatohepatitis. In some embodiments, the steatohepatitis is non-alcoholic steatohepatitis. In some embodiments, the steatohepatitis is nonalcoholic fatty liver disease. In some embodiments, the disorder is alcoholic liver disease. In some embodiments, the disorder is fatty liver. In some embodiments, the disorder is liver steatosis, liver cirrhosis, or liver fibrosis. In some embodiments, the disorder is acute fatty liver of pregnancy. In some embodiments, the patient is an adult human.

In certain embodiments, the present invention provides a method for treating or preventing aging, Alzheimer's disease, cancer, cardiovascular disease, diabetic nephropathy, diabetic retinopathy, a disorder of glucose metabolism, dyslipidemia, dyslipoproteinemia, enhancing bile production, enhancing reverse lipid transport, hypertension, impotence, inflammation, insulin resistance, lipid elimination in bile, modulating C reactive protein, obesity, oxysterol elimination in bile, pancreatitis, pancreaticitis, Parkinson's disease, a peroxisome proliferator activated receptor-associated disorder, phospholipid elimination in bile, renal disease, septicemia, metabolic syndrome disorders (e.g., Syndrome X), or a thrombotic disorder.

In certain embodiments, the disorder is selected from the group consisting of lipodystrophy, lysosomal acid lipase deficiency, and a glycogen storage disease. In some embodiments, the patient is an adult human.

In certain embodiments, the disorder is selected from the group consisting of hepatitis C, an infection by human immunodeficiency virus, an alpha 1-antitrypsin deficiency, Bassen-Kornzweig syndrome, hypobetalipoproteinemia, Celiac disease, Wilson's disease, and Weber-Christian syndrome. In some embodiments, the disorder is hepatitis B. In some embodiments, the disorder is hepatitis C. In some embodiments, the disorder is an infection by human immunodeficiency virus. In some embodiments, the disorder is an alpha 1-antitrypsin deficiency. In some embodiments, the disorder is Bassen-Kornzweig syndrome. In some embodiments, the disorder is hypobetalipoproteinemia. In some embodiments, the disorder is Celiac disease or Wilson's disease. In some embodiments, the disorder is Weber-Christian syndrome. In some embodiments, the patient is an adult human.

In certain embodiments, the condition is selected from the group consisting of toxic liver injury, total parenteral nutrition, severe surgical weight loss, environmental toxicity, malnutrition, and starvation. In some embodiments, the condition is toxic liver injury. In some embodiments, the condition is total parenteral nutrition or severe surgical weight loss. In some embodiments, the condition is environmental toxicity. In some embodiments, the condition is malnutrition or starvation. In some embodiments, the patient is an adult human.

In certain embodiments, in order to prolong the effect of a drug, the compound of formula (V), or a pharmaceutically acceptable salt thereof, is administered by subcutaneous or intramuscular injection, or by dissolving or suspending the drug in an oil vehicle.

In certain embodiments, the actual dosage level of the compound of formula (V), or a pharmaceutically acceptable

salt thereof, in the pharmaceutical compositions of the present invention may be varied so as to obtain an amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, which is effective to achieve the desired therapeutic response for a particular patient, composition, and mode of administration, without being toxic to the patient.

In certain embodiments, the selected dosage level is dependent upon a variety of factors including the route of administration, the time of administration, the rate of excretion or metabolism of the particular compound being employed, the rate and extent of absorption, the duration of the treatment, other drugs, compounds and/or materials used in combination with the particular compound employed, the age, sex, weight, condition, general health and prior medical history of the patient being treated, and like factors well known in the medical arts. In certain embodiments, a physician or veterinarian having ordinary skill in the art can readily determine and prescribe the effective amount of the pharmaceutical composition as required.

In certain embodiments, a suitable daily dose of the compound of formula (V) or a pharmaceutically acceptable salt thereof, will be an amount that corresponds to the lowest dose effective to produce a therapeutic effect. In certain embodiments, the compound of formula (V), or a pharmaceutically acceptable salt thereof, is administered at about 0.01 mg/kg to about 200 mg/kg. In certain embodiments, when compound of formula (V), or a pharmaceutically acceptable salt thereof, is co-administered with another therapeutic agent, the effective amount may be less than when the compound of formula (V), or a pharmaceutically acceptable salt thereof, is used in isolation.

In certain embodiments, the effective daily dose of the compound of formula (V), or a pharmaceutically acceptable salt thereof, may be administered as two, three, four, five, six or more sub-doses. In certain embodiments, the two, three, four, five, six or more sub-doses are administered separately at appropriate intervals throughout the day, optionally, in unit dosage forms. In some embodiments, dosing is one administration per day. In some embodiments, the compound of formula (V), or a pharmaceutically acceptable salt thereof, is administered to a patient for 1 day, 5 days, 10 days, 20 days, 30 days, 1 week, 2 weeks, 3 weeks, 4 weeks, 1 month, 2 months, 3 months, 4 months, 5 months, 6 months, 7 months, 8 months, 9 months, 10 months, 11 months, 12 months, 1 year, 2 years, 3 years, 4 years, or 5 years. In some embodiments, the compound of formula (V), or a pharmaceutically acceptable salt thereof, is administered to a patient for the duration of the patient's life span.

### VII. Combination Therapy

In various embodiments, bempedoic acid, or a pharmaceutically acceptable salt thereof, as described and/or made herein, including pharmaceutical materials and pharmaceutical compositions of the present invention, can be part of a combination therapy. In certain embodiments, the combination therapy comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof; and a second therapeutic agent. In certain embodiments, the combination therapy comprises a pharmaceutical material comprising a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof; and a second therapeutic agent.

In some embodiments, the second therapeutic agent is selected from the group comprising a lovastatin, a thiazolidinedione or fibrate, a bile-acid-binding-resin, a niacin, an

anti-obesity drug, a hormone, an antiviral agent (e.g., to treat an underlying hepatitis C infection causing liver disease in the patient), anticancer agents (e.g., to treat hepatocellular carcinoma or other cancer causing liver disease or fatty liver), antioxidants, medications that decrease insulin resistance, or medications that improve lipid metabolism (e.g., treatments for hyperlipidemia), a tyroprophostine, a sulfonylurea-based drug, a biguanide, an  $\alpha$ -glucosidase inhibitor, an apolipoprotein A-I agonist, apolipoprotein E, a cardiovascular drug, an HDL-raising drug, an HDL enhancer, or a regulator of the apolipoprotein A-I, apolipoprotein A-IV and/or apolipoprotein genes. In some embodiments, the purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, is greater than 99.0% by weight of the total weight of the pharmaceutical material.

In various embodiments, the second therapeutic agent can be a statin and/or ezetimibe. See, e.g., U.S. Patent Application Publication Nos. 2018/0078518 (combination of bempedoic acid with a statin), 2018/0064671 and 2018/0338922 (combination of bempedoic acid with ezetimibe); International Publication No. WO 2018/218147 (combination of bempedoic acid with ezetimibe); and International Publication No. WO 2018/148417 (combination of bempedoic acid with ezetimibe and a statin).

In certain embodiments, administering a pharmaceutical material or a pharmaceutical composition of the present invention comprising the compound of formula (V), or a pharmaceutically acceptable salt thereof, and a second therapeutic agent is intended to provide a beneficial effect from the co-action of the compound of formula (V), or a pharmaceutically acceptable salt thereof, and a second therapeutic agent. In some embodiments, the beneficial effect of the combination therapy may include pharmacokinetic or pharmacodynamic co-action resulting from the combination of the compound of formula (V), or a pharmaceutically acceptable salt thereof, and a second therapeutic agent.

### VIII. Kits

In various embodiments, the invention provides kits for treating a condition, disease or disorder described herein. In some embodiments, a kit comprises: i) instructions for treating a condition, disease or disorder, for example, as described herein, and ii) the compound of formula (V), or a pharmaceutically acceptable salt thereof (e.g., a pharmaceutical material comprising a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof). In some embodiments, the kit may comprise one or more unit dosage forms containing an amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof, that is effective for treating the condition, disease or disorder.

The description herein includes multiple aspects and embodiments of the present invention, including methods of making the compound of formula (V), or a pharmaceutically acceptable salt thereof; methods of using a compound of formula (V), or pharmaceutically acceptable salt thereof, for example, a purified amount of the compound of formula (V), or a pharmaceutically acceptable salt thereof; compositions comprising a purified amount of the compound of formula (V), or pharmaceutically acceptable salt thereof; and kits. The patent application specifically includes all combinations and permutations of the aspects and embodiments as described herein. In particular, it should be understood that the pharmaceutical materials, pharmaceutical compositions, methods of treating a disorder or a condition, and kits can

include and/or use bempedoic acid, or a pharmaceutically acceptable salt thereof, as made by the methods described herein.

#### EXAMPLES

In order that the invention described herein may be more fully understood, the following example is set forth. The synthetic and analytical protocols described in this application are offered to illustrate the compounds, pharmaceutical compositions, and methods provided herein and are not to be construed in any way as limiting their scope.

##### Example 1: Manufacturing Process for Preparing a Pharmaceutical Material Comprising a Purified Amount of the Compound of Formula (V)

In this example, the synthesis of purified bempedoic acid refers to FIG. 1.

##### Step 1—Preparation of Compound of Formula (I)

###### Lithium Diisopropylamide (LDA) Preparation

A reaction vessel was charged with diisopropylamine (317±3 kg, 1.1 eq.) and tetrahydrofuran (THF, 2,102±105 L) and the mixture then cooled to ≤-10° C. n-Butyllithium (n-BuLi, 757±8 kg, 1.2 eq.) was then dosed over ≥1 hour while the temperature was maintained at ≤-10° C. The charge line was rinsed with THF. The addition was highly exothermic. Finally, the batch was then cooled back to ≤-10° C. while being stirred.

###### Alkylation Reaction

Ethyl isobutyrate (317±3 kg, 1.1 eq.) was added to the reactor over ≥1 hour at ≤-10° C. (FIG. 1). The batch was stirred while maintaining the temperature at ≤-10° C. 1-Bromo-5-chloropentane (460±5 kg, 1.00 eq.) was dosed over ≥1 hour at ≤-10° C. The line was rinsed with THF. The addition was highly exothermic. The reaction mixture was then stirred for ≥10 hours at ≤-10° C. This stage of the reaction was confirmed to be complete using gas chromatography (1-bromo-5-chloropentane: ≤3% area). The reaction mixture was then warmed to 0±5° C. and this temperature maintained until the conversion was complete. The reaction was confirmed to be complete using gas chromatography (1-bromo-5-chloropentane: ≤0.5% area).

###### Quench and Phase Separation

A solution of 9% hydrochloric acid (HCl, 1337±50 kg) was added to the reaction mixture over ≥1 hour while the temperature was maintained at ≤30° C. to quench the reaction. After dosing, the reaction mixture was stirred at ≤30° C. for ≥15 minutes. The pH of the aqueous layer was measured (range: pH 6 to 10). The agitator was stopped and the layers were allowed to settle for ≤30 minutes. The lower aqueous phase was removed for disposal.

###### Distillation and Removal of THF

The solvent was removed by distillation under vacuum at ≤40° C. to the desired volume of approximately 950 L.

The concentration of the compound of formula (I) was measured using gas chromatography (GC) (target compound of formula (I) concentration range: 57-62% wt). If necessary, to adjust the concentration of the compound of formula (I) to be within the target range, THE was added or further removed via distillation.

The batch was then cooled to ≤30° C. and the crude compound of formula (I) concentrate was drummed. The

process for obtaining the compound of formula (I) was repeated in an identical manner to obtain a second batch.

##### Step 2—Preparation of Compound of Formula (II)

###### Additional Aqueous Wash

Two individual batches of the compound of formula (I) (57-62% w/w solution in THF) were charged to a vessel. While stirring, 5% HCl (1,767±79 kg) was charged at ≤25° C. The addition was exothermic. The mixture was agitated for ≥15 minutes. Agitation was stopped and the phases were allowed to settle for ≥30 minutes. The lower aqueous phase was removed, leaving compound of formula (I)/THF in the reactor.

###### Iodide Exchange Reaction

Methyl-ethyl-ketone (MEK, 4,384±227 L) and sodium iodide (NaI, 831±9 kg, 1.16 eq.) were charged while stirring (FIG. 1). The batch was heated to reflux (75-80° C.). After approximately 30 hours, GC was used to measure reaction completion (compound of formula (I) ≤1.0% area). If the reaction had not completed, additional time was allowed (expected reaction time: 25 to 35 hours) and NaI was recharged as needed. The mixture was then cooled to approximately 20° C.

###### Solvent Exchange and Aqueous Work-up

The batch was concentrated via vacuum distillation at ≤60° C. until no more distillate was collected. The mixture was then cooled to 20±5° C. and n-heptane (3,624±187 L) was charged. Then, 5% aqueous sodium bisulfite (NaHSO<sub>3</sub>, 2,121±104 kg) was added and the mixture stirred for ≥60 minutes. Agitation was stopped and the phases allowed to settle for ≥60 minutes. The lower aqueous phase was removed for disposal. Water (1980±102 L) was added and the mixture agitated for ≥60 minutes. The phases were allowed to settle for ≥60 minutes and the lower aqueous phase removed for disposal. An optional second water wash was performed if required.

###### Final Concentration

The batch was concentrated using vacuum distillation at ≤50° C. until no more distillate was collected. The batch was then cooled to 20° C. and the compound of formula (II) was drummed and sampled for assay analysis. The expected yield range was 80-120% (w/w %).

##### Step 3—Preparation of Compound of Formula (IV)

###### Sodium t-Pentoxide DMAc Preparation

The following intermediate/compound of formula (IV) sequence was based on a charge of 700 kg of compound of formula (II)/n-heptane with assay of 94.9% wt/wt, which represented a contained charge of 665 kg of compound of formula (II).

A solution of N,N-dimethylacetamide (DMAc, 1,476±37 kg) and sodium t-pentoxide (271±3 kg, 2.10 eq.) was prepared in a vessel and the mixture was agitated for approximately 30 minutes until nearly all of the solids were dissolved.

##### Preparation of the First Intermediate

Compound of formula (II) (700 kg, 1 eq.), DMAc (1,272±27 kg), and TosMIC (219±1 kg) were charged to a vessel (see FIG. 1). The mixture was cooled to ≤-5° C. and well agitated. To this solution, the sodium t-pentoxide/DMAc mixture was added over approximately 1 hour at ≤-5° C. The transfer line was rinsed with DMAc (181±9 kg). The reaction was strongly exothermic. The reaction

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mixture was agitated for  $\geq 30$  minutes at  $\leq -5^\circ$  C. The conversion was confirmed to be complete using high-performance liquid chromatography with ultraviolet detection (HPLC-UV) (monoalkylated TosMIC  $\leq 1\%$  area and compound of formula (II)  $\leq 1.4\%$  area). Optional kicker charges of compound of formula (II), TosMIC and sodium t-pentoxide were employed as required, to ensure completion of the reaction, based on the following instructions (Table 18):

TABLE 18

| Instructions for Determining Kicker Charge Action                         |                                                 |
|---------------------------------------------------------------------------|-------------------------------------------------|
| IPC Test Result Criteria                                                  | Kicker Charge Action                            |
| $\leq 1\%$ Monoalkylated TosMIC and<br>$> 1.4\%$ compound of formula (II) | Charge additional TosMIC and sodium t-pentoxide |
| $> 1\%$ Monoalkylated TosMIC and<br>$\leq 1.4\%$ compound of formula (II) | Charge additional compound of formula (II)      |
| $> 1\%$ Monoalkylated TosMIC and<br>$> 1.4\%$ compound of formula (II)    | Charge additional sodium t-pentoxide            |

#### Quench and Phase Separation

n-Heptane (2,407 $\pm$ 120 L) and water (3,061 $\pm$ 153 L) were charged to another reactor. The first intermediate reaction mixture was transferred to the n-heptane/water mixture under temperature control between  $0^\circ$  C. and  $40^\circ$  C. (target  $20^\circ$  C.). The reaction was slightly exothermic. The transfer line was rinsed with n-heptane (470 $\pm$ 24 L). The mixture was then agitated for  $\geq 1$  hour. Agitation was stopped and the mixture allowed to settle for  $\geq 1$  hour. The lower aqueous phase was removed for disposal. A solution of 5% aqueous sodium chloride (NaCl, 3,106 $\pm$ 147 kg) was charged and the mixture agitated for  $\geq 1$  hour at approximately  $20^\circ$  C. (range:  $0^\circ$  C. to  $40^\circ$  C.). The agitator was stopped and the mixture allowed to settle for  $\geq 60$  minutes. The lower aqueous phase was removed for disposal. The remaining solution of the first intermediate in n-heptane was transferred to another vessel. Compound of Formula (IV) Reaction

Isopropyl acetate (IPAc, 451 $\pm$ 23 L) was added to the solution of the first intermediate in n-heptane and the mixture cooled to  $-10\pm 10^\circ$  C. Concentrated HCl (115 $\pm$ 2 kg) was added while maintaining the temperature at  $\leq 25^\circ$  C. The reaction was exothermic and the reaction mixture allowed to warm, if needed, to  $20\pm 5^\circ$  C. The mixture was agitated for  $\geq 30$  minutes during the warming period. The reaction conversion was measured using HPLC-UV (intermediate  $\leq 2\%$  area).

#### Quench and Phase Separation

In a separate vessel, a sodium hydroxide (NaOH) solution (50% wt/wt, 175 $\pm$ 2 kg) was combined with water (1927 $\pm$ 96 L). The resulting aqueous NaOH solution was added to the reaction mixture at approximately  $20^\circ$  C. (range:  $10^\circ$  C. to  $40^\circ$  C.). The line was rinsed with water. The mixture was stirred for  $\geq 3$  hours. The neutralization endpoint is pH 9 to 12. Agitation was stopped and the phases allowed to settle for  $\geq 60$  minutes. The lower aqueous phase was removed for disposal. A dilute aqueous solution, containing NaCl (55 $\pm$ 3 kg), water (1,572 $\pm$ 79 L) and 50% sodium hydroxide (4.6 $\pm$ 0.2 kg), was prepared in a separate vessel and charged to the compound of formula (IV) product mixture. A water rinse (128 $\pm$ 6 L) was applied and the mixture was agitated for  $\geq 60$  minutes. Agitation was stopped and the phases were allowed to settle for  $\geq 60$  minutes. The lower aqueous phase was removed for disposal.

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#### Concentration

The mixture was concentrated under vacuum at  $\leq 80^\circ$  C. until no more distillate was collected. The distillation was monitored using GC (compound of formula (IV) 75% area). The batch was cooled to approximately  $20^\circ$  C. and the compound of formula (IV) concentrate was held until the second batch was prepared. The process for preparing the compound of formula (IV) was repeated in an identical manner to provide a second batch of the compound of formula (IV) in n-heptane. The second batch (in n-heptane) was then combined with the first batch for final distillation and packaging. The product was weighed and sampled for assay. The expected yield range is 80-120% w/w.

#### Step 4—Preparation of Compound of Formula (V) (Crude Bempedoic Acid)

##### Reaction 1 (Ketone Reduction)

Compound of formula (IV) (545 $\pm$ 5 kg) and ethanol (EtOH, 1090 $\pm$ 55 kg) were charged to a vessel. While maintaining the batch at  $\leq 35^\circ$  C., sodium borohydride (NaBH<sub>4</sub>, 12 wt % in 40% NaOH, 155 $\pm$ 2 kg, 0.35 eq.) was charged over approximately 2-3 hours (FIG. 1). The addition was exothermic. The charging line was rinsed with water (155 $\pm$ 8 kg). After holding at  $25\pm 10^\circ$  C. for 1 hour the conversion was measured using HPLC-UV (compound of formula (IV) 0.5% area).

##### Reaction 2 (Saponification)

An aqueous solution of NaOH (50% wt/wt, 435 $\pm$ 4 kg) was charged to the vessel at  $50^\circ$  C. The addition was exothermic. The charging line was rinsed with water (155 $\pm$ 8 kg) and the reaction mixture was warmed to  $50\pm 5^\circ$  C. for 6 hours. The saponification was measured using HPLC-UV (compound of formula (V) monoester 0.5% area). Water (1873 $\pm$ 94 kg) was charged to the reaction mixture. EtOH and water were distilled under vacuum and at  $\leq 50^\circ$  C. until the batch volume reached the target level (approximately 2184 L). The mixture was transferred to another reactor and the transfer line was rinsed with water (273 $\pm$ 14 kg).

##### pH Adjustment, Phase Separation, and Extraction

Methyl tert-butyl ether (MTBE, 1628 $\pm$ 81 kg) was added and the batch cooled to  $10\text{--}15^\circ$  C. Concentrated HCl (647 $\pm$ 6 kg) was added slowly at  $10\text{--}20^\circ$  C. (the addition was exothermic) and the batch stirred  $\geq 1$  hour. A sample was taken for pH analysis and the pH was adjusted with HCl or NaOH as needed (target pH range: 5 to 6). The formation of hydrogen gas was observed. Agitation was stopped and the phases were allowed to settle for  $\geq 60$  minutes at  $10\text{--}20^\circ$  C. The lower aqueous phase was removed for disposal. The batch was transferred to another vessel and rinsed forward with MTBE. The concentration of compound of formula (V) in MTBE was measured using HPLC-UV (compound of formula (V): 17% to 20% weight).

#### Step 5—Purification of Compound of Formula (V)/Preparation of Crystalline Form of Compound of Formula (V)

##### Silica Gel Preparations

The diameter $\times$ height ratio for the silica gel plug varied from 1 $\times$ 1 to 1 $\times$ 3. Silica gel (60 $\pm$ 2 kg) was charged to a filter and wetted with ethyl acetate (EtOAc) that was charged into the reactor and then drained to the filter. The silica gel bed was preheated by recirculating EtOAc (1,173 $\pm$ 59 kg) at  $50\pm 5^\circ$  C. Excess EtOAc was removed immediately prior to filtration of the compound of formula (V) batch.

## Solvent Exchange to Ethyl Acetate

The compound of formula (V) in MTBE was charged to a reactor. The batch was concentrated under vacuum at  $\leq 50^\circ$  C. to 30% to 35% of the initial volume. EtOAc (2,002 $\pm$ 100 kg) was charged and the batch was concentrated again to 30% to 35% of the initial volume. EtOAc (1601 $\pm$ 80 kg) was charged and distillation is repeated. EtOAc (1601 $\pm$ 80 kg) was charged and the batch was sampled. The solvent exchange was measured by GC (MTBE $\leq$ 0.1% weight). Additional EtOAc charges and distillations were performed as required.

## Silica Gel Filtration

When the solvent exchange was complete, the batch was warmed to  $50\pm 5^\circ$  C. Then the batch was filtered through the preheated silica gel plug into another reactor. To rinse the line and the silica gel, EtOAc (964 $\pm$ 20 kg) was charged to the reactor, warmed to  $50^\circ$  C., and then a portion of the warm EtOAc was transferred through the silica gel plug. Loss of product due to retention on the silica gel filter was measured using HPLC (compound of formula (V) $\leq$ 0.5% weight in the eluate). Additional flushes with EtOAc were performed as required. Purified compound of formula (V) in EtOAc was partially concentrated by distillation under vacuum, at  $\leq 50^\circ$  C., to a final volume of approximately 1700 L.

## Crystallization

The temperature of the concentrated compound of formula (V) in EtOAc was adjusted to approximately  $50\pm 5^\circ$  C. Water (24 $\pm$ 1 kg) was charged, the line was rinsed with EtOAc (74 $\pm$ 10 kg) and the solution was maintained at  $50\pm 5^\circ$  C. for  $\geq 1$  hour. The solution was then cooled to  $22\pm 5^\circ$  C. using a cooling rate of  $14^\circ$  C./hour, with slow agitation, and then held at this temperature for  $\geq 2$  hours to initiate crystallization. Once slurry formation was confirmed, the solution was stirred at approximately  $20$ - $25^\circ$  C. for  $\geq 6$  hours. The batch was then cooled to  $0\pm 5^\circ$  C. using a cooling rate of  $11^\circ$  C./hour and then stirred at this temperature for  $\geq 6$  hours.

## Isolation and Drying

The crude crystalline form of the compound of formula (V) was isolated by centrifugation at  $0\pm 5^\circ$  C. and then washed with chilled EtOAc at  $0\pm 5^\circ$  C. The wet cake was dried under vacuum at  $45^\circ$  C. Drying was monitored by loss-on drying (LOD) (LOD $\leq$ 0.5%).

## Optional In Situ Filtration

For vessels designed with in-reactor (in situ) filtration, the slurry was allowed to settle for  $\geq 1$  hour at  $0\pm 5^\circ$  C. The batch was filtered and the wet cake was left in the reactor. EtOAc (1,064 $\pm$ 53 kg) was charged to another vessel, chilled to  $0\pm 5^\circ$  C. and transferred, backwards through the decant filters, to the reactor containing the wet cake. The batch was agitated for  $\geq 1$  hour and then allowed to settle for  $\geq 1$  hour. Filtration was repeated. The slurry wash and filtration process was repeated three times in an identical manner.

Step 6—Preparation of Pharmaceutical Material  
Including Purified Amount of Compound of  
Formula (V)

## Recrystallization (Ethyl Acetate Water)

Following in situ filtration, the amount of the crystalline form of the compound of formula (V) produced in Step 5 was estimated based on the assumption that there was 100% conversion of the compound of formula (IV) (charge amount—545 kg) into the compound of formula (V) in Step 4 (FIG. 1).

In a reactor, EtOAc was charged to the crystalline form of the compound of formula (V) wetcake until the volume

reaches the 1433 L mark (approximately 619 kg of EtOAc) and the suspension was then heated to  $55$ - $60^\circ$  C. until all solids dissolved. To the clear solution, water (16 $\pm$ 1 kg) was added and the batch agitated at  $55$ - $70^\circ$  C. for  $\geq 1$  hour. The temperature was adjusted to  $55\pm 5^\circ$  C. and the batch was then transferred to another reactor via polish filtration. The reactor, filter and line were rinsed with EtOAc (162 $\pm$ 12 kg). The temperature was adjusted to  $55\pm 5^\circ$  C. The hot solution was then cooled over  $\geq 1$  hour to  $30\pm 5^\circ$  C. and agitated for  $\geq 2$  hours. The batch was then heated over  $\geq 1$  hour to  $40\pm 5^\circ$  C. and then maintained at  $40\pm 5^\circ$  C. for  $\geq 1$  hour. The batch was then cooled over  $\geq 1$  hour to  $35\pm 5^\circ$  C. and then maintained at  $35\pm 5^\circ$  C. for  $\geq 2$  hours. The batch was then cooled over  $\geq 5$  hours to  $5\pm 5^\circ$  C. and then maintained at  $5\pm 5^\circ$  C. for  $\geq 4$  hours. The resulting solids were isolated by centrifugation. The wash solvent was acetonitrile (ACN), stored for use at  $20\pm 10^\circ$  C.

## Isolation, Drying, Compound Identification of the Compound of Formula (V) and IPC Testing

The purified compound of formula (V) solids were collected by centrifugation and then washed with acetonitrile (2 $\times$ 2 kg/kg of the compound of formula (V)) at  $20\pm 10^\circ$  C. to remove all residual mother liquor. The wet cake was dried in vacuum at  $\leq 45^\circ$  C. Yield: 324.2 kg (84.9%) of a pharmaceutical material comprising a purified amount of the compound of formula (V).

The purified compound of formula (V) was prepared for  $^1\text{H}$ - and  $^{13}\text{C}$ -NMR analysis by preparing 10 mg/mL and 50 mg/mL solutions of the purified compound of formula (V) in  $\text{CDCl}_3$ .  $^1\text{H}$ - and  $^{13}\text{C}$ -NMR spectra were obtained using an Inova 500 MHz NMR spectrometer. To ensure better relative quantitation of the integrals and that all signals were captured the window was expanded from 5000 to 8000 Hz and the wait time extended between acquisitions from 1 to 25 sec. The resulting  $^1\text{H}$ - and  $^{13}\text{C}$ -NMR spectra of the compound of formula (V) (FIGS. 2(a) and 2(b), respectively) are consistent with known  $^1\text{H}$ - and  $^{13}\text{C}$ -NMR spectra of bempedoic acid (Table 19).

TABLE 19

| Bempedoic acid $^1\text{H}$ - and $^{13}\text{C}$ -NMR assignments |                       |        |
|--------------------------------------------------------------------|-----------------------|--------|
| Signal Assignment                                                  | Signal Location (ppm) |        |
| Nucleus                                                            | Proton (mult)         | Carbon |
| $\text{CO}_2\text{H}$                                              | 11.99 (singlet)       | 179.05 |
| C2                                                                 | N/A                   | 41.43  |
| $(\text{CH}_3)_2$                                                  | 1.05 (singlet)        | 25.23  |
| C3/H3                                                              | 1.41 (multiplet)      | 40.46  |
| C4/H4                                                              | 1.15-1.34 (multiplet) | 24.78  |
| C5/H5                                                              | 1.15-1.34 (multiplet) | 29.99  |
| C6/H6                                                              | 1.15-1.34 (multiplet) | 25.42  |
| C7/H7                                                              | 1.41 (multiplet)      | 37.39  |
| C8/H8                                                              | 3.32 (singlet)        | 69.69  |
| OH                                                                 | 4.19 (singlet)        | N/A    |

A sample of the purified compound of formula (V) was run on an Agilent 1100 HPLC coupled to a Thermo LTQ-XL Mass Spectrometer electrospray running in positive electrospray mode. The capillary temperature was  $200^\circ$  C. The column was a Waters X-Bridge C18, 4.6 $\times$ 75 mm, 2.5  $\mu\text{m}$ . Mobile phase A was 0.05% formic acid and mobile phase B was 0.04% formic acid in acetonitrile. The experimental mass of the compound of formula (V) was found to be 344.38 Da, which is in good agreement with the calculated mass for bempedoic acid of 344.49 Da.

The expected yield range was 66-91%. The residual solvent was measured using GC (ACN $\leq$ 350 ppm) to deter-



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mine completion of drying. When drying was complete, impurities were measured using HPLC with charged aerosol detection (CAD) (unknown impurities  $\leq 0.08\%$  by weight and known impurities  $\leq 0.13\%$  by weight). If the impurity profile criterion was met, the product was treated as the final Active Pharmaceutical Ingredient (API). If the impurity profile criterion was not met, another recrystallization was performed as described above.

Using the HPLC assay described in Example 3, the purity of the purified compound of formula (V) was determined to be 99.6% (w/w).

X-ray powder diffraction (XRPD) data for the crystalline form of the compound of formula (V) were collected using a Panalytical X'Pert<sup>3</sup> Powder diffractometer (Cu, K $\alpha$  radiation; X-ray tube setting—45 kV, 40 mA; divergence slit—fixed  $\frac{1}{8}^\circ$ ; scan mode—continuous; scan range—3 to  $40^\circ$  (2 $\theta$ ); scan step time—18.87 seconds; step size— $0.0131^\circ$  (2 $\theta$ )). Samples of the crystalline form of the compound of formula (V) were placed on a Si zero-background holder. The 2 theta position was calibrated against a Panalytical Si reference standard disc. An XRPD pattern of the crystalline form of the compound of formula (V) is provided in FIG. 4. Tabulated characteristics of the XRPD pattern in FIG. 4 are provided below in Table 20, which lists diffraction angle 2 $\theta$  and relative intensity (expressed as a percentage with respect to the most intense peak).

Differential Scanning Calorimetry (DSC) data for the crystalline form of the compound of formula (V) were collected using a TA Q2000 DSC instrument. The DSC instrument was calibrated using an indium reference standard. Samples of the crystalline form of the compound of formula (V) were placed inside crimped aluminum sample pans and heated at a rate of  $10^\circ$  C./minute from ambient temperature ( $\sim 25^\circ$  C.) to  $300^\circ$  C. A DSC curve for the crystalline form of the compound of formula (V) is provided in FIG. 5. The DSC curve displayed an endothermic event with an onset value of about  $92.4^\circ$  C.

Thermogravimetric analysis (TGA) data for the crystalline form of the compound of formula (V) were collected using a TA Discovery 550 TGA instrument. The TGA instrument was calibrated using a nickel reference standard. Samples of the crystalline form of the compound of formula (V) were placed in open platinum sample pans and heated at a rate of  $10^\circ$  C./minute from about ambient temperature ( $\sim 25^\circ$  C.) to about  $315^\circ$  C. A TGA curve for the crystalline form of the compound of formula (V) is provided in FIG. 6. The TGA curve displayed negligible weight loss prior to decomposition occurring.

TABLE 20

| X-ray Powder Diffraction Pattern Data of the Crystalline Form of the Compound of Formula (V) |                        |
|----------------------------------------------------------------------------------------------|------------------------|
| Angle [2 $\theta$ ]                                                                          | Relative Intensity [%] |
| 5.2                                                                                          | 2.33                   |
| 10.3                                                                                         | 70.75                  |
| 10.4                                                                                         | 78.65                  |
| 11.8                                                                                         | 2.88                   |
| 13.7                                                                                         | 2.72                   |
| 15.5                                                                                         | 8.08                   |
| 15.6                                                                                         | 7.16                   |
| 17.3                                                                                         | 8.20                   |
| 17.6                                                                                         | 18.72                  |
| 17.9                                                                                         | 100.00                 |
| 18.8                                                                                         | 42.30                  |
| 19.5                                                                                         | 21.42                  |

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TABLE 20-continued

| X-ray Powder Diffraction Pattern Data of the Crystalline Form of the Compound of Formula (V) |                        |
|----------------------------------------------------------------------------------------------|------------------------|
| Angle [2 $\theta$ ]                                                                          | Relative Intensity [%] |
| 19.7                                                                                         | 15.07                  |
| 20.4                                                                                         | 16.93                  |
| 20.7                                                                                         | 23.95                  |
| 21.1                                                                                         | 5.78                   |
| 22.0                                                                                         | 13.87                  |
| 22.6                                                                                         | 17.54                  |
| 23.1                                                                                         | 7.78                   |
| 23.6                                                                                         | 4.97                   |
| 23.9                                                                                         | 6.19                   |
| 24.7                                                                                         | 1.98                   |
| 25.8                                                                                         | 3.04                   |
| 26.3                                                                                         | 2.10                   |
| 27.5                                                                                         | 13.36                  |
| 29.2                                                                                         | 3.86                   |
| 30.2                                                                                         | 1.27                   |
| 30.8                                                                                         | 5.34                   |
| 31.3                                                                                         | 1.40                   |
| 31.9                                                                                         | 2.95                   |
| 32.9                                                                                         | 1.27                   |
| 34.4                                                                                         | 5.98                   |
| 35.1                                                                                         | 2.07                   |
| 36.2                                                                                         | 3.16                   |
| 37.2                                                                                         | 2.37                   |
| 37.9                                                                                         | 1.79                   |

In addition, single crystals of the crystalline form of the compound of formula (V) were analyzed by single crystal X-ray diffraction. The unit cell parameters of the crystalline form of the compound of formula (V) and the data collection and structure refinement methods are shown in Tables 21 and 22, respectively.

TABLE 21

| Unit Cell Parameters of the Crystalline Form of the Compound of Formula (V) |                                                                                                                              |
|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Empirical formula                                                           | C <sub>19</sub> H <sub>36</sub> O <sub>5</sub>                                                                               |
| Formula weight                                                              | 344.48                                                                                                                       |
| Temperature                                                                 | 297(2) K                                                                                                                     |
| Wavelength                                                                  | 0.71073 Å                                                                                                                    |
| Crystal size                                                                | 0.400 × 0.140 × 0.090 mm                                                                                                     |
| Crystal system                                                              | Monoclinic                                                                                                                   |
| Space group                                                                 | P2 <sub>1</sub> /c                                                                                                           |
| Unit cell dimensions                                                        | a = 17.9209(8) Å $\alpha = 90^\circ$<br>b = 9.8547(5) Å $\beta = 106.8340(10)^\circ$<br>c = 12.2775(6) Å $\gamma = 90^\circ$ |
| Volume                                                                      | 2075.35(17) Å <sup>3</sup>                                                                                                   |
| Z                                                                           | 4                                                                                                                            |
| Density (calculated)                                                        | 1.102 Mg/m <sup>3</sup>                                                                                                      |
| Absorption coefficient                                                      | 0.078 mm <sup>-1</sup>                                                                                                       |
| F(000)                                                                      | 760                                                                                                                          |

TABLE 22

| Data Collection and Structure Refinement Methods<br>for the Crystalline Form of the Compound of Formula (V) |                                                                      |
|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Diffractometer                                                                                              | Bruker D8 Quest PHOTON 100 CMOS                                      |
| Radiation source                                                                                            | Incoatec Microfocus Source (I $\mu$ S)<br>monochromated MoK $\alpha$ |
| Data collection method                                                                                      | omega/phi scans                                                      |
| Theta range for data collection                                                                             | 2.384° to 25.243°                                                    |
| Limiting indices                                                                                            | -21<= $h$ <=21, -11<= $k$ <=11, -14<= $l$ <=14                       |
| Reflections collected/unique                                                                                | 51013/3745 [R(int) = 0.0514]                                         |
| Completeness to theta = 25.242                                                                              | 99.8%                                                                |
| Absorption correction                                                                                       | Semi-empirical from equivalents                                      |
| Max. and min. transmission                                                                                  | 0.7454 and 0.6534                                                    |
| Refinement method                                                                                           | Full-matrix least-squares on F <sup>2</sup>                          |
| Data/restraints/parameters                                                                                  | 3745/3/228                                                           |
| Goodness-of-fit on F <sup>2</sup>                                                                           | 1.027                                                                |
| Final R indices [I > 2 sigma(I)]                                                                            | R1 = 0.0684, wR2 = 0.1658                                            |
| R indices (all data)                                                                                        | R1 = 0.0858, wR2 = 0.1780                                            |
| Extinction coefficient                                                                                      | n/a                                                                  |
| Largest diff. peak and hole                                                                                 | 0.956 and -0.379 e $\cdot$ A <sup>-3</sup>                           |

Atomic coordinates ( $\times 10^4$ ) and equivalent isotropic displacement parameters ( $\text{\AA}^2 \times 10^3$ ) are shown in Table 23, below. U(eq) is defined as one third of the trace of the orthogonalized U<sup>ij</sup> tensor.

TABLE 23

| Atomic Coordinates and Equivalent Isotropic Atomic Displacement<br>Parameters for the Crystalline Form of the Compound of Formula (V) |         |         |          |       |
|---------------------------------------------------------------------------------------------------------------------------------------|---------|---------|----------|-------|
| Atom                                                                                                                                  | x       | y       | z        | U(eq) |
| O1                                                                                                                                    | 8702(1) | 759(2)  | 9437(2)  | 54(1) |
| O2                                                                                                                                    | 7773(1) | 838(2)  | 10289(2) | 61(1) |
| O3                                                                                                                                    | 7792(1) | 9467(2) | 7690(2)  | 64(1) |
| O4                                                                                                                                    | 3323(1) | 6206(2) | 3606(2)  | 61(1) |
| O5                                                                                                                                    | 3141(1) | 8412(2) | 3400(2)  | 58(1) |
| C1                                                                                                                                    | 8438(2) | 1070(2) | 10303(2) | 39(1) |
| C2                                                                                                                                    | 9051(1) | 1748(3) | 11258(2) | 40(1) |
| C3                                                                                                                                    | 8742(2) | 1960(4) | 12278(2) | 65(1) |
| C4                                                                                                                                    | 9778(2) | 833(3)  | 11597(3) | 62(1) |
| C5                                                                                                                                    | 9260(2) | 3111(3) | 10806(2) | 43(1) |
| C6                                                                                                                                    | 8566(2) | 4011(3) | 10253(3) | 50(1) |
| C7                                                                                                                                    | 8782(2) | 5364(3) | 9841(2)  | 53(1) |
| C8                                                                                                                                    | 8072(2) | 6113(3) | 9111(3)  | 55(1) |
| C9                                                                                                                                    | 8256(2) | 7413(3) | 8618(3)  | 65(1) |
| C10                                                                                                                                   | 7539(2) | 8176(3) | 7903(3)  | 57(1) |
| C11                                                                                                                                   | 7131(2) | 7409(3) | 6807(2)  | 51(1) |
| C12                                                                                                                                   | 6389(2) | 8053(3) | 6074(2)  | 52(1) |
| C13                                                                                                                                   | 5964(2) | 7180(3) | 5081(3)  | 53(1) |
| C14                                                                                                                                   | 5205(2) | 7760(3) | 4343(2)  | 49(1) |
| C15                                                                                                                                   | 4828(1) | 6862(3) | 3332(2)  | 44(1) |
| C16                                                                                                                                   | 4040(1) | 7331(3) | 2540(2)  | 42(1) |
| C17                                                                                                                                   | 3762(2) | 6284(3) | 1581(3)  | 61(1) |
| C18                                                                                                                                   | 4109(2) | 8719(3) | 2026(3)  | 60(1) |
| C19                                                                                                                                   | 3454(1) | 7401(3) | 3216(2)  | 41(1) |

Bond lengths ( $\text{\AA}$ ) are shown in Table 24, below.

TABLE 24

| Selected Bond Lengths ( $\text{\AA}$ ) for the Crystalline<br>Form of the Compound of Formula (V) |                              |
|---------------------------------------------------------------------------------------------------|------------------------------|
| Bond                                                                                              | Bond length ( $\text{\AA}$ ) |
| O(1)—H(1)                                                                                         | 0.862(18)                    |
| O(1)—C(1)                                                                                         | 1.318(3)                     |
| O(2)—C(1)                                                                                         | 1.209(3)                     |
| O(3)—H(3)                                                                                         | 0.910(18)                    |
| O(3)—C(10)                                                                                        | 1.400(3)                     |
| O(4)—H(4)                                                                                         | 0.859(18)                    |
| O(4)—C(19)                                                                                        | 1.318(3)                     |
| O(5)—C(19)                                                                                        | 1.197(3)                     |

TABLE 24-continued

| Selected Bond Lengths ( $\text{\AA}$ ) for the Crystalline<br>Form of the Compound of Formula (V) |                              |
|---------------------------------------------------------------------------------------------------|------------------------------|
| Bond                                                                                              | Bond length ( $\text{\AA}$ ) |
| C(1)—C(2)                                                                                         | 1.511(4)                     |
| C(2)—C(4)                                                                                         | 1.540(4)                     |
| C(2)—C(3)                                                                                         | 1.523(4)                     |
| C(2)—C(5)                                                                                         | 1.540(4)                     |
| C(5)—C(6)                                                                                         | 1.519(4)                     |
| C(6)—C(7)                                                                                         | 1.514(4)                     |
| C(7)—C(8)                                                                                         | 1.519(4)                     |
| C(8)—C(9)                                                                                         | 1.494(4)                     |
| C(9)—C(10)                                                                                        | 1.528(4)                     |
| C(10)—C(11)                                                                                       | 1.532(4)                     |
| C(11)—C(12)                                                                                       | 1.512(4)                     |
| C(12)—C(13)                                                                                       | 1.507(4)                     |
| C(13)—C(14)                                                                                       | 1.511(4)                     |
| C(14)—C(15)                                                                                       | 1.515(4)                     |
| C(15)—C(16)                                                                                       | 1.535(4)                     |
| C(16)—C(19)                                                                                       | 1.516(4)                     |
| C(16)—C(17)                                                                                       | 1.536(4)                     |
| C(16)—C(18)                                                                                       | 1.527(4)                     |

Bond angles ( $^\circ$ ) are shown in Table 25, below.

TABLE 25

| Selected Bond Angles ( $^\circ$ ) for the Crystalline<br>Form of the Compound of Formula (V) |                            |                   |                            |
|----------------------------------------------------------------------------------------------|----------------------------|-------------------|----------------------------|
|                                                                                              | Bond<br>angle ( $^\circ$ ) |                   | Bond<br>angle ( $^\circ$ ) |
| H(1)—O(1)—C(1)                                                                               | 109(2)                     | O(3)—C(10)—C(9)   | 106.9(2)                   |
| H(3)—O(3)—C(10)                                                                              | 107(2)                     | O(3)—C(10)—C(11)  | 112.5(2)                   |
| H(4)—O(4)—C(19)                                                                              | 114(3)                     | C(9)—C(10)—C(11)  | 111.7(3)                   |
| O(2)—C(1)—O(1)                                                                               | 121.8(2)                   | C(12)—C(11)—C(10) | 115.4(2)                   |
| O(2)—C(1)—C(2)                                                                               | 125.9(2)                   | C(11)—C(12)—C(13) | 113.1(2)                   |
| O(1)—C(1)—C(2)                                                                               | 112.2(2)                   | C(12)—C(13)—C(14) | 115.2(2)                   |
| C(1)—C(2)—C(4)                                                                               | 108.6(2)                   | C(15)—C(14)—C(13) | 112.4(2)                   |
| C(1)—C(2)—C(3)                                                                               | 110.1(2)                   | C(14)—C(15)—C(16) | 117.0(2)                   |
| C(4)—C(2)—C(3)                                                                               | 109.8(2)                   | C(19)—C(16)—C(15) | 108.8(2)                   |
| C(1)—C(2)—C(5)                                                                               | 107.70(19)                 | C(19)—C(16)—C(17) | 109.1(2)                   |
| C(4)—C(2)—C(5)                                                                               | 109.4(2)                   | C(15)—C(16)—C(17) | 108.6(2)                   |
| C(3)—C(2)—C(5)                                                                               | 111.2(2)                   | C(19)—C(16)—C(18) | 109.7(2)                   |
| C(6)—C(5)—C(2)                                                                               | 114.7(2)                   | C(15)—C(16)—C(18) | 111.2(2)                   |
| C(1)—C(5)—C(2)                                                                               | 114.1(2)                   | C(17)—C(16)—C(18) | 109.4(2)                   |
| C(6)—C(7)—C(8)                                                                               | 112.0(2)                   | O(5)—C(19)—O(4)   | 122.2(2)                   |
| C(7)—C(8)—C(9)                                                                               | 114.2(3)                   | O(5)—C(19)—C(16)  | 125.3(2)                   |
| C(8)—C(9)—C(10)                                                                              | 114.0(3)                   | O(4)—C(19)—C(16)  | 112.5(2)                   |

Torsion angles (°) are shown in Table 26, below.

TABLE 26

| Selected Torsion Angles (°) for the Crystalline Form of the Compound of Formula (V) |                   |                         |                   |
|-------------------------------------------------------------------------------------|-------------------|-------------------------|-------------------|
|                                                                                     | Torsion angle (°) |                         | Torsion angle (°) |
| H(1)—O(1)—C(1)—O(2)                                                                 | −3(3)             | C(8)—C(9)—C(10)—C(11)   | −67.6(4)          |
| H(1)—O(1)—C(1)—C(2)                                                                 | 178(3)            | O(3)—C(10)—C(11)—C(12)  | −62.9(4)          |
| O(2)—C(1)—C(2)—C(4)                                                                 | 126.7(3)          | C(9)—C(10)—C(11)—C(12)  | 176.9(3)          |
| O(1)—C(1)—C(2)—C(4)                                                                 | −54.3(3)          | C(10)—C(11)—C(12)—C(13) | −173.4(3)         |
| O(2)—C(1)—C(2)—C(3)                                                                 | 6.5(4)            | C(11)—C(12)—C(13)—C(14) | 178.2(3)          |
| O(1)—C(1)—C(2)—C(3)                                                                 | −174.5(2)         | C(12)—C(13)—C(14)—C(15) | 177.3(3)          |
| O(2)—C(1)—C(2)—C(5)                                                                 | −114.9(3)         | C(13)—C(14)—C(15)—C(16) | 178.3(2)          |
| O(1)—C(1)—C(2)—C(5)                                                                 | 64.2(3)           | C(14)—C(15)—C(16)—C(19) | −61.2(3)          |
| C(1)—C(2)—C(5)—C(6)                                                                 | 52.0(3)           | C(14)—C(15)—C(16)—C(17) | −179.8(2)         |
| C(4)—C(2)—C(5)—C(6)                                                                 | 169.9(2)          | C(14)—C(15)—C(16)—C(18) | 59.7(3)           |
| C(3)—C(2)—C(5)—C(6)                                                                 | −68.7(3)          | H(4)—O(4)—C(19)—O(5)    | 4(3)              |
| C(2)—C(5)—C(6)—C(7)                                                                 | 178.4(2)          | H(4)—O(4)—C(19)—C(16)   | −176(3)           |
| C(5)—C(6)—C(7)—C(8)                                                                 | 169.7(2)          | C(15)—C(16)—C(19)—O(5)  | 116.1(3)          |
| C(6)—C(7)—C(8)—C(9)                                                                 | −176.1(3)         | C(17)—C(16)—C(19)—O(5)  | −125.6(3)         |
| C(7)—C(8)—C(9)—C(10)                                                                | −178.9(3)         | C(18)—C(16)—C(19)—O(5)  | −5.8(4)           |
| H(3)—O(3)—C(10)—C(9)                                                                | −174(2)           | C(15)—C(16)—C(19)—O(4)  | −63.2(3)          |
| H(3)—O(3)—C(10)—C(11)                                                               | 63(2)             | C(17)—C(16)—C(19)—O(4)  | 55.1(3)           |
| C(8)—C(9)—C(10)—O(3)                                                                | 168.9(3)          | C(18)—C(16)—C(19)—O(4)  | 174.9(2)          |

Anisotropic Displacement Parameters ( $\text{\AA}^2$ ) are shown in Table 27, below. The anisotropic displacement factor exponent may be expressed in the form:  $-2\pi^2[h^2a^{*2}U^{11} + \dots + 2hka^*b^*U^{12}]$ .

TABLE 27

| Anisotropic Displacement Parameters ( $\text{\AA}^2$ ) for the Crystalline Form of the Compound of Formula (V) |       |       |       |        |        |        |
|----------------------------------------------------------------------------------------------------------------|-------|-------|-------|--------|--------|--------|
| Atom                                                                                                           | U11   | U22   | U33   | U23    | U13    | U12    |
| O1                                                                                                             | 52(1) | 60(1) | 48(1) | −18(1) | 13(1)  | −4(1)  |
| O2                                                                                                             | 48(1) | 70(1) | 70(1) | −12(1) | 23(1)  | −12(1) |
| O3                                                                                                             | 70(1) | 38(1) | 67(1) | −1(1)  | −7(1)  | 3(1)   |
| O4                                                                                                             | 67(1) | 40(1) | 92(2) | 12(1)  | 47(1)  | 7(1)   |
| O5                                                                                                             | 57(1) | 39(1) | 83(2) | 0(1)   | 26(1)  | 10(1)  |
| C1                                                                                                             | 44(1) | 31(1) | 42(1) | 4(1)   | 12(1)  | 5(1)   |
| C2                                                                                                             | 44(1) | 42(1) | 32(1) | 3(1)   | 6(1)   | 4(1)   |
| C3                                                                                                             | 76(2) | 80(2) | 39(2) | 3(2)   | 17(2)  | 0(2)   |
| C4                                                                                                             | 56(2) | 60(2) | 61(2) | 14(2)  | 1(1)   | 15(2)  |
| C5                                                                                                             | 43(1) | 40(1) | 39(1) | −2(1)  | 1(1)   | −3(1)  |
| C6                                                                                                             | 52(2) | 44(2) | 55(2) | 6(1)   | 15(1)  | 6(1)   |
| C7                                                                                                             | 61(2) | 37(2) | 50(2) | −1(1)  | −4(1)  | −2(1)  |
| C8                                                                                                             | 61(2) | 40(2) | 59(2) | 4(1)   | 10(1)  | 5(1)   |
| C9                                                                                                             | 63(2) | 36(2) | 74(2) | 8(1)   | −14(2) | −9(1)  |
| C10                                                                                                            | 54(2) | 44(2) | 60(2) | 8(1)   | −3(1)  | −6(1)  |
| C11                                                                                                            | 49(2) | 36(1) | 59(2) | 0(1)   | 1(1)   | 0(1)   |
| C12                                                                                                            | 48(2) | 44(2) | 57(2) | −4(1)  | 3(1)   | 1(1)   |
| C13                                                                                                            | 45(2) | 47(2) | 59(2) | −8(1)  | 5(1)   | 3(1)   |
| C14                                                                                                            | 44(2) | 44(2) | 55(2) | −6(1)  | 8(1)   | 2(1)   |
| C15                                                                                                            | 40(1) | 40(1) | 54(2) | −7(1)  | 15(1)  | 0(1)   |
| C16                                                                                                            | 39(1) | 40(1) | 45(1) | −2(1)  | 11(1)  | −4(1)  |
| C17                                                                                                            | 60(2) | 67(2) | 55(2) | −16(2) | 16(1)  | −13(2) |
| C18                                                                                                            | 62(2) | 55(2) | 63(2) | 14(2)  | 16(2)  | −6(2)  |
| C19                                                                                                            | 35(1) | 36(1) | 48(2) | −1(1)  | 7(1)   | 2(1)   |

Hydrogen atom coordinates and isotropic atomic displacement parameters ( $\text{\AA}^2$ ) are shown in Table 28, below.

TABLE 28

| Hydrogen Atom Coordinates and Isotropic Displacement Parameters ( $\text{\AA}^2$ ) for the Crystalline Form of the Compound of Formula (V) |          |          |          |        |    |
|--------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|--------|----|
| Atom                                                                                                                                       | x        | y        | z        | U      |    |
| H(1)                                                                                                                                       | 8339(16) | 350(30)  | 8930(20) | 81(12) | 35 |
| H(3)                                                                                                                                       | 7357(15) | 9950(30) | 7330(30) | 85     |    |
| H(4)                                                                                                                                       | 2972(18) | 6210(40) | 3960(30) | 90(13) |    |
| H(3B)                                                                                                                                      | 9136     | 2391     | 12880    | 97     | 40 |
| H(3C)                                                                                                                                      | 8287     | 2525     | 12061    | 97     |    |
| H(3D)                                                                                                                                      | 8610     | 1098     | 12537    | 97     |    |
| H(4A)                                                                                                                                      | 10173    | 1252     | 12205    | 93     | 45 |
| H(4B)                                                                                                                                      | 9642     | −32      | 11845    | 93     |    |
| H(4C)                                                                                                                                      | 9973     | 708      | 10953    | 93     |    |
| H(5A)                                                                                                                                      | 9609     | 3604     | 11434    | 52     | 50 |
| H(5B)                                                                                                                                      | 9541     | 2928     | 10255    | 52     |    |
| H(6A)                                                                                                                                      | 8224     | 3531     | 9611     | 60     |    |
| H(6B)                                                                                                                                      | 8277     | 4176     | 10797    | 60     | 55 |
| H(7A)                                                                                                                                      | 9146     | 5214     | 9403     | 64     |    |
| H(7B)                                                                                                                                      | 9039     | 5921     | 10493    | 64     |    |
| H(8A)                                                                                                                                      | 7725     | 6306     | 9569     | 66     | 60 |
| H(8B)                                                                                                                                      | 7795     | 5522     | 8494     | 66     |    |
| H(9A)                                                                                                                                      | 8594     | 7218     | 8147     | 78     |    |
| H(9B)                                                                                                                                      | 8541     | 7997     | 9234     | 78     | 65 |
| H(10A)                                                                                                                                     | 7171     | 8278     | 8352     | 68     |    |
| H(11A)                                                                                                                                     | 7493     | 7322     | 6357     | 62     |    |
| H(11B)                                                                                                                                     | 7010     | 6500     | 7009     | 62     | 70 |
| H(12A)                                                                                                                                     | 6516     | 8915     | 5790     | 63     |    |
| H(12B)                                                                                                                                     | 6046     | 8235     | 6540     | 63     |    |
| H(13A)                                                                                                                                     | 5858     | 6307     | 5370     | 63     | 75 |
| H(13B)                                                                                                                                     | 6305     | 7021     | 4609     | 63     |    |
| H(14A)                                                                                                                                     | 4848     | 7873     | 4799     | 59     |    |
| H(14B)                                                                                                                                     | 5302     | 8648     | 4073     | 59     | 80 |
| H(15A)                                                                                                                                     | 4758     | 5967     | 3616     | 53     |    |
| H(15B)                                                                                                                                     | 5189     | 6770     | 2881     | 53     |    |
| H(17A)                                                                                                                                     | 4131     | 6235     | 1151     | 91     | 85 |
| H(17B)                                                                                                                                     | 3718     | 5410     | 1904     | 91     |    |
| H(17C)                                                                                                                                     | 3262     | 6552     | 1090     | 91     |    |
| H(18A)                                                                                                                                     | 4480     | 8671     | 1599     | 90     | 90 |
| H(18B)                                                                                                                                     | 3610     | 8983     | 1531     | 90     |    |
| H(18C)                                                                                                                                     | 4279     | 9376     | 2624     | 90     | 95 |

Selected hydrogen bond information (Å and °) shown in Table 29, below.

TABLE 29

| Selected Hydrogen Bond Formation (Å and °) for the Crystalline Form of the Compound of Formula (V) |           |              |              |        |
|----------------------------------------------------------------------------------------------------|-----------|--------------|--------------|--------|
| D—H . . . A                                                                                        | d(D—H)    | d(H . . . A) | d(D . . . A) | <(DHA) |
| O(1)—H(1) . . . O(3) #1                                                                            | 0.862(18) | 1.78(2)      | 2.618(3)     | 165(3) |
| O(3)—H(3) . . . O(5) #2                                                                            | 0.910(18) | 1.94(3)      | 2.768(3)     | 151(3) |
| O(4)—H(4) . . . O(2) #3                                                                            | 0.859(18) | 1.87(2)      | 2.716(3)     | 168(4) |

Symmetry transformations used to generate equivalent atoms: #1 x,y-1,z #2-x+1, -y+2, -z+1 #3-x+1,y+1/2, -z+3/2

Example 2: Alternative Manufacturing Process for Preparing a Pharmaceutical Material Comprising a Purified Amount of the Compound of Formula (V)

#### Step 1—Preparation of Compound of Formula (I)

##### Lithium Diisopropylamide (LDA) Preparation

A reaction vessel was charged with approximately 321 kg of diisopropylamine and approximately 1870 L of tetrahydrofuran (THF) and the mixture was then cooled to -18° C. to -5° C. Approximately 794 kg of n-butyllithium (n-BuLi, solution in heptane) was slowly dosed while maintaining the temperature at -18° C. to -5° C. The batch was held to -18° C. to -5° C. with stirring.

##### Alkylation Reaction

Approximately 317 kg of ethyl isobutyrate was added to the reactor containing the LDA over a target ≥1 hour with the temperature controlled at -18° C. to -5° C. The line was then rinsed with approximately 100 L THF. The batch was stirred while maintaining the temperature at -18° C. to -5° C. Approximately 460 kg of 1-bromo-5-chloropentane was dosed over a target ≥1 hour at -18° C. to -5° C. The line was then rinsed with approximately 100 L THF. The reaction mixture was stirred at -18° C. to -5° C. and then warmed to 0° C.: 5° C. The reaction was confirmed to be complete using gas chromatography (GC) (1-bromo-5-chloropentane: ≤0.4% area).

##### Quench and Phase Separation

Approximately 1337 kg of a solution of 9% aqueous hydrochloric acid (HCl) was added to the reaction mixture while maintaining the temperature at ≤30° C. to quench the reaction. After dosing, the reaction mixture was stirred at 20° C.: 5° C. for ≥15 minutes. The layers are allowed to settle. The pH of the aqueous layer was then measured (range: pH 6 to 10). If the pH range is not met, additional sodium hydroxide (NaOH) or HCl may be added. The lower aqueous phase was removed for disposal.

##### Distillation and Removal of THE

The solvent was removed by distillation under vacuum at ≤40° C. to the desired volume of approximately 950 L.

The crude compound of formula (I) concentrate was temporarily stored in a reaction vessel or drummed until processing continues to make the compound of formula (II). The compound of formula (I) process was repeated in an identical manner to obtain a second batch.

#### Step 2—Preparation of Compound of Formula (II)

##### Additional Aqueous Wash

Two individual batches of the compound of formula (I) in THE are charged to a vessel. While stirring, approximately 1767 kg of a 5% aqueous HCl solution was then charged at 20° C.±5° C. The mixture was agitated for ≥15 minutes. Agitation was stopped, and the phases were allowed to settle. The lower aqueous phase was removed, leaving the compound of formula (I)/THF in the reactor.

##### Iodide Exchange Reaction

Approximately 4386 L of methyl-ethyl-ketone (MEK) and approximately 824 kg of sodium iodide (NaI) were charged while stirring. The batch was heated to reflux. After approximately 30 hours, GC was used to measure reaction completion (the compound of formula (I) ≤3.0% area). If the reaction has not completed, additional time is allowed and additional NaI may be charged if needed. The mixture was then cooled to approximately 20° C.±10° C.

##### Solvent Exchange and Aqueous Work-up

The batch was concentrated via vacuum distillation at ≤60° C. until no more distillate was collected. Approximately 3000 L of n-heptane was then charged. Approximately 2115 kg of 5% aqueous sodium bisulfite (NaHSO<sub>3</sub>) was prepared and the compound of formula (II) reaction mixture was added. An n-heptane rinse of approximately 612 L was charged. The mixture was stirred at 20° C.±5° C. Agitation was stopped, and the phases were allowed to settle. The lower aqueous phase was removed for disposal. About 1976 L of water was added, the mixture was agitated, the phases were allowed to settle, and the lower aqueous phase was then removed for disposal. The water wash was repeated one more time.

##### Final Concentration

The batch was concentrated using vacuum distillation at ≤50° C. until no more distillate was collected. The compound of formula (II) was then drummed and sampled for intermediate testing. The expected yield range is 80% to 100%.

#### Step 3—Preparation of Compound of Formula (IV)

##### Sodium t-Pentoxide DMAc Preparation

The following first intermediate/compound of formula (IV) sequence is based on a charge of approximately 722 kg of compound of formula (II)/heptane with assay of 90.0% wt/wt, which represents a contained charge of 650 kg of compound of formula (II).

A solution of approximately 1450 kg of N, N-Dimethylacetamide (DMAc) and approximately 267.3 kg of sodium t-pentoxide was prepared in a vessel and the mixture agitated at ≤30° C.

##### Preparation of First Intermediate

The compound of formula (II) in heptane (approximately 722 kg), DMAc (approximately 1259 kg), and TosMIC (approximately 213.8 kg) were charged to a vessel. The mixture was then cooled to -15° C. to 0° C. and the mixture well agitated. To this solution, the sodium t-pentoxide/DMAc mixture was added while at -15° C. to 0° C. The transfer line was rinsed with approximately 178 kg DMAc. The reaction mixture was agitated at -15° C. to 0° C. The conversion was confirmed to be complete using high-performance liquid chromatography (HPLC) with ultraviolet detection (HPLC-UV) (monoalkylated TosMIC ≤3.0% area and compound of formula (II) ≤3.0% area). Optional kicker

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(additional) charges of the compound of formula (II), TosMIC, and sodium t-pentoxide can be employed as required, to ensure completion of the reaction, based on the information presented in Table 30.

TABLE 30

| Optional Kicker Charges for Preparation of Intermediate       |                                                 |
|---------------------------------------------------------------|-------------------------------------------------|
| IPC Test Result Criteria                                      | Kicker Charge Action                            |
| ≤3.0% Monoalkylated TosMIC and >3.0% compound of formula (II) | Charge additional TosMIC and sodium t-pentoxide |
| >3.0% Monoalkylated TosMIC and ≤3.0% compound of formula (II) | Charge additional compound of formula (II)      |
| >3.0% Monoalkylated TosMIC and >3.0% compound of formula (II) | Charge additional sodium t-pentoxide            |

IPC = In-process control;

TosMIC = p-Toluenesulfonylmethyl isocyanide.

#### Quench and Phase Separation

Approximately 2344 L of n-heptane and approximately 2993 L of water were charged to another reactor. The first intermediate reaction mixture was transferred to the heptane/water mixture under temperature control between 0° C. and 40° C. (target 20° C.). The transfer line was then rinsed with approximately 456 L n-heptane. The mixture was agitated for 1 to 3 hours while between 0° C. and 40° C. Agitation was then stopped, and the mixture allowed to settle. The lower aqueous phase was removed for disposal. Approximately 3036 kg of a solution of about 5% aqueous sodium chloride (NaCl) was charged and the mixture agitated. The agitator was then stopped, and the mixture allowed to settle. The lower aqueous phase was removed for disposal.

#### Compound of Formula (IV) Reaction

Approximately 440 L of isopropyl acetate (IPAc) was added to the solution of the first intermediate in heptane and the mixture cooled to -15° C. to 0° C. Concentrated HCl (approximately 112 kg) was then added while maintaining the temperature at -15° C. to 25° C. The reaction mixture was allowed to warm, if needed, to 10° C. to 25° C. The mixture was agitated for 30 to 60 minutes once 10° C. to 25° C. was reached. The reaction conversion was measured using HPLC-UV (the first intermediates ≤20% area).

#### Quench and Phase Separation

In a separate vessel, approximately 177 kg of NaOH (50% wt/wt,) was combined with about 1884 L of water. The resulting aqueous NaOH solution was then added to the reaction mixture at approximately 20° C. (range: 10° C. to 40° C.). The line was rinsed with approximately 124 L water. The mixture was stirred. The neutralization endpoint is pH 9 to 12. Agitation was stopped, and the phases allowed to settle. The lower aqueous phase was removed for disposal. A dilute aqueous solution, containing about 54 kg of NaCl, about 1535 L of water, and about 4.5 kg of 50% NaOH, was prepared in a separate vessel and charged to the compound of formula (IV) product mixture. A water rinse of approximately 126 L was then applied. The mixture was agitated, the phases were settled, and the lower aqueous phase removed for disposal.

#### Concentration

The mixture was concentrated under vacuum at ≤80° C. to a reduced volume. The batch was then cooled to approximately 20° C. and the compound of formula (IV) concentrate held until the second batch was prepared. The compound of formula (IV) process was repeated in an identical manner to provide a second batch of compound of formula (IV) in heptane. The second batch (in heptane) was then combined with the first batch for final distillation. The

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distillation was monitored using GC (compound of formula (IV) ≥75% weight). The packaging of product was performed. The product was weighed and sampled for intermediate testing. The expected yield range is 85% to 105%.

#### Step 4—Preparation of Compound of Formula (V) (Crude Bempedoic Acid)

##### Reaction 1 (Ketone Reduction)

Approximately 710 kg the compound of formula (IV) and approximately 1420 kg ethanol (EtOH) were charged to a vessel. While maintaining the batch at 25° C. ±10° C., approximately 202 kg sodium borohydride (NaBH<sub>4</sub>, 12 wt % in 40% NaOH, approximately 0.35 eq.) was charged. The charging line was then rinsed with approximately 202 kg water. After holding at 25° C. ±5° C. for ≥1 hour the conversion was measured using HPLC-UV (compound of formula (IV) ≤0.9% area).

##### Reaction 2 (Saponification)

Approximately 567 kg of a solution of NaOH (50% wt/wt) is charged at 15° C. to 50° C. The charging line is rinsed with approximately 202 kg water, and the reaction mixture is warmed at 50° C. ±5° C. for ≥6 hours. The saponification is measured using HPLC-UV (compound of formula (V) monoethyl ester ≤1.3% area). Approximately 2440 kg water is charged to the reaction mixture. EtOH and water are distilled under vacuum and at ≤50° C. until the batch volume reaches the target level (approximately 2845 L).

##### pH Adjustment, Phase Separation, and Extraction

The mixture was diluted with approximately 356 L of water and 2121 kg methyl tert-butyl ether (MTBE) was then added while maintaining the batch at 15° C. to 50° C. The batch was cooled to 10° C. to 20° C. Concentrated HCl (approximately 912 kg) was added slowly at 10° C. to 20° C. A sample was taken for pH analysis and the pH was adjusted with HCl or NaOH as needed (target pH range: 5.0 to 6.0). Agitation was stopped, and the phases allowed to settle. The lower aqueous phase was removed for disposal. The concentration of the compound of formula (V) in MTBE was measured using HPLC-UV. The batch was then transferred to another vessel and rinsed forward with approximately 629 kg of MTBE.

#### Step 5—Purification of Compound of Formula (V)/Preparation of Crystalline Form of Compound of Formula (V)

##### Silica Gel Preparations

The diameter×height ratio for the silica gel plug can vary from 1×0.8 to 1×3. Approximately 78 kg of silica gel was charged to a filter. The silica gel bed was prepared by recirculating approximately 1173 kg of EtOAc at 50° C. ±5° C. Excess EtOAc was removed immediately prior to filtration of the compound of formula (V) batch.

##### Solvent Exchange to Ethyl Acetate

The compound of formula (V) in MTBE was concentrated under vacuum at ≤50° C. to approximately 1148 L. Approximately 2608 kg of EtOAc was charged and the batch was then concentrated again to approximately 1148 L. Approximately 2086 kg of EtOAc was charged and distillation repeated. EtOAc (approximately 2086 kg) was charged and the batch was then sampled. The solvent exchange was measured with GC (MTBE ≤0.1% weight). Additional EtOAc charges and distillations may be performed if necessary.

## Silica Gel Filtration

When the solvent exchange was complete, the batch was warmed to 45° C. to 55° C. Then the batch was filtered through the preheated silica gel plug into another reactor. To rinse the line and the silica gel, approximately 521 kg of EtOAc was charged to the reactor, warmed to 50° C.±5° C., and then warm EtOAc was transferred through the silica gel plug. The purified compound of formula (V) in EtOAc mixture was then partially concentrated by distillation under vacuum, at ≤50° C., to a final volume of approximately 2321 L.

## Crystallization

The concentrated compound of formula (V) in EtOAc was adjusted to approximately 50° C.±5° C. Approximately 31.3 kg of water was charged, and the solution was maintained at 50° C.±5° C. for ≥1 hour. The solution was then slowly cooled to 22° C.±5° C., with agitation, over ≥2 hours to start the crystallization of the compound of formula (V). The mixture was stirred at approximately 22° C.±5° C. for ≥6 hours and then slurry formation confirmed. If a slurry is not present, additional stirring at 20° C. to 25° C. and seeding may be performed if necessary. The batch was then slowly cooled to 0° C.±5° C. over ≥2 hours and stirred for ≥6 hours at 0° C.±5° C.

## Optional In Situ Filtration

For vessels designed with in-reactor (in situ) filtration, the slurry was allowed to settle at approximately 0° C. The batch was then filtered, and the wet cake left in the reactor. Approximately 1386 kg of EtOAc was charged to another vessel, chilled to 0° C.±5° C. and transferred to the reactor containing wet cake. The batch was agitated at 0° C.±5° C. and then allowed to settle. The solids were filtered. The slurry wash and filtration processes were repeated 3 times in an identical manner.

Step 6—Preparation of Pharmaceutical Material  
Including Purified Amount of Compound of  
Formula (V)

## Recrystallization (Ethyl Acetate Water)

Following in situ filtration, the amount of the compound of formula (V) is assumed to be approximately 488 kg, based on 100% conversion of the compound of formula (IV) charge amount (710 kg) used to prepare the compound of formula (V).

In a reactor, EtOAc was charged to the compound of formula (V) solids until the volume reaches the 1867 L mark and the suspension was then heated to 55° C. to 60° C. with stirring. To the mixture, approximately 20.5 kg of water was added, and the batch agitated at 55° C. to 70° C. for ≥1 hour. A check for the formation of a solution was performed. The batch was then transferred to another reactor via polish filtration. The reactor, filter, and line were rinsed with EtOAc. The temperature was then adjusted to 55° C.±5° C. The hot solution was then cooled over ≥2 hours to 30° C.±5° C. and then agitated at 30° C.±5° C. for ≥2 hours. If a slurry was not present, additional stirring at 30° C.±5° C. and seeding may be performed as required. The batch was then heated to 40° C. 5° C., over ≥1 hour and held at 40° C.±5° C. for ≥1 hour. The batch was then cooled to 35° C.±5° C. for over ≥1 hour and held at 35° C.±5° C. for ≥2 hours. The batch was then cooled slowly over ≥5 hours to 5° C.±5° C. and held at 5° C.±5° C. for ≥4 hours.

## Isolation, Drying, and IPC Testing

The resulting solids were isolated by centrifugation and washed with ≤2000 kg acetonitrile. The wet cake was then dried under vacuum at ≤45° C. (jacket). The residual solvent

was measured using GC (ACN≤410 ppm and EtOAc≤5000 ppm) to determine completion of drying. The expected yield range is 66% to 91%. If the pharmaceutical material release specification criteria were met, the product was treated as the final pharmaceutical material. If the pharmaceutical material release specifications were not met, a second recrystallization was conducted.

## Optional Second Recrystallization (Ethyl Acetate Water)

The following procedure describes the second recrystallization of the compound of formula (V) for a batch size of approximately 430 kg.

Compound of formula (V) solids (approximately 430 kg) were charged to a vessel, followed by EtOAc (approximately 1238 kg). The suspension was then heated to 55° C. to 60° C. with stirring. To the mixture, approximately 18 kg of water was added, and the mixture was then agitated at 55° C. to 70° C. for ≥1 hour. A check for the formation of a solution was performed. The temperature was then adjusted to 55° C.±5° C. and the batch transferred to another reactor via polish filtration. The reactor, filter, and line were rinsed with EtOAc and the temperature then adjusted to 55° C.: 5° C. The hot solution was then cooled over ≥2 hour to 30° C.±5° C. and then agitated at 30° C.: 5° C. for ≥2 hours. If a slurry was not present, additional stirring at 30° C.±5° C. and seeding may be performed as required. The batch was then heated to 40° C.±5° C., over ≥1 hour and held at 40° C.±5° C. for ≥1 hour. The batch was then cooled to 35° C.: 5° C. for over ≥1 hour and held at 35° C.: 5° C. for ≥2 hours. The batch was then cooled slowly over ≥5 hours to 5° C.±5° C. and held at 5° C.±5° C. for ≥4 hours.

## Optional Isolation, Drying, and IPC Testing

The resulting solids were isolated by centrifugation and washed with ≤2000 kg acetonitrile. The wet cake was dried under vacuum at ≤45° C. (jacket). The residual solvent was then measured using GC (ACN≤410 ppm and EtOAc≤5000 ppm) to determine completion of drying. The expected yield range is 75% to 100%. If the impurity profile criteria were met, the product is treated as the final pharmaceutical material.

Example 3: Analytical Method for Determining the  
Purity of the Compound of Formula (V)

## Determining the Amount of Impurities

The amount of impurities present in the purified form of the compound of formula (V) was determined using a high performance liquid chromatograph equipped with gradient capability, a thermostatic column compartment and a charged aerosol detection (CAD) detector.

The amount of impurities within the purified form of the compound of formula (V) was determined to be in the range of 0.05-0.50% w/w.

Column: Waters XBridge BEH C18 (4.6 mm i.d.×150 mm, 2.5 μm)

Mobile Phase: A: 0.05% Formic acid (HCOOH) in water (H<sub>2</sub>O)

Mobile Phase: B: 0.05% HCOOH in acetonitrile (ACN)

Sample temperature: Ambient

Column temperature: 40° C.

Gradient (time: A:B): (0 min: 90:10; 8.5 min., 56:44; 20 min, 45:55; 32 min., 5:95; 36 min., 5:95).

Flow rate: 1.2 mL/min

Retention time: ~15.2 min (purified form of bempedoic acid)

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Determining the Purity of the Compound of Formula (V) (Bempedoic Acid)

The level of purity of the purified form of the compound of formula (V) was determined using a high performance liquid chromatograph equipped with a UV detector.

The assay for the purified form of the compound of formula (V) was determined to be in the range of 98-102% (anhydrous, solvent-free basis).

Column: Waters XBridge BEH C18 (4.6 mm i.d.×150 mm, 2.5 μm)

Mobile Phase: A: 0.05% Phosphoric acid (H<sub>3</sub>PO<sub>4</sub>) in H<sub>2</sub>O:ACN (50:50)

Sample temperature: Ambient

Column temperature: 40° C.

Detection: 215 nm

Flow rate: 1.2 mL/min

Analysis time: 16 min.

Gradient: Isocratic

Retention time: ~4.6 min (purified form of bempedoic acid)

#### INCORPORATION BY REFERENCE

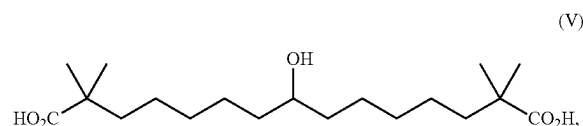
The entire disclosure of each of the patent documents and scientific articles referred to herein is incorporated by reference for all purposes.

#### EQUIVALENTS

The invention may be embodied in other specific forms without departing from the spirit or essential characteristics thereof. The foregoing embodiments are therefore to be considered in all respects illustrative rather than limiting the invention described herein. Scope of the invention is thus indicated by the appended claims rather than by the foregoing description, and all changes that come within the meaning and range of equivalency of the claims are intended to be embraced therein.

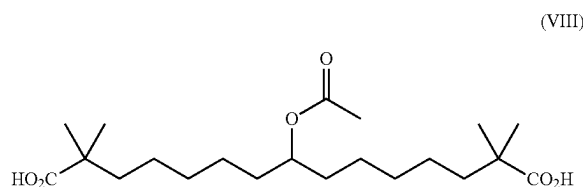
We claim:

1. A pharmaceutical material comprising a crystalline form of the compound of formula (V):



or a pharmaceutically acceptable salt thereof;

wherein the pharmaceutical material comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 99.0% by weight based on the total weight of the pharmaceutical material; and the pharmaceutical material comprises 0.0001% to 0.15%



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or a pharmaceutically acceptable salt thereof, based on the total weight of the pharmaceutical material.

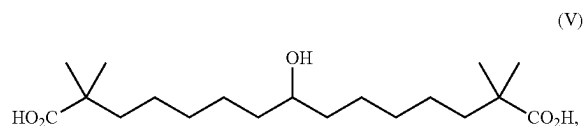
2. The pharmaceutical material of claim 1, wherein the crystalline form of the compound of formula (V) exhibits an X-ray powder diffraction pattern comprising peaks at the following diffraction angles (2θ): 10.4±0.2, 17.91±0.2, 18.8±0.2, 19.5±0.2, and 20.7±0.2.

3. The pharmaceutical material of claim 1, wherein the crystalline form of the compound of formula (V) is characterized by an X-ray powder diffraction pattern substantially the same as shown in FIG. 4.

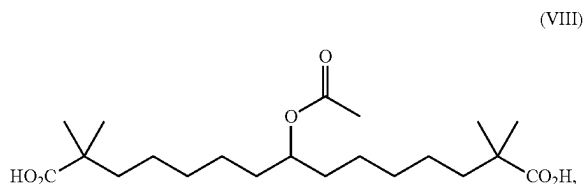
4. The pharmaceutical material claim 1, wherein the crystalline form of the compound of formula (V) has a melting point onset as determined by differential scanning calorimetry in the range of from about 90° C. to about 94° C.

5. The pharmaceutical material of claim 1, wherein the pharmaceutical material comprises the compound of formula (V) in an amount greater than 99.5% by weight based on the total weight of the pharmaceutical material.

6. A pharmaceutical material comprising a crystalline form of the compound of formula (V):



wherein the pharmaceutical material comprises the compound of formula (V) in an amount of from about 98% to about 102% by weight based on the total weight of the pharmaceutical material, as determined by a high performance liquid chromatography (HPLC) assay; and the pharmaceutical material comprises 0.0001% to 0.15% of a compound of formula (VIII):



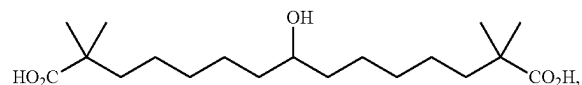
or a pharmaceutically acceptable salt thereof, based on the total weight of the pharmaceutical material.

7. The pharmaceutical material of claim 6, wherein the HPLC assay uses a C18 column (4.6 mm i.d.×150 mm, 2.5 μm) at a temperature of about 40° C., with isocratic elution of a mobile phase comprising about 0.05% phosphoric acid in water/acetonitrile (about 50:50) at a flow rate of about 1.2 mL/minute, and detection at 215 nm, wherein the retention time of the compound of formula (V) is about 4.6 minutes.

8. A method of treating primary hyperlipidemia in a patient in need thereof, the method comprising administering to a patient a therapeutically-effective amount of the pharmaceutical material of claim 1.

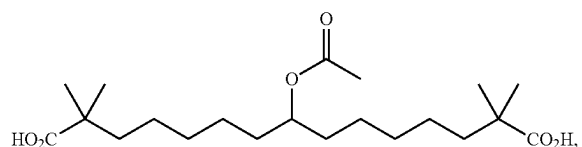
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9. A pharmaceutical material comprising a crystalline form of the compound of formula (V):



or a pharmaceutically acceptable salt thereof;

wherein the pharmaceutical material comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 85% by weight based on the total weight of the pharmaceutical material, and the pharmaceutical material comprises 0.0001% to 0.15% of a compound of formula (VIII):



or a pharmaceutically acceptable salt thereof, based on the total weight of the pharmaceutical material.

10. The pharmaceutical material of claim 9 wherein the crystalline form of the compound of formula (V) exhibits an X-ray powder diffraction pattern comprising peaks at the

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following diffraction angles (2θ): 10.4±0.2, 17.9±0.2, 18.8±0.2, 19.5±0.2, and 20.7±0.2.

11. The pharmaceutical material of claim 9, wherein the pharmaceutical material has a melting point onset as determined by differential scanning calorimetry in a range from 90° C. to 94° C.

12. The pharmaceutical material of claim 9 wherein the pharmaceutical material comprises the compound of formula (V), or a pharmaceutically acceptable salt thereof, in an amount greater than 90% by weight based on the total weight of the pharmaceutical material.

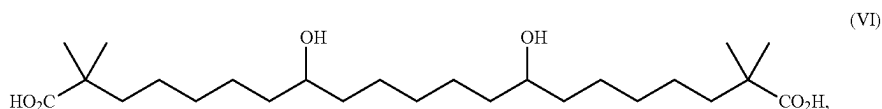
13. The pharmaceutical material of claim 9, wherein the pharmaceutical material comprises the compound of formula (V) or a pharmaceutically acceptable salt thereof, in an amount greater than 95% by weight based on the total weight of the pharmaceutical material.

14. The pharmaceutical material of claim 9, wherein the pharmaceutical material comprises the compound of formula (V) or a pharmaceutically acceptable salt thereof, in an amount greater than 96% by weight based on the total weight of the pharmaceutical material.

15. The pharmaceutical material of claim 9, wherein the pharmaceutical material comprises the compound of formula (V) or a pharmaceutically acceptable salt thereof, in an amount greater than 97% by weight based on the total weight of the pharmaceutical material.

16. The pharmaceutical material of claim 9, wherein the pharmaceutical material comprises the compound of formula (V) or a pharmaceutically acceptable salt thereof, in an amount greater than 98% by weight based on the total weight of the pharmaceutical material.

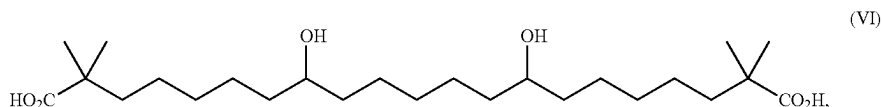
17. The pharmaceutical material of claim 9, further comprising a compound of formula (VI):



or a pharmaceutically acceptable salt thereof.

18. The pharmaceutical material of claim 17, wherein the pharmaceutical material comprises 0.0001% to 0.15% of a compound of formula (VI), based on the total weight of the pharmaceutical material.

19. The pharmaceutical material of claim 1, further comprising a compound of formula (VI):



or a pharmaceutically acceptable salt thereof.

20. The pharmaceutical material of claim 19, wherein the pharmaceutical material comprises 0.0001% to 0.15% of a compound of formula (VI), based on the total weight of the pharmaceutical material.

\* \* \* \* \*